EXPONENTIAL SUMS OVER THE RIEMANN ZEROS VERSUS THE RIEMANN–VON MANGOLDT CLOCK, WITH FINITE MÖBIUS INTERPOLATION AND PRIME-POWER SPECTRAL IDENTITIES

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Remark 0.1. This manuscript develops the analytic identities in full theorem–proof form. Standard and literature-based material is cited at the point of use. Original finite identities are proved directly. Heuristics and conjectures are explicitly labelled. Numerical evidence is confined to the final part, immediately before the conclusions, and is never used as a substitute for an analytic proof.

1. Introduction: the exponential zero sum versus the Riemann–von Mangoldt clock

The point of departure is not the finite interpolation introduced later. It is the direct comparison between the exponential walk formed by the actual ordinates of the nontrivial zeros and the walk formed by the smooth Riemann–von Mangoldt clock. Let

$$0 < \gamma_1 \le \gamma_2 \le \cdots \tag{1.1}$$

denote the positive ordinates, repeated according to multiplicity, and fix a frequency $\omega > 0$. The primary object is

$$S_N(\omega) = \sum_{k=1}^{N} e^{i\omega\gamma_k}. \tag{1.2}$$

The smooth comparison sequence is obtained by inverting the principal Riemann–von Mangoldt counting function. Writing

$$N_0(T) = \frac{T}{2\pi}\log\left(\frac{T}{2\pi e}\right) + \frac{7}{8}, \tag{1.3}$$

we define $\widetilde{\gamma}_k$ by

$$N_0(\widetilde{\gamma}_k) = k - \frac{1}{2} \tag{1.4}$$

and form

$$S_N^{\mathrm{RVM}}(\omega) = \sum_{k=1}^{N} e^{i\omega\widetilde{\gamma}_k}. \tag{1.5}$$

The discrepancy

$$\mathcal{E}_N(\omega) = S_N(\omega) - S_N^{\mathrm{RVM}}(\omega) \tag{1.6}$$

is the object from which the subsequent prime-power investigation arose. The smooth clock reproduces the main zero density. For admissible regularized test functions, the arithmetic side of the explicit formula is supported on prime powers; this does not by itself assert that every component of the sharp residual is a prime-power mode. The two branches with frequencies

$$\omega - \log n \qquad \text{and} \qquad \omega + \log n \tag{1.7}$$

are therefore not an auxiliary model imposed on the data. They are the Fourier displacements of the two prime terms in the Guinand–Weil explicit formula.

A second construction was found while studying how a finite Möbius section can be completed by the reciprocal zeta function. It begins with an affine interpolation of the terminal prime coefficients. The smallest numerical example already contains the complete mechanism. For $N = 10$,

$$A_{10}(s) = 1 - 2^{-s} - 3^{-s} - 5^{-s} + 6^{-s} + 10^{-s}, \tag{1.8}$$

$$P_{10}(s) = 7^{-s}, \tag{1.9}$$

$$B_{10} = A_{10}(0) = 0. \tag{1.10}$$

If the unknown terminal value is denoted by the letter m, the first interpolation is

$$\mathcal{I}_{10}(s; m) = A_{10}(s) + m7^{-s}. \tag{1.11}$$

Solving

$$\mathcal{I}_{10}(s; m) = \zeta(s)^{-1} \tag{1.12}$$

for the number m produces

$$\Phi_{10}(s) = 7^s \left(\zeta(s)^{-1} - A_{10}(s)\right). \tag{1.13}$$

The general symbolic form is

$$\mathcal{I}_N(s; m) = A_N(s) + \frac{m - B_N}{h_N} P_N(s), \tag{1.14}$$

and solving $\mathcal{I}_N(s; m) = \zeta(s)^{-1}$ yields

$$\Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)}. \tag{1.15}$$

This finite interpolation is analytically independent of the phase-walk experiment, but it uses the same reciprocal-zeta and prime-support architecture. It is included because it gives exact finite identities, not because it is the historical origin of the project.

The principal rigorous results established below are the following. First, the Riemann–von Mangoldt comparison is written exactly as a Stieltjes integral and its smooth antiderivative is derived. Second, the finite interpolation is solved completely: its value at the origin, its limit on the positive real axis, its exponential rate, its rounding criterion, its poles, and its finite parts are proved. Third, the two prime-power branches are derived from a regularized explicit formula and their signed invariant

$$\Delta_\omega(Q) = \sum_{n=p^r>Q} \left(|c_n^-|^2 - |c_n^+|^2\right) \tag{1.16}$$

is shown to satisfy

$$\Delta_\omega(Q) \sim \frac{\omega}{\pi^2 \log Q}. \tag{1.17}$$

Fourth, the complete-zero walk and the ordinate-only walk are separated by an atomic positive measure. This gives several exact routes to the Riemann Hypothesis. None is claimed to be completed: each route is stated with the precise missing uniform or local estimate.

Numerical material is placed only in the final part. It supports the analytic structure but is not used in any proof.

Part 1. The exponential walk and the Riemann–von Mangoldt benchmark

2. The zero-counting function and its smooth inverse

Definition 2.1 (Midpoint zero count)**.** Let $N(T)$ be the number of nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \le T$, counted with multiplicity. At a zero ordinate $T = \tau$, let

$$N^{\#}(\tau) = \frac{1}{2}\left(N(\tau^-) + N(\tau^+)\right). \tag{2.1}$$

For a simple ordered ordinate γ_k, this equals $k - 1/2$.

Definition 2.2 (Midpoint argument function and exact smooth remainder)**.** For T that is not a zero ordinate, let

$$S(T) = \frac{1}{\pi} \arg \zeta\left(\frac{1}{2} + iT\right), \tag{2.2}$$

with the standard path from 2 to $2 + iT$ and then to $1/2 + iT$. At a zero ordinate τ, define $S^{\#}(\tau)$ as the average of the two one-sided limits. Define

$$R_\vartheta(T) = 1 + \frac{\vartheta(T)}{\pi} - N_0(T), \tag{2.3}$$

where

$$\vartheta(T) = \Im \log \Gamma\left(\frac{1}{4} + \frac{iT}{2}\right) - \frac{T}{2} \log \pi. \tag{2.4}$$

Theorem 2.3 (Riemann–von Mangoldt formula)**.** *For every $T > 0$, with midpoint interpretation at zero ordinates,*

$$\boxed{N^{\#}(T) = N_0(T) + S^{\#}(T) + R_\vartheta(T).} \tag{2.5}$$

Moreover,

$$R_\vartheta(T) = O(T^{-1}) \qquad (T \to \infty). \tag{2.6}$$

The exact counting identity and the asymptotic expansion are standard; see [12, Chapter 9]*.*

Proof. For T that is not a zero ordinate, the argument principle applied to the completed zeta function

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) \tag{2.7}$$

gives the exact identity

$$N(T) = 1 + \frac{\vartheta(T)}{\pi} + S(T). \tag{2.8}$$

At a zero ordinate, the average of the two one-sided identities gives the same formula with $N^{\#}$ and $S^{\#}$. Subtracting $N_0(T)$ and using the definition (2.3) proves (2.5) exactly.

It remains to estimate R_ϑ. Stirling's formula in a fixed sector gives

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + O(|z|^{-1}). \tag{2.9}$$

With $z = 1/4 + iT/2$, taking imaginary parts yields

$$\vartheta(T) = \frac{T}{2} \log\left(\frac{T}{2\pi}\right) - \frac{T}{2} - \frac{\pi}{8} + O(T^{-1}). \tag{2.10}$$

Consequently,

$$1 + \frac{\vartheta(T)}{\pi} = \frac{T}{2\pi} \log\left(\frac{T}{2\pi e}\right) + \frac{7}{8} + O(T^{-1}) \tag{2.11}$$

$$= N_0(T) + O(T^{-1}), \tag{2.12}$$

which is exactly (2.6). □

Theorem 2.4 (Principal Lambert–W inverse of the smooth clock)**.** *Let $x > -1/8$. The equation $N_0(T) = x$ has a unique solution $T > 2\pi$, namely*

$$\boxed{T = \frac{2\pi(x - 7/8)}{W_0((x - 7/8)/e)}.} \tag{2.13}$$

Here W_0 is the principal real branch, and the value at $x = 7/8$ is understood by continuity as $T = 2\pi e$. In particular, for every $k \geq 1$,

$$\widetilde{\gamma}_k = \frac{2\pi(k - 11/8)}{W_0((k - 11/8)/e)}. \tag{2.14}$$

Proof. Put

$$y = x - \frac{7}{8}, \qquad u = \frac{T}{2\pi e}. \tag{2.15}$$

The equation $N_0(T) = x$ is

$$y = e\, u \log u. \tag{2.16}$$

Writing $v = \log u$ gives

$$v e^v = \frac{y}{e}. \tag{2.17}$$

Since $x > -1/8$, one has $y/e > -1/e$, so the principal branch $W_0(y/e)$ is real and greater than -1. Thus

$$v = W_0(y/e), \qquad u = e^v = \frac{y/e}{W_0(y/e)}, \tag{2.18}$$

and hence (2.13). The inequality $W_0(y/e) > -1$ implies $u > e^{-1}$, hence $T > 2\pi$. Conversely, $N_0'(T) = (2\pi)^{-1} \log(T/2\pi) > 0$ on $(2\pi, \infty)$, so this solution is unique in that interval. At $y = 0$, $W_0(y/e) \sim y/e$, giving the continuous value $T = 2\pi e$. Finally, $x = k - 1/2 > -1/8$ for every $k \geq 1$, and substitution gives (2.14). □

3. Stieltjes representation and the smooth RVM walk

Definition 3.1 (Symmetric phase sum)**.** Define

$$S_\omega^{\#}(T) = \sum_{0<\gamma<T} m(\gamma) e^{i\omega\gamma} + \frac{1}{2} \sum_{\gamma=T} m(\gamma) e^{i\omega\gamma}. \tag{3.1}$$

Proposition 3.2 (Atomic representation and integration by parts)**.** *Let*

$$\mathrm{d}N = \sum_{\gamma>0} m(\gamma)\delta_\gamma. \tag{3.2}$$

Then

$$\boxed{S_\omega^{\#}(T) = \int_{(0,T)} e^{i\omega t}\ \mathrm{d}N(t) + \frac{1}{2} e^{i\omega T} \Delta N(T).} \tag{3.3}$$

If $0 < T_0 < T$ and neither endpoint is a zero ordinate, then

$$\int_{(T_0,T]} e^{i\omega t}\ \mathrm{d}N(t) = e^{i\omega T} N(T) - e^{i\omega T_0} N(T_0) \tag{3.4}$$

$$- i\omega \int_{T_0}^{T} e^{i\omega t} N(t)\ \mathrm{d}t. \tag{3.5}$$

The midpoint formula at a zero endpoint is obtained by averaging the two one-sided identities.

Proof. The first identity is the direct evaluation of the atomic measure $\mathrm{d}N$: atoms strictly below T receive full weight and the atom at T receives half weight. Merely changing the point value of the counting function would not change its Stieltjes measure, which is why the explicit endpoint term is required.

For endpoints away from atoms, N is a right-continuous bounded-variation function and $f(t) = e^{i\omega t}$ is continuously differentiable. The Riemann–Stieltjes integration-by-parts formula gives

$$\int_{(T_0,T]} f \, \mathrm{d}N = f(T)N(T) - f(T_0)N(T_0) - \int_{T_0}^{T} N(t) f'(t) \, \mathrm{d}t. \tag{3.6}$$

Substitution of $f'(t) = i\omega e^{i\omega t}$ proves the displayed formula. Averaging the limits from the two sides supplies half of any endpoint atom. □

Theorem 3.3 (Smooth Riemann–von Mangoldt contribution)**.** *The Stieltjes integral of the smooth density*

$$N_0'(t) = \frac{1}{2\pi} \log\!\left(\frac{t}{2\pi}\right) \tag{3.7}$$

has the antiderivative

$$\boxed{A_{\mathrm{smooth},\omega}(T) = \frac{e^{i\omega T}\log(T/2\pi) - \mathrm{Ei}(i\omega T)}{2\pi i\omega} + C_\omega.} \tag{3.8}$$

Consequently, for any $T_0 > 0$,

$$\int_{T_0}^{T} e^{i\omega t} N_0'(t) \, \mathrm{d}t = A_{\mathrm{smooth},\omega}(T) - A_{\mathrm{smooth},\omega}(T_0). \tag{3.9}$$

Proof. Differentiate the numerator in (3.8). The derivative of the first term is

$$\frac{\mathrm{d}}{\mathrm{d}T}\left[e^{i\omega T}\log(T/2\pi)\right] = i\omega e^{i\omega T}\log(T/2\pi) + \frac{e^{i\omega T}}{T}. \tag{3.10}$$

Since $\frac{\mathrm{d}}{\mathrm{d}z}\,\mathrm{Ei}(z) = e^z/z$, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}T}\,\mathrm{Ei}(i\omega T) = \frac{e^{i\omega T}}{T}. \tag{3.11}$$

The two T^{-1} terms cancel. Division by $2\pi i\omega$ leaves exactly $e^{i\omega T}N_0'(T)$. The definite identity follows from the fundamental theorem of calculus on any interval avoiding the chosen branch cut; a change of branch alters only the constant C_ω. □

Corollary 3.4 (Exact RVM remainder decomposition)**.** *Let $0 < T_0 < T$ and suppose first that neither endpoint is a zero ordinate. Then*

$$\int_{(T_0,T]} e^{i\omega t} \, \mathrm{d}N(t) = A_{\mathrm{smooth},\omega}(T) - A_{\mathrm{smooth},\omega}(T_0) \tag{3.12}$$

$$+ \int_{(T_0,T]} e^{i\omega t} \, \mathrm{d}S^{\#}(t) + \int_{T_0}^{T} e^{i\omega t} R_\vartheta'(t) \, \mathrm{d}t. \tag{3.13}$$

At a zero endpoint, the corresponding symmetric identity is obtained by averaging the one-sided limits.

Proof. The exact identity (2.5) is an equality of locally bounded-variation functions:

$$N^{\#}(t) = N_0(t) + S^{\#}(t) + R_\vartheta(t). \tag{3.14}$$

Away from endpoint atoms, taking Stieltjes differentials gives

$$\mathrm{d}N = N_0'(t) \, \mathrm{d}t + \mathrm{d}S^{\#}(t) + R_\vartheta'(t) \, \mathrm{d}t. \tag{3.15}$$

Integrating against $e^{i\omega t}$ over $(T_0, T]$ and invoking Theorem 3.3 for the N_0'-term proves the formula. Midpoint endpoint weights follow from the same one-sided averaging used in Proposition 3.2. □

4. Exact comparison of the indexed zero walk with the smooth clock

Definition 4.1. For each k, define the RVM displacement

$$\delta_k = \gamma_k - \widetilde{\gamma}_k. \tag{4.1}$$

Lemma 4.2 (Exact phase perturbation identity)**.** *For real x and h,*

$$e^{i(x+h)} - e^{ix} = ih \int_0^1 e^{i(x+th)}\, \mathrm{d}t. \tag{4.2}$$

In particular,

$$|e^{i(x+h)} - e^{ix}| \le |h|, \tag{4.3}$$

$$\left|e^{i(x+h)} - e^{ix} - ihe^{ix}\right| \le \frac{h^2}{2}. \tag{4.4}$$

Proof. Define $F(t) = e^{i(x+th)}$ on $[0,1]$. Since $F'(t) = ihe^{i(x+th)}$, the fundamental theorem of calculus gives

$$e^{i(x+h)} - e^{ix} = F(1) - F(0) = ih \int_0^1 e^{i(x+th)}\, \mathrm{d}t. \tag{4.5}$$

The integrand has modulus one, so taking absolute values yields the Lipschitz estimate. For the quadratic estimate, subtract ihe^{ix}:

$$e^{i(x+h)} - e^{ix} - ihe^{ix} = ihe^{ix} \int_0^1 (e^{ith} - 1)\, \mathrm{d}t. \tag{4.6}$$

Apply the already proved Lipschitz bound to $e^{ith} - 1$, obtaining $|e^{ith} - 1| \le t|h|$. Integration of t from zero to one gives $h^2/2$. □

Theorem 4.3 (Exact and first-order decompositions of the discrepancy)**.** *For every N,*

$$\mathcal{E}_N(\omega) = i\omega \sum_{k=1}^{N} \delta_k \int_0^1 e^{i\omega(\widetilde{\gamma}_k + t\delta_k)}\, \mathrm{d}t. \tag{4.7}$$

Moreover,

$$\mathcal{E}_N(\omega) = i\omega \sum_{k=1}^{N} \delta_k e^{i\omega\widetilde{\gamma}_k} + \mathcal{R}_N(\omega), \tag{4.8}$$

where

$$|\mathcal{R}_N(\omega)| \le \frac{\omega^2}{2} \sum_{k=1}^{N} \delta_k^2. \tag{4.9}$$

Proof. For each k, apply Lemma 4.2 to

$$x = \omega\widetilde{\gamma}_k, \qquad h = \omega(\gamma_k - \widetilde{\gamma}_k) = \omega\delta_k. \tag{4.10}$$

The resulting identity is

$$e^{i\omega\gamma_k} - e^{i\omega\widetilde{\gamma}_k} = i\omega\delta_k \int_0^1 e^{i\omega(\widetilde{\gamma}_k + t\delta_k)}\, \mathrm{d}t. \tag{4.11}$$

Summing from 1 to N gives the exact formula. For the first-order form, the quadratic estimate in the lemma gives a remainder r_k with

$$|r_k| \le \frac{\omega^2\delta_k^2}{2}. \tag{4.12}$$

The triangle inequality then gives the stated bound for the sum of all remainders. No restriction on the signs or dependence of the displacements is needed. □

Definition 4.4 (Midpoint rank of a distinct ordinate)**.** Let

$$0 < \tau_1 < \tau_2 < \cdots$$

be the distinct positive zero ordinates, and let m_j be the total multiplicity at τ_j. Define the midpoint rank

$$\kappa_j = N(\tau_j^-) + \frac{m_j}{2} = N^{\#}(\tau_j), \tag{4.13}$$

the smooth comparison ordinate

$$\widetilde{\tau}_j = N_0^{-1}(\kappa_j), \tag{4.14}$$

and the displacement

$$\delta_j^{\#} = \tau_j - \widetilde{\tau}_j. \tag{4.15}$$

Proposition 4.5 (Displacement in terms of the Riemann–von Mangoldt remainder)**.** *For every distinct positive zero ordinate τ_j, there is a point ξ_j between τ_j and $\widetilde{\tau}_j$ such that*

$$\boxed{\delta_j^{\#} = -\frac{S^{\#}(\tau_j) + R_\vartheta(\tau_j)}{N_0'(\xi_j)}}. \tag{4.16}$$

Equivalently,

$$\delta_j^{\#} = -\frac{2\pi\,[S^{\#}(\tau_j) + R_\vartheta(\tau_j)]}{\log(\xi_j/2\pi)}. \tag{4.17}$$

If τ_j is simple and is the j-th ordinate when ordinates are listed with multiplicity, then $\kappa_j = j - \frac{1}{2}$, $\widetilde{\tau}_j = \widetilde{\gamma}_j$, and $\delta_j^{\#} = \delta_j$.

Proof. By Definitions 2.1 and 4.4,

$$N_0(\widetilde{\tau}_j) = \kappa_j = N^{\#}(\tau_j). \tag{4.18}$$

The exact Riemann–von Mangoldt identity (2.5) therefore gives

$$N_0(\widetilde{\tau}_j) - N_0(\tau_j) = S^{\#}(\tau_j) + R_\vartheta(\tau_j). \tag{4.19}$$

Both τ_j and $\widetilde{\tau}_j$ exceed 2π, so N_0 is differentiable on the closed interval joining them and $N_0'(t) = (2\pi)^{-1}\log(t/2\pi) > 0$ there. The mean-value theorem provides ξ_j between the two ordinates such that

$$N_0(\widetilde{\tau}_j) - N_0(\tau_j) = N_0'(\xi_j)(\widetilde{\tau}_j - \tau_j) \tag{4.20}$$

$$= -N_0'(\xi_j)\delta_j^{\#}. \tag{4.21}$$

Solving for $\delta_j^{\#}$ gives (4.16); substitution of the explicit derivative gives (4.17). In the simple case $m_j = 1$ and $N(\tau_j^-) = j - 1$, so $\kappa_j = j - \frac{1}{2}$, proving the final assertion. □

Remark 4.6. Equations (4.8) and (4.17) explain why a comparison with the RVM clock is sensitive to the argument of zeta. The explicit formula then rewrites that argument contribution in terms of prime powers. This is the analytic route by which the prime-power branches entered the project.

Part 2. Finite Möbius interpolation

5. The first affine interpolation: numerical and symbolic forms

Definition 5.1 (Terminal affine interpolation)**.** Let $Q_N = \{q_1, \dots, q_{h_N}\}$ be the primes in $(N/2, N]$, and let A_N, P_N, B_N be defined in Section 6. For a prescribed terminal sum $m \in \mathbb{C}$, define

$$\boxed{\mathcal{I}_N(s; m) = A_N(s) + \frac{m - B_N}{h_N} P_N(s).} \tag{5.1}$$

This is the symmetric affine interpolation in which all terminal-prime coefficients are assigned their common mean

$$\lambda_N(m) = \frac{m - B_N}{h_N}. \tag{5.2}$$

Proposition 5.2 (Recovery of the true finite Möbius polynomial)**.** *For the true Mertens value $m = M(N)$,*

$$\mathcal{I}_N(s; M(N)) = M_s(N). \tag{5.3}$$

Proof. The set Q_N contains only primes, so every terminal Möbius coefficient is exactly -1. Therefore its total contribution at $s = 0$ is $-h_N$, which proves $M(N) = B_N - h_N$. Hence the affine mean coefficient for the true terminal value is

$$\frac{M(N) - B_N}{h_N} = \frac{-h_N}{h_N} = -1. \tag{5.4}$$

Substitution assigns coefficient -1 to every term in $P_N(s)$, so

$$\mathcal{I}_N(s; M(N)) = A_N(s) - P_N(s). \tag{5.5}$$

The right side is exactly the decomposition of $M_s(N)$. □

Example 5.3 (The numerical model $N = 10$)**.** The Möbius values up to ten are

$$(1, -1, -1, 0, -1, 1, -1, 0, 0, 1). \tag{5.6}$$

The only terminal prime is 7, so

$$Q_{10} = \{7\}, \qquad h_{10} = 1, \qquad P_{10}(s) = 7^{-s}, \tag{5.7}$$

$$A_{10}(s) = 1 - 2^{-s} - 3^{-s} - 5^{-s} + 6^{-s} + 10^{-s}, \qquad B_{10} = 0. \tag{5.8}$$

Thus

$$\mathcal{I}_{10}(s; m) = A_{10}(s) + m7^{-s}. \tag{5.9}$$

The actual terminal value is $M(10) = -1$, which restores the coefficient -1 of 7^{-s}.

Theorem 5.4 (Completion equation)**.** *At every point where $P_N(s) \neq 0$, the unique value of m satisfying*

$$\mathcal{I}_N(s; m) = \zeta(s)^{-1} \tag{5.10}$$

is

$$\boxed{m = \Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)}.} \tag{5.11}$$

Proof. Starting from the affine interpolation, the completion equation is

$$A_N(s) + \frac{m - B_N}{h_N} P_N(s) = \zeta(s)^{-1}. \tag{5.12}$$

Because $h_N \neq 0$ and $P_N(s) \neq 0$, this is a nondegenerate linear equation in m. Subtract $A_N(s)$, multiply by $h_N/P_N(s)$, and add B_N to obtain the displayed solution. If two values m_1, m_2 solved the equation, subtracting their equations would give

$$\frac{m_1 - m_2}{h_N} P_N(s) = 0, \tag{5.13}$$

so $m_1 = m_2$. This proves both existence and uniqueness. □

Remark 5.5. The function $\Phi_N(s)$ is therefore not introduced by an arbitrary quotient. It is the terminal value selected by requiring the affine finite interpolation to coincide with the completed reciprocal-zeta Dirichlet series at the spectral parameter s.

6. Terminal half-interval decomposition

Definition 6.1. For an integer $N \geq 2$, define

$$Q_N = \left\{ q \text{ prime} : \frac{N}{2} < q \leq N \right\}, \tag{6.1}$$

and put

$$h_N = |Q_N|, \qquad q_N = \min Q_N. \tag{6.2}$$

Bertrand's postulate guarantees $h_N \geq 1$.

Definition 6.2. Let

$$M_s(N) = \sum_{n \leq N} \frac{\mu(n)}{n^s}, \tag{6.3}$$

$$P_N(s) = \sum_{q \in Q_N} q^{-s}, \tag{6.4}$$

$$A_N(s) = \sum_{\substack{n \leq N \\ n \notin Q_N}} \frac{\mu(n)}{n^s}, \tag{6.5}$$

$$B_N = A_N(0) = \sum_{\substack{n \leq N \\ n \notin Q_N}} \mu(n). \tag{6.6}$$

Since every $q \in Q_N$ is prime and $\mu(q) = -1$,

$$M_s(N) = A_N(s) - P_N(s) \tag{6.7}$$

and

$$M(N) = B_N - h_N. \tag{6.8}$$

Definition 6.3. For $\Re s > 1$, define the exact tail

$$R_N(s) = \frac{1}{\zeta(s)} - M_s(N) = \sum_{n > N} \frac{\mu(n)}{n^s}. \tag{6.9}$$

The first equality supplies a meromorphic continuation of R_N.

Definition 6.4. Define

$$\Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)} \tag{6.10}$$

at points at which the quotient is defined, and meromorphically elsewhere.

7. Canonical identity and endpoint values

Theorem 7.1 (Canonical orientation)**.** *For every $N \geq 2$,*

$$\boxed{\Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)} = M(N) + h_N \frac{R_N(s)}{P_N(s)}.} \tag{7.1}$$

Proof. The functions $A_N(s)$ and $P_N(s)$ are finite Dirichlet polynomials and hence entire. In the half-plane $\Re s > 1$, absolute convergence of the reciprocal-zeta Dirichlet series gives

$$\zeta(s)^{-1} = M_s(N) + R_N(s). \tag{7.2}$$

The terminal decomposition $M_s(N) = A_N(s) - P_N(s)$ therefore yields

$$\zeta(s)^{-1} - A_N(s) = M_s(N) + R_N(s) - A_N(s) \tag{7.3}$$

$$= R_N(s) - P_N(s). \tag{7.4}$$

Divide by $P_N(s)$ at a point where it is nonzero, multiply by h_N, and add B_N:

$$\Phi_N(s) = B_N + h_N \left(\frac{R_N(s)}{P_N(s)} - 1 \right) \tag{7.5}$$

$$= B_N - h_N + h_N \frac{R_N(s)}{P_N(s)}. \tag{7.6}$$

Because the terminal primes all have Möbius value -1, $M(N) = B_N - h_N$. This proves the identity on $\Re s > 1$. Both sides are meromorphic continuations of the same function, so the identity extends to every connected component of their common domain and hence meromorphically to the plane. □

Corollary 7.2 (Value at the origin)**.** *The meromorphic interpolation satisfies*

$$\boxed{\Phi_N(0) = -2.} \tag{7.7}$$

Proof. The point $s = 0$ is regular for every finite Dirichlet polynomial in the formula. The classical special value $\zeta(0) = -1/2$ gives

$$\zeta(0)^{-1} = -2. \tag{7.8}$$

Moreover, by definition,

$$A_N(0) = B_N, \qquad P_N(0) = \sum_{q \in Q_N} 1 = h_N. \tag{7.9}$$

Since $h_N \geq 1$, the denominator does not vanish. Direct substitution in the completion solution gives

$$\Phi_N(0) = B_N + h_N \frac{-2 - B_N}{h_N} \tag{7.10}$$

$$= B_N - 2 - B_N = -2. \tag{7.11}$$

No limiting argument is involved. □

Corollary 7.3 (Large positive real limit)**.** *Along the positive real axis,*

$$\boxed{\lim_{\sigma \to +\infty} \Phi_N(\sigma) = M(N).} \tag{7.12}$$

Proof. For $\sigma > 1$, the tail satisfies

$$|R_N(\sigma)| \leq \sum_{n > N} n^{-\sigma} \leq \int_N^\infty x^{-\sigma} \, \mathrm{d}x = \frac{N^{1-\sigma}}{\sigma - 1}. \tag{7.13}$$

The term $q_N^{-\sigma}$ occurs in $P_N(\sigma)$, and all terms are positive, hence

$$P_N(\sigma) \geq q_N^{-\sigma}. \tag{7.14}$$

Consequently,

$$\left|h_N \frac{R_N(\sigma)}{P_N(\sigma)}\right| \le \frac{h_N N}{\sigma - 1}\left(\frac{q_N}{N}\right)^{\sigma}. \tag{7.15}$$

For $N \ge 3$, Bertrand's theorem gives a prime in $(N/2, N)$, so $q_N < N$, and the exponential factor tends to zero. For $N = 2$, $q_N/N = 1$, but the same bound is $2/(\sigma - 1)$, which also tends to zero. Hence the second expression in Theorem 7.1 tends to $M(N)$ for every $N \ge 2$. □

Example 7.4 (The $N = 10$ orientation check)**.** For $N = 10$,

$$Q_{10} = \{7\}, \qquad h_{10} = 1, \qquad M(10) = -1, \qquad B_{10} = 0. \tag{7.16}$$

The inverted quotient P_N/R_N would give $-1/2$ at $s = 0$. The quotient in (7.1) gives

$$0 + \frac{-2}{1} = -2. \tag{7.17}$$

This finite check confirms the orientation of the quotient in Theorem 7.1.

8. Exponential approach, rounding, and recovery of the first tail term

Definition 8.1. Let

$$a_N = \min\{n > N : \mu(n) \ne 0\}. \tag{8.1}$$

Theorem 8.2 (Leading exponential term)**.** *As $\sigma \to +\infty$,*

$$\boxed{\Phi_N(\sigma) - M(N) = h_N \mu(a_N)\left(\frac{q_N}{a_N}\right)^{\sigma}(1 + o(1)).} \tag{8.2}$$

Proof. Factor the least terminal prime from the denominator:

$$P_N(\sigma) = q_N^{-\sigma}\left[1 + \sum_{\substack{q \in Q_N \\ q > q_N}}\left(\frac{q_N}{q}\right)^{\sigma}\right]. \tag{8.3}$$

Every ratio in the bracket is strictly smaller than one, and there are only finitely many of them. Thus the bracket is $1 + o(1)$.

For the numerator, factor the first nonzero Möbius term after N:

$$R_N(\sigma) = \mu(a_N) a_N^{-\sigma}\left[1 + \sum_{n > a_N} \frac{\mu(n)}{\mu(a_N)}\left(\frac{a_N}{n}\right)^{\sigma}\right]. \tag{8.4}$$

For $\sigma \ge 2$, the absolute value of the summand is bounded by $(a_N/n)^2$, whose sum over $n > a_N$ is finite. Each fixed summand tends to zero as $\sigma \to \infty$. Dominated convergence therefore shows that the bracket is $1 + o(1)$. Division gives

$$\frac{R_N(\sigma)}{P_N(\sigma)} = \mu(a_N)\left(\frac{q_N}{a_N}\right)^{\sigma}(1 + o(1)). \tag{8.5}$$

Multiply by h_N and use the canonical identity. □

Remark 8.3. The factor is $(q_N/a_N)^{\sigma}$, not $(a_N/q_N)^{\sigma}$. Since $q_N \le N < a_N$, the correct factor decays. The inverted factor would contradict Corollary 7.3.

Theorem 8.4 (Uniform real-axis error bound)**.** *For every real $\sigma > 1$,*

$$\boxed{|\Phi_N(\sigma) - M(N)| \le \frac{h_N N}{\sigma - 1}\left(\frac{q_N}{N}\right)^{\sigma}.} \tag{8.6}$$

Proof. The reciprocal-zeta tail is absolutely convergent for $\sigma > 1$, so

$$|R_N(\sigma)| \le \sum_{n>N} |\mu(n)| n^{-\sigma} \tag{8.7}$$

$$\le \sum_{n>N} n^{-\sigma} \tag{8.8}$$

$$\le \int_N^\infty x^{-\sigma}\, \mathrm{d}x = \frac{N^{1-\sigma}}{\sigma - 1}. \tag{8.9}$$

The denominator is a sum of positive real terms. Retaining only its largest term gives

$$P_N(\sigma) = \sum_{q \in Q_N} q^{-\sigma} \ge q_N^{-\sigma}. \tag{8.10}$$

The canonical identity now gives

$$|\Phi_N(\sigma) - M(N)| = h_N \frac{|R_N(\sigma)|}{P_N(\sigma)} \tag{8.11}$$

$$\le \frac{h_N N^{1-\sigma} q_N^\sigma}{\sigma - 1} \tag{8.12}$$

$$= \frac{h_N N}{\sigma - 1} \left(\frac{q_N}{N} \right)^\sigma. \tag{8.13}$$

□

Corollary 8.5 (Finite rounding criterion)**.** *If*

$$\boxed{\frac{h_N N}{\sigma - 1} \left(\frac{q_N}{N} \right)^\sigma < \frac{1}{2},} \tag{8.14}$$

then $M(N)$ is the unique nearest integer to the real number $\Phi_N(\sigma)$.

Proof. Theorem 8.4 and (8.14) give

$$|\Phi_N(\sigma) - M(N)| < \frac{1}{2}. \tag{8.15}$$

For any integer $k \neq M(N)$, the reverse triangle inequality yields

$$|\Phi_N(\sigma) - k| \ge |k - M(N)| - |\Phi_N(\sigma) - M(N)| \tag{8.16}$$

$$> 1 - \frac{1}{2} = \frac{1}{2}. \tag{8.17}$$

Thus $M(N)$ is strictly closer to $\Phi_N(\sigma)$ than every other integer. □

Definition 8.6. Define

$$E_N(s) = \frac{\Phi_N(s) - M(N)}{h_N} = \frac{R_N(s)}{P_N(s)}. \tag{8.18}$$

Corollary 8.7 (Recovery of a_N)**.** *The first nonzero Möbius term after N is recovered by*

$$\boxed{a_N = q_N \lim_{\sigma \to +\infty} |E_N(\sigma)|^{-1/\sigma}.} \tag{8.19}$$

Proof. By Theorem 8.2 and Definition 8.6,

$$E_N(\sigma) = \mu(a_N) \left(\frac{q_N}{a_N} \right)^\sigma (1 + o(1)). \tag{8.20}$$

Since $|\mu(a_N)| = 1$, taking absolute values and then $-1/\sigma$-th powers gives

$$|E_N(\sigma)|^{-1/\sigma} = \frac{a_N}{q_N} |1 + o(1)|^{-1/\sigma} \longrightarrow \frac{a_N}{q_N}. \tag{8.21}$$

Multiplication by q_N proves (8.19). □

9. Feasible terminal values and non-distinction in $\Re s > 1$

Definition 9.1. For an integer parameter m, define

$$F_N(s;m) = M_s(N) + P_N(s)\left(1 + \frac{m - B_N}{h_N}\right). \tag{9.1}$$

A value m is called terminally feasible when

$$-1 \le \frac{m - B_N}{h_N} \le 1. \tag{9.2}$$

Proposition 9.2 (Exact substitution identity)**.** *The expression* (9.1) *is the exact result of substituting*

$$A_N(s) = M_s(N) + P_N(s) \tag{9.3}$$

into the terminal-value interpolation.

Proof. From $M_s(N) = A_N(s) - P_N(s)$ one has $A_N(s) = M_s(N) + P_N(s)$. Substitution in (5.1) gives

$$\mathcal{I}_N(s;m) = M_s(N) + P_N(s) + \frac{m - B_N}{h_N} P_N(s) \tag{9.4}$$

$$= F_N(s;m), \tag{9.5}$$

which is the claimed identity. □

Theorem 9.3 (The half-plane limit does not select $M(N)$)**.** *Fix s with $\Re s > 1$. For every sequence of feasible integers $m(N)$,*

$$\boxed{F_N(s;m(N)) \longrightarrow \frac{1}{\zeta(s)} \qquad (N \to \infty).} \tag{9.6}$$

Thus convergence in the absolutely convergent half-plane does not distinguish $m(N) = M(N)$ from other feasible terminal values.

Proof. Fix s with $\sigma = \Re s > 1$. Absolute convergence of $\sum \mu(n) n^{-s}$ gives

$$M_s(N) \longrightarrow \zeta(s)^{-1}. \tag{9.7}$$

For the terminal prime polynomial,

$$|P_N(s)| \le \sum_{N/2 < q \le N} q^{-\sigma} \tag{9.8}$$

$$\le \sum_{N/2 < n \le N} n^{-\sigma} \tag{9.9}$$

$$\le \int_{N/2-1}^{N} x^{-\sigma}\, \mathrm{d}x = O_\sigma(N^{1-\sigma}). \tag{9.10}$$

Feasibility means $|(m(N) - B_N)/h_N| \le 1$, so the multiplier $1 + (m(N) - B_N)/h_N$ has absolute value at most two. The entire terminal correction is therefore $O_\sigma(N^{1-\sigma})$, uniformly over all feasible choices. Adding it to $M_s(N)$ proves the common limit. □

Remark 9.4. The selection of the true value $M(N)$ is arithmetical and combinatorial. It is not produced by convergence to $1/\zeta(s)$ in $\Re s > 1$, because the entire feasible family has the same limit.

10. Poles and finite parts at zeta zeros

Definition 10.1. If a meromorphic function has a Laurent expansion

$$f(s) = \frac{a_{-1}}{s-\rho} + a_0 + O(s-\rho), \tag{10.1}$$

its finite part at ρ is

$$\mathrm{FP}_{s=\rho} f(s) = a_0. \tag{10.2}$$

Proposition 10.2 (Finite part of the reciprocal zeta function)**.** *Let ρ be a simple zero of ζ. Then*

$$\frac{1}{\zeta(s)} = \frac{1}{\zeta'(\rho)(s-\rho)} - \frac{\zeta''(\rho)}{2\zeta'(\rho)^2} + O(s-\rho), \tag{10.3}$$

and hence

$$\mathrm{FP}_{s=\rho} \frac{1}{\zeta(s)} = -\frac{\zeta''(\rho)}{2\zeta'(\rho)^2}. \tag{10.4}$$

Proof. Write $z = s-\rho$. Since the zero is simple,

$$\zeta(\rho+z) = \zeta'(\rho)z + \frac{1}{2}\zeta''(\rho)z^2 + O(z^3). \tag{10.5}$$

Factor the linear term:

$$\zeta(\rho+z) = \zeta'(\rho)z\left[1 + \frac{\zeta''(\rho)}{2\zeta'(\rho)}z + O(z^2)\right]. \tag{10.6}$$

Using $(1+w)^{-1} = 1 - w + O(w^2)$,

$$\frac{1}{\zeta(\rho+z)} = \frac{1}{\zeta'(\rho)z}\left[1 - \frac{\zeta''(\rho)}{2\zeta'(\rho)}z + O(z^2)\right] \tag{10.7}$$

$$= \frac{1}{\zeta'(\rho)z} - \frac{\zeta''(\rho)}{2\zeta'(\rho)^2} + O(z). \tag{10.8}$$

The constant Laurent coefficient is the finite part. □

Proposition 10.3 (Finite part of Φ_N)**.** *Assume that ρ is a simple zero of ζ and $P_N(\rho) \neq 0$. Then*

$$\mathrm{FP}_{s=\rho} \Phi_N(s) = B_N + h_N \frac{-\frac{\zeta''(\rho)}{2\zeta'(\rho)^2} - A_N(\rho)}{P_N(\rho)} \tag{10.9}$$

$$- h_N \frac{P_N'(\rho)}{\zeta'(\rho)P_N(\rho)^2}. \tag{10.10}$$

Proof. Set $z = s-\rho$. Proposition 10.2 and the Taylor expansion of the entire function A_N give

$$\zeta(s)^{-1} - A_N(s) = \frac{a_{-1}}{z} + a_0 + O(z), \tag{10.11}$$

where

$$a_{-1} = \frac{1}{\zeta'(\rho)}, \qquad a_0 = -\frac{\zeta''(\rho)}{2\zeta'(\rho)^2} - A_N(\rho). \tag{10.12}$$

Likewise,

$$P_N(s) = p_0 + p_1 z + O(z^2), \qquad p_0 = P_N(\rho), \quad p_1 = P_N'(\rho). \tag{10.13}$$

Because $p_0 \neq 0$,

$$\frac{1}{P_N(s)} = \frac{1}{p_0} - \frac{p_1}{p_0^2}z + O(z^2). \tag{10.14}$$

Multiplication yields

$$\frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)} = \frac{a_{-1}}{p_0 z} + \frac{a_0}{p_0} - \frac{a_{-1}p_1}{p_0^2} + O(z). \tag{10.15}$$

Multiplying by h_N and adding B_N gives the asserted finite part. □

Definition 10.4. For a simple zero ρ, define the normalized finite-part observable

$$\mathcal{F}_N(\rho) = N^{-\rho}\,\mathrm{FP}_{s=\rho}\,\Phi_N(s). \tag{10.16}$$

Part 3. Prime-power phase walks over zeta ordinates

11. Phase walks, endpoint conventions, and a universal half

Definition 11.1. Fix $q > 1$ and write

$$\omega = \log q. \tag{11.1}$$

For the positive ordinates, counted with multiplicity, define

$$S_q(T) = \sum_{0<\gamma\le T} m(\gamma) e^{i\omega\gamma}. \tag{11.2}$$

At an ordinate γ_k, the right, left, and midpoint values are

$$S_k^{\mathrm{right}} = \sum_{j\le k} b_j, \tag{11.3}$$

$$S_k^{\mathrm{left}} = \sum_{j<k} b_j, \tag{11.4}$$

$$S_k^{\mathrm{mid}} = \sum_{j<k} b_j + \frac{1}{2} b_k, \qquad b_k = e^{i\omega\gamma_k}. \tag{11.5}$$

Remark 11.2. The quantities $S_k^{\mathrm{right}}, S_k^{\mathrm{left}}$, and S_k^{mid} are indexwise: an ordinate of multiplicity m appears m times in the ordered list. The symmetric midpoint of the whole atom at a distinct ordinate τ instead contains $\frac{1}{2}m(\tau)e^{i\omega\tau}$.

Proposition 11.3 (Endpoint conversion)**.** *At every index k,*

$$\overline{b_k} S_k^{\mathrm{right}} = \overline{b_k} S_k^{\mathrm{mid}} + \frac{1}{2}, \tag{11.6}$$

$$\overline{b_k} S_k^{\mathrm{left}} = \overline{b_k} S_k^{\mathrm{mid}} - \frac{1}{2}. \tag{11.7}$$

Thus changing the endpoint convention translates the carrier coefficient by $\pm 1/2$ and does not change a centered variance.

Proof. The right sum differs from the indexwise midpoint sum by $\frac{1}{2}b_k$, while the left sum differs by $-\frac{1}{2}b_k$. Multiplication by $\overline{b}_k$ and $|b_k| = 1$ gives the two identities. Translating all sampled values by a fixed constant does not alter their centered variance. □

Theorem 11.4 (Exact triangular half identity)**.** *Let $|b_k| = 1$ and*

$$S_k = \sum_{j\le k} b_j. \tag{11.8}$$

Then

$$\boxed{\Re\left(\frac{1}{N}\sum_{k=1}^{N} \overline{b_k} S_k\right) = \frac{1}{2} + \frac{|S_N|^2}{2N}.} \tag{11.9}$$

Proof. Let $b = (b_1, \dots, b_N)^{\mathsf{T}}$, and let L be the lower triangular matrix with $L_{kj} = 1$ for $j \le k$. Then the vector of partial sums is Lb, and

$$\sum_{k=1}^{N} \overline{b_k} S_k = b^* L b. \tag{11.10}$$

Let J denote the all-ones matrix. Entrywise, $L + L^*$ equals $J + I$: off the diagonal exactly one of L_{jk}, L_{kj} equals one, while on the diagonal both do. Hence

$$2\Re(b^* L b) = b^*(L + L^*)b \tag{11.11}$$

$$= b^* J b + b^* b \tag{11.12}$$

$$= \left| \sum_{k=1}^{N} b_k \right|^2 + N, \tag{11.13}$$

because $|b_k| = 1$. Division by $2N$ proves the identity. □

Corollary 11.5. *For midpoint evaluation,*

$$\Re\left(\frac{1}{N} \sum_{k=1}^{N} \overline{b_k} S_k^{\mathrm{mid}} \right) = \frac{|S_N|^2}{2N}. \tag{11.14}$$

Proof. By Proposition 11.3,

$$\overline{b}_k S_k^{\mathrm{mid}} = \overline{b}_k S_k^{\mathrm{right}} - \frac{1}{2}. \tag{11.15}$$

Average over k, take real parts, and apply Theorem 11.4. □

Warning 11.6. Theorem 11.4 is universal for unit phases. It does not identify prime-power structure and does not imply the Riemann Hypothesis. It applies exactly to the raw full-sample cumulative walk, not automatically to a residual after smooth and prime-power subtraction.

12. Complete-zero weights and the positive horizontal defect

Definition 12.1. For a positive ordinate τ, define

$$W_q(\tau) = \sum_{\rho : \Im\rho = \tau} q^{\beta - \frac{1}{2}}, \tag{12.1}$$

where zeros are counted with multiplicity.

Definition 12.2. Define

$$D_q(\tau) = \sum_{\substack{\rho = \beta + i\tau \\ \beta > 1/2}} \left[\cosh\left((\beta - \tfrac{1}{2}) \log q \right) - 1 \right]. \tag{12.2}$$

Theorem 12.3 (Positive defect at each ordinate)**.** *For every $q > 1$ and every positive ordinate τ,*

$$\boxed{W_q(\tau) = m(\tau) + 2D_q(\tau), \qquad D_q(\tau) \ge 0.} \tag{12.3}$$

Moreover,

$$D_q(\tau) = 0 \tag{12.4}$$

if and only if every zero at ordinate τ lies on the critical line.

Proof. The functional equation and conjugation symmetry imply that, at a fixed positive ordinate τ, every zero $\beta + i\tau$ off the critical line is accompanied with equal multiplicity by $1 - \beta + i\tau$; see [12, Chapter 2]. Put $\delta = \beta - 1/2 > 0$. The pair contributes to $W_q(\tau)$

$$q^{\beta-1/2} + q^{1/2-\beta} = q^{\delta} + q^{-\delta} \tag{12.5}$$

$$= 2\cosh(\delta \log q) \tag{12.6}$$

$$= 2 + 2\left[\cosh(\delta \log q) - 1\right]. \tag{12.7}$$

The first term 2 is exactly the contribution of two unit weights to the multiplicity $m(\tau)$; the second is the excess. A zero on the critical line has $\delta = 0$ and contributes one to both quantities. Summing all pairs proves $W_q = m + 2D_q$. Since $q > 1$, $\log q \neq 0$, and $\cosh x - 1 \geq 0$ with equality only at $x = 0$, the defect vanishes precisely when no positive δ occurs. □

Definition 12.4. Define the locally finite complex measures

$$\mu_q = \sum_{\tau>0} m(\tau) e^{i\tau \log q} \delta_\tau, \tag{12.8}$$

$$\nu_q = \sum_{\tau>0} W_q(\tau) e^{i\tau \log q} \delta_\tau. \tag{12.9}$$

Corollary 12.5 (Atomic positive criterion)**.** *The exact measure identity*

$$\boxed{\frac{1}{2} e^{-iT \log q}(\nu_q - \mu_q) = \sum_{\tau>0} D_q(\tau)\delta_\tau} \tag{12.10}$$

holds. For any one fixed $q > 1$,

$$\nu_q = \mu_q \quad \Longleftrightarrow \quad \text{RH}. \tag{12.11}$$

Proof. At the atom $T = \tau$, the coefficient of $\nu_q - \mu_q$ is

$$\left[W_q(\tau) - m(\tau)\right] e^{i\tau \log q} = 2D_q(\tau) e^{i\tau \log q}. \tag{12.12}$$

Multiplication by $\frac{1}{2} e^{-iT \log q}$ cancels the phase at each atom and gives the positive measure in (12.10). If $\nu_q = \mu_q$, this measure is zero, so every nonnegative mass $D_q(\tau)$ is zero. Theorem 12.3 then places every zero on the critical line. Conversely, under RH, every weight $q^{\beta-1/2}$ equals one, so $W_q(\tau) = m(\tau)$ and the two measures coincide. □

Remark 12.6. No value of $q > 1$ is logically privileged. For a small displacement $\delta = \beta - 1/2$,

$$\cosh(\delta \log q) - 1 = \frac{1}{2}\delta^2 (\log q)^2 + O\left(\delta^4 (\log q)^4\right). \tag{12.13}$$

Larger q amplifies small horizontal displacements but also enlarges the analytic weights.

13. The explicit-formula origin of the two prime-power branches

The Guinand explicit formula and its later formulations relate suitable test-function sums over zeta zeros to the von Mangoldt function $\Lambda(n)$, hence to prime powers [10, 13]. Landau's formula and Gonek's uniform refinements isolate the same prime-power resonances in exponential sums over zeros [9, 8, 7].

Definition 13.1. Let $n = p^r$ be a prime power, put

$$\ell_n = \log n, \tag{13.1}$$

and suppose $\ell_n \neq \omega$. Define

$$c_n^- = \frac{i\Lambda(n)}{2\pi\sqrt{n}(\omega - \ell_n)}, \tag{13.2}$$

$$c_n^+ = \frac{i\Lambda(n)}{2\pi\sqrt{n}(\omega+\ell_n)}. \tag{13.3}$$

The associated oscillators are

$$P_n^-(T) = c_n^- e^{i(\omega-\ell_n)T}, \tag{13.4}$$

$$P_n^+(T) = c_n^+ e^{i(\omega+\ell_n)T}. \tag{13.5}$$

Warning 13.2 (Resonant carrier). The coefficient model (13.2)–(13.3) is nonresonant. All infinite sums built from these coefficients, including D_∞ and $\Delta_\omega(Q)$, are understood under the standing assumption

$$e^\omega \neq p^r \qquad \text{for every prime power } p^r. \tag{13.6}$$

If $e^\omega = p^r$, the quotient $(1-e^{-i(\ell_n-\omega)T})/(\ell_n-\omega)$ must be replaced by its limit iT; the corresponding branch is polynomial in T, and the nonresonant formulas below do not apply without a separate subtraction. The numerical case $\omega = 1$ is nonresonant.

Heuristic 13.3 (Sharp-window prime-spectral ansatz)**.** For a sharp counting window, formal integration of the explicit formula suggests a decomposition

$$S_q(T) = C_q + A_{\mathrm{smooth},q}(T) + \frac{1}{2}e^{i\omega T} + \sum_{n=p^r}\left(P_n^-(T)+P_n^+(T)\right), \tag{13.7}$$

where the infinite prime-power sum requires a specified truncation or regularization. This is a heuristic argument when used with the discontinuous indicator $\mathbf{1}_{\gamma\le T}$.

Definition 13.4 (Smooth term used in the computation)**.** The MATLAB program uses

$$A_{\mathrm{smooth}}(T) = \frac{e^{i\omega T}\log(T/2\pi) - \mathrm{Ei}(i\omega T)}{2\pi i\omega}, \tag{13.8}$$

up to a branch-dependent additive constant. The constant is treated by a joint constant–carrier fit.

Warning 13.5. Equation (13.8) is the antiderivative of the smooth Riemann–von Mangoldt density multiplied by $e^{i\omega T}$. It is not, by itself, a complete sharp-window explicit formula. Gamma-factor, pole, endpoint, branch, and regularization conventions must be retained when an exact global identity is claimed.

14. Exact algebra of each prime-power mode

Proposition 14.1 (Sum, difference, and ratio identities)**.** *For every nonresonant prime power n,*

$$c_n^- + c_n^+ = \frac{i\Lambda(n)\omega}{\pi\sqrt{n}(\omega^2-\ell_n^2)}, \tag{14.1}$$

$$c_n^- - c_n^+ = \frac{i\Lambda(n)\ell_n}{\pi\sqrt{n}(\omega^2-\ell_n^2)}, \tag{14.2}$$

$$\frac{c_n^-}{c_n^+} = \frac{\omega+\ell_n}{\omega-\ell_n}. \tag{14.3}$$

Proof. Starting from the definitions, combine denominators:

$$c_n^- + c_n^+ = \frac{i\Lambda(n)}{2\pi\sqrt{n}}\left[\frac{1}{\omega-\ell_n}+\frac{1}{\omega+\ell_n}\right] \tag{14.4}$$

$$= \frac{i\Lambda(n)}{2\pi\sqrt{n}}\frac{2\omega}{\omega^2-\ell_n^2}, \tag{14.5}$$

which is (14.1). Replacing the plus by a minus in the bracket gives $2\ell_n/(\omega^2-\ell_n^2)$, proving (14.2). Finally the common factor cancels in the quotient:

$$\frac{c_n^-}{c_n^+} = \frac{\omega+\ell_n}{\omega-\ell_n}. \tag{14.6}$$

All identities are exact whenever $\ell_n \neq \omega$. □

Definition 14.2. Write

$$c_n^- = ia_n^-, \qquad c_n^+ = ia_n^+, \qquad a_n^\pm \in \mathbb{R}, \tag{14.7}$$

and define

$$u_n = a_n^- + a_n^+, \tag{14.8}$$

$$v_n = a_n^- - a_n^+, \tag{14.9}$$

$$d_n = |c_n^-|^2 - |c_n^+|^2. \tag{14.10}$$

Proposition 14.3 (Signed mode invariant)**.** *The quantities in Definition 14.2 satisfy*

$$u_n = -\frac{\omega\Lambda(n)}{\pi\sqrt{n}(\ell_n^2-\omega^2)}, \tag{14.11}$$

$$v_n = -\frac{\Lambda(n)\ell_n}{\pi\sqrt{n}(\ell_n^2-\omega^2)}, \tag{14.12}$$

$$\boxed{d_n = u_n v_n = \frac{\omega\Lambda(n)^2\ell_n}{\pi^2 n(\ell_n^2-\omega^2)^2}.} \tag{14.13}$$

Proof. Because the denominators in $c_n^\pm$ are real, the coefficients are purely imaginary. Taking their real scalar factors and using the sum and difference identities gives

$$u_n = \frac{\Lambda(n)}{2\pi\sqrt{n}}\left[\frac{1}{\omega-\ell_n}+\frac{1}{\omega+\ell_n}\right] \tag{14.14}$$

$$= -\frac{\omega\Lambda(n)}{\pi\sqrt{n}(\ell_n^2-\omega^2)}, \tag{14.15}$$

$$v_n = \frac{\Lambda(n)}{2\pi\sqrt{n}}\left[\frac{1}{\omega-\ell_n}-\frac{1}{\omega+\ell_n}\right] \tag{14.16}$$

$$= -\frac{\ell_n\Lambda(n)}{\pi\sqrt{n}(\ell_n^2-\omega^2)}. \tag{14.17}$$

The algebraic factorization

$$(a_n^-)^2-(a_n^+)^2 = (a_n^-+a_n^+)(a_n^--a_n^+) = u_n v_n \tag{14.18}$$

then gives the displayed formula for d_n. Its positivity follows from $\omega>0$, $\Lambda(n)^2\geq 0$, $\ell_n>0$, and the squared denominator. □

Proposition 14.4 (Elliptic form and anisotropy)**.** *After demodulation by $e^{-i\omega T}$, the mode is*

$$g_n(T) = c_n^- e^{-i\ell_n T} + c_n^+ e^{i\ell_n T} = \frac{\Lambda(n)}{\pi\sqrt{n}(\omega^2-\ell_n^2)}\left[\ell_n\sin(\ell_n T)+i\omega\cos(\ell_n T)\right]. \tag{14.19}$$

Its unsigned elliptic area is

$$\pi d_n. \tag{14.20}$$

For uniform phase, its anisotropy is

$$\mathcal{A}_n = \frac{|\ell_n^2-\omega^2|}{\ell_n^2+\omega^2} \longrightarrow 1 \qquad (n\to\infty). \tag{14.21}$$

Proof. Using $e^{ix} = \cos x + i \sin x$, expand $c_n^- e^{-i\ell_n T} + c_n^+ e^{i\ell_n T}$ and collect its real and imaginary parts. This gives

$$g_n(T) = A_n \sin(\ell_n T) + iB_n \cos(\ell_n T), \tag{14.22}$$

with

$$A_n = \frac{\Lambda(n)\ell_n}{\pi\sqrt{n}(\omega^2 - \ell_n^2)}, \qquad B_n = \frac{\Lambda(n)\omega}{\pi\sqrt{n}(\omega^2 - \ell_n^2)}. \tag{14.23}$$

This is an ellipse with semiaxes $|A_n|$ and $|B_n|$. Their product is $|d_n| = d_n$, so its unsigned area is πd_n. Under uniform phase, the coordinate variances are $A_n^2/2$ and $B_n^2/2$. The normalized difference of principal variances is therefore

$$\frac{|A_n^2 - B_n^2|}{A_n^2 + B_n^2} = \frac{|\ell_n^2 - \omega^2|}{\ell_n^2 + \omega^2}, \tag{14.24}$$

which tends to one as $\ell_n \to \infty$. □

Lemma 14.5 (Dominance of $n = 3$ in the minus branch for $\omega = 1$)**.** *For $\omega = 1$, the largest coefficient $|c_n^-|$ among prime powers $n \geq 2$ occurs at $n = 3$.*

Proof. For a prime power $n = p^r$, $\Lambda(n) = \log p \leq \log n$. When $n \geq 3$ and $\omega = 1$, $\log n > 1$, so

$$|c_n^-| \leq f(n) := \frac{\log n}{2\pi\sqrt{n}(\log n - 1)}. \tag{14.25}$$

Let $L = \log x > 1$. Logarithmic differentiation gives

$$x\frac{f'(x)}{f(x)} = \frac{1}{L} - \frac{1}{2} - \frac{1}{L-1} \tag{14.26}$$

$$= -\frac{1}{L(L-1)} - \frac{1}{2} < 0. \tag{14.27}$$

Thus f is strictly decreasing on (e, ∞). At $n = 3$, the number is prime and $\Lambda(3) = \log 3$, so equality holds in (14.25); hence $n = 3$ dominates every $n \geq 4$. The remaining comparison with $n = 2$ is direct. Using the certified bounds $0.6931 < \log 2 < 0.6932$ and $1.0986 < \log 3 < 1.0987$ in the two explicit formulas gives $|c_3^-| > |c_2^-|$. Therefore $n = 3$ is the unique maximum. □

15. The convergent signed tail

Definition 15.1. Define

$$D_\infty = \sum_{n=p^r} d_n, \tag{15.1}$$

$$\Delta_\omega(Q) = \sum_{\substack{n=p^r \\ n>Q}} d_n = D_\infty - \sum_{\substack{n=p^r \\ n\leq Q}} d_n. \tag{15.2}$$

Theorem 15.2 (Asymptotic signed tail)**.** *For fixed $\omega > 0$ satisfying* (13.6)*,*

$$\boxed{\Delta_\omega(Q) \sim \frac{\omega}{\pi^2 \log Q} \qquad (Q \to \infty).} \tag{15.3}$$

Proof. Separate primes from proper prime powers. For a prime p, $\Lambda(p) = \ell_p = \log p$, and the exact formula becomes

$$d_p = \frac{\omega}{\pi^2 p \log p}\left(1 - \frac{\omega^2}{(\log p)^2}\right)^{-2}. \tag{15.4}$$

For $p > e^{2\omega}$, expand $(1-z)^{-2} = 1 + O(z)$ with $z = \omega^2/(\log p)^2$. Uniformly in that range,

$$d_p = \frac{\omega}{\pi^2 p \log p} + O_\omega\left(\frac{1}{p(\log p)^3}\right). \tag{15.5}$$

The error tail is $O_\omega((\log Q)^{-3})$ by the prime number theorem and partial summation.

For the main term, write the prime sum as a Stieltjes integral:

$$\sum_{p>Q} \frac{1}{p\log p} = \int_Q^\infty \frac{1}{x\log x}\,\mathrm{d}\pi(x). \tag{15.6}$$

The prime number theorem $\pi(x) = x/\log x + o(x/\log x)$, followed by partial summation, gives

$$\int_Q^\infty \frac{1}{x\log x}\,\mathrm{d}\pi(x) = \int_Q^\infty \frac{\mathrm{d}x}{x(\log x)^2} + o\left(\frac{1}{\log Q}\right) = \frac{1}{\log Q} + o\left(\frac{1}{\log Q}\right). \tag{15.7}$$

It remains to bound $n = p^r$ with $r \geq 2$. Outside a fixed finite set depending on ω, the denominator satisfies $(r^2(\log p)^2 - \omega^2)^2 \gg_\omega r^4(\log p)^4$. Hence

$$d_{p^r} \ll_\omega \frac{1}{r^3 p^r \log p}. \tag{15.8}$$

Summing over $p^r > Q$, $r \geq 2$, gives $O_\omega(Q^{-1/2})$, after separating the finitely many small primes. This is $o(1/\log Q)$. Combining the prime and proper-power parts proves the theorem. □

Remark 15.3. For $\omega = 1$,

$$\pi\Delta_1(Q) \sim \frac{1}{\pi\log Q}, \tag{15.9}$$

$$\frac{3}{4}\Delta_1(Q) \sim \frac{3}{4\pi^2\log Q}. \tag{15.10}$$

These constants match two finite numerical quantities, but the matching relations themselves remain conjectural.

16. Finite zero-sampled carrier identities

Definition 16.1. Fix a finite multiset of positive ordinates

$$\Gamma_H = \{\gamma_k : 0 < \gamma_k \leq H\}, \qquad N(H) = |\Gamma_H|, \tag{16.1}$$

and let Y be any complex-valued function on Γ_H. For a cutoff Q, define

$$R_Q(\gamma) = Y(\gamma) - \sum_{\substack{n=p^r \\ n\leq Q}} \left(P_n^-(\gamma) + P_n^+(\gamma)\right). \tag{16.2}$$

The raw carrier coefficient and zero kernel are

$$\widehat{c}_{Q,H} = \frac{1}{N(H)} \sum_{\gamma\in\Gamma_H} e^{-i\omega\gamma} R_Q(\gamma), \tag{16.3}$$

$$M_H(x) = \frac{1}{N(H)} \sum_{\gamma\in\Gamma_H} x^{i\gamma}, \qquad x > 0. \tag{16.4}$$

Also put

$$\widehat{c}_{Y,H} = \frac{1}{N(H)} \sum_{\gamma\in\Gamma_H} e^{-i\omega\gamma} Y(\gamma). \tag{16.5}$$

Theorem 16.2 (Exact finite-sample carrier formula)**.** *For the objects in Definition 16.1,*

$$\boxed{\widehat{c}_{Q,H} = \widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} \left(c_n^- \overline{M_H(n)} + c_n^+ M_H(n)\right).} \tag{16.6}$$

Proof. For each sampled ordinate, demodulation gives

$$e^{-i\omega\gamma_k} P_n^-(\gamma_k) = c_n^- e^{-i\ell_n\gamma_k} = c_n^- n^{-i\gamma_k}, \tag{16.7}$$

$$e^{-i\omega\gamma_k} P_n^+(\gamma_k) = c_n^+ e^{i\ell_n\gamma_k} = c_n^+ n^{i\gamma_k}. \tag{16.8}$$

Insert the definition of the residual into the sample average defining $\widehat{c}_{Q,H}$. Finite sums may be interchanged without any convergence issue. The average of $n^{i\gamma_k}$ is $M_H(n)$, and the average of $n^{-i\gamma_k}$ is its complex conjugate because $n > 0$. Collecting the terms yields the formula exactly. □

Theorem 16.3 (Exact imaginary-part formula)**.** *Write*

$$M_H(n) = x_n + iy_n. \tag{16.9}$$

Then

$$\Im\left(c_n^- \overline{M_H(n)} + c_n^+ M_H(n)\right) = u_n x_n, \tag{16.10}$$

and consequently

$$\boxed{\Im\widehat{c}_{Q,H} = \Im\widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} u_n \Re M_H(n).} \tag{16.11}$$

Proof. Write $M_H(n) = x_n + iy_n$ and $c_n^\pm = ia_n^\pm$ with real $a_n^\pm$. Then

$$c_n^- \overline{M_H(n)} + c_n^+ M_H(n) = ia_n^-(x_n - iy_n) + ia_n^+(x_n + iy_n) \tag{16.12}$$

$$= (a_n^- - a_n^+)y_n + i(a_n^- + a_n^+)x_n \tag{16.13}$$

$$= v_n y_n + iu_n x_n. \tag{16.14}$$

Its imaginary part is therefore $u_n x_n = u_n \Re M_H(n)$. Taking imaginary parts in the exact carrier formula and summing proves the result. □

Definition 16.4. Define the Q-independent quantity

$$C_H = \Im\widehat{c}_{Y,H} - \pi D_\infty. \tag{16.15}$$

Theorem 16.5 (Exact global defect identity)**.** *For the finite-sample raw coefficient,*

$$\boxed{\Im\widehat{c}_{Q,H} - \pi\Delta_\omega(Q) = C_H - \sum_{\substack{n=p^r \\ n\le Q}} u_n \left[\Re M_H(n) - \pi v_n\right].} \tag{16.16}$$

Proof. The exact carrier identity gives

$$\Im\widehat{c}_{Q,H} = \Im\widehat{c}_{Y,H} - \sum_{n\le Q} u_n \Re M_H(n). \tag{16.17}$$

On the other hand, $d_n = u_n v_n$ and $\Delta_\omega(Q) = D_\infty - \sum_{n\le Q} d_n$, so

$$\pi\Delta_\omega(Q) = \pi D_\infty - \pi \sum_{n\le Q} u_n v_n. \tag{16.18}$$

Subtract the second equation from the first and group the cutoff-independent terms:

$$\Im\widehat{c}_{Q,H} - \pi\Delta_\omega(Q) = \left(\Im\widehat{c}_{Y,H} - \pi D_\infty\right) \tag{16.19}$$

$$-\sum_{n\le Q} u_n\left[\Re M_H(n)-\pi v_n\right]. \tag{16.20}$$

The parenthesis is exactly C_H. Every manipulation is finite except for D_∞, whose convergence follows from Theorem 15.2. □

Corollary 16.6 (Equivalent raw transport target)**.** *Along any specified joint sequence $H = H(Q)$ with $Q \to \infty$, the conjectural raw relation*

$$\Im\widehat{c}_{Q,H} = \pi\Delta_\omega(Q) + o\left(\frac{1}{\log Q}\right) \tag{16.21}$$

is exactly equivalent to

$$\boxed{C_H - \sum_{n\le Q} u_n\left[\Re M_H(n)-\pi v_n\right] = o\left(\frac{1}{\log Q}\right).} \tag{16.22}$$

Proof. Subtract $\pi\Delta_\omega(Q)$ from $\Im\widehat{c}_{Q,H}$ and apply Theorem 16.5. The dependence $H = H(Q)$ must be specified because C_H and every kernel value $M_H(n)$ depend on the sample height. No limit interchange is involved: the equivalence holds pointwise along the chosen sequence. □

Theorem 16.7 (Exact increment identity)**.** *When Q crosses a prime power n,*

$$\Delta\Im\widehat{c}_{Q,H} = -u_n\Re M_H(n), \tag{16.23}$$

$$\Delta\left(\pi\Delta_\omega(Q)\right) = -\pi u_n v_n. \tag{16.24}$$

If $u_n v_n \neq 0$, then

$$\boxed{\frac{\Delta\Im\widehat{c}_{Q,H}}{\Delta(\pi\Delta_\omega(Q))} = \frac{\Re M_H(n)}{\pi v_n}.} \tag{16.25}$$

Proof. Let Q_- and Q_+ be cutoffs immediately before and immediately after the prime power n, so that no other prime power lies between them. The exact raw coefficient formula (16.6) gives

$$\widehat{c}_{Q_+,H} - \widehat{c}_{Q_-,H} = -\left(c_n^-\overline{M_H(n)} + c_n^+ M_H(n)\right). \tag{16.26}$$

Taking imaginary parts and using (16.10) yields (16.23). By definition, $\Delta_\omega(Q)$ loses exactly the single summand $d_n = u_n v_n$ when the cutoff crosses n, which proves (16.24). Division of the two nonzero increments gives (16.25). □

17. Exact joint constant–carrier identities

The coefficient reported by the corrected numerical program is not the raw coefficient $\widehat{c}_{Q,H}$. It is the coefficient obtained by jointly projecting onto the constant vector and the carrier. The projection correction can be written exactly in the same zero kernels.

Definition 17.1 (Joint carrier data)**.** Put

$$q = e^\omega, \qquad \mu_H = M_H(q), \qquad \mathfrak{d}_H = 1 - |\mu_H|^2. \tag{17.1}$$

Assume $\mathfrak{d}_H > 0$. For a sampled function Z, define its joint carrier coefficient by

$$\mathfrak{c}_H[Z] = \frac{\langle e^{i\omega\gamma}, Z\rangle_H - \overline{\mu_H}\langle 1, Z\rangle_H}{\mathfrak{d}_H}. \tag{17.2}$$

Thus the coefficient reported at cutoff Q is

$$c_{Q,H} = \mathfrak{c}_H[R_Q], \qquad c_{Y,H} = \mathfrak{c}_H[Y]. \tag{17.3}$$

Definition 17.2 (Joint contribution of one prime power)**.** For a prime power n, define

$$A_H(n) = c_n^- \overline{M_H(n)} + c_n^+ M_H(n), \tag{17.4}$$

$$B_H(n) = c_n^- M_H(q/n) + c_n^+ M_H(qn), \tag{17.5}$$

$$K_H(n) = \frac{A_H(n) - \overline{\mu_H} B_H(n)}{\mathfrak{d}_H}. \tag{17.6}$$

Also define the real quantity

$$J_H(n) = \Im K_H(n). \tag{17.7}$$

Theorem 17.3 (Exact reported-carrier formula)**.** *For every finite cutoff Q,*

$$\boxed{c_{Q,H} = c_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} K_H(n).} \tag{17.8}$$

Moreover,

$$\boxed{J_H(n) = \frac{u_n \Re M_H(n) - \Re\left[\overline{\mu_H}\left(a_n^- M_H(q/n) + a_n^+ M_H(qn)\right)\right]}{\mathfrak{d}_H}.} \tag{17.9}$$

Consequently,

$$\boxed{\Im c_{Q,H} = \Im c_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} J_H(n).} \tag{17.10}$$

Proof. The joint carrier functional $\mathfrak{c}_H$ is complex linear in its argument. The residual decomposition

$$R_Q = Y - \sum_{\substack{n=p^r \\ n\le Q}} \left(P_n^- + P_n^+\right) \tag{17.11}$$

therefore gives

$$c_{Q,H} = c_{Y,H} - \sum_{n\le Q} \mathfrak{c}_H[P_n^- + P_n^+]. \tag{17.12}$$

For a single prime power,

$$\begin{aligned} \left\langle e^{i\omega\gamma}, P_n^- + P_n^+ \right\rangle_H &= c_n^- \overline{M_H(n)} + c_n^+ M_H(n) \\ &= A_H(n), \end{aligned} \tag{17.13}$$

whereas

$$\begin{aligned} \left\langle 1, P_n^- + P_n^+ \right\rangle_H &= c_n^- M_H(q/n) + c_n^+ M_H(qn) \\ &= B_H(n). \end{aligned} \tag{17.14}$$

Substitution in (17.2) proves $\mathfrak{c}_H[P_n^- + P_n^+] = K_H(n)$, and hence (17.8).

To compute the imaginary part, write $c_n^\pm = i a_n^\pm$. The first projection satisfies

$$\Im A_H(n) = u_n \Re M_H(n) \tag{17.15}$$

by Theorem 16.3. Furthermore,

$$B_H(n) = i\left[a_n^- M_H(q/n) + a_n^+ M_H(qn)\right], \tag{17.16}$$

and for any complex number Z, $\Im(iZ) = \Re Z$. Therefore

$$\Im(\overline{\mu_H} B_H(n)) = \Re\Big[\overline{\mu_H}\left(a_n^- M_H(q/n) + a_n^+ M_H(qn)\right)\Big]. \tag{17.17}$$

Since $\mathfrak{d}_H$ is real and positive, taking imaginary parts in (17.6) gives (17.9). Finally, taking imaginary parts in (17.8) proves (17.10). □

Definition 17.4 (Joint global constant)**.** Define

$$C_H^{\mathrm{joint}} = \Im c_{Y,H} - \pi D_\infty. \tag{17.18}$$

Theorem 17.5 (Exact joint defect and increment identities)**.** *For every finite cutoff Q,*

$$\boxed{\Im c_{Q,H} - \pi\Delta_\omega(Q) = C_H^{\mathrm{joint}} - \sum_{\substack{n=p^r\\ n\le Q}} \left[J_H(n) - \pi u_n v_n\right].} \tag{17.19}$$

When the cutoff crosses a single prime power n,

$$\Delta\Im c_{Q,H} = -J_H(n), \tag{17.20}$$

$$\Delta(\pi\Delta_\omega(Q)) = -\pi u_n v_n. \tag{17.21}$$

If $u_n v_n \neq 0$, then

$$\boxed{\frac{\Delta\Im c_{Q,H}}{\Delta(\pi\Delta_\omega(Q))} = \frac{J_H(n)}{\pi u_n v_n}.} \tag{17.22}$$

Proof. Equation (17.10) gives

$$\Im c_{Q,H} = \Im c_{Y,H} - \sum_{n\le Q} J_H(n). \tag{17.23}$$

The signed tail identity remains

$$\pi\Delta_\omega(Q) = \pi D_\infty - \pi \sum_{n\le Q} u_n v_n. \tag{17.24}$$

Subtracting these two equations proves (17.19). Crossing a single prime power adds exactly one term to each finite cutoff sum, yielding (17.20) and (17.21). Their quotient is (17.22). □

Corollary 17.6 (Correct transport target for the reported coefficient)**.** *Along any specified joint sequence $H = H(Q)$ with $Q \to \infty$, the assertion*

$$\Im c_{Q,H} = \pi\Delta_\omega(Q) + o\Big(\frac{1}{\log Q}\Big) \tag{17.25}$$

is equivalent to

$$\boxed{C_H^{\mathrm{joint}} - \sum_{n\le Q} \left[J_H(n) - \pi u_n v_n\right] = o\Big(\frac{1}{\log Q}\Big).} \tag{17.26}$$

Proof. This is an immediate reformulation of (17.19). The specification of a joint sequence is necessary because C_H^{joint}, $J_H(n)$, and the sample kernel all depend on H. □

Warning 17.7 (Raw and reported increments are different). The raw ratio (16.25) is not the increment statistic of the coefficient printed by the corrected program. The reported statistic is (17.22). They coincide only when the exact constant-projection correction in (17.9) is negligible and $\mathfrak{d}_H = 1 + o(1)$.

18. Landau scaling and the conditional obstruction to termwise area transport

Landau's formula concerns sums of x^ρ, not sums of $x^{i\gamma}$ without horizontal weights. Passing from the former to the latter requires RH, or a separate control of the real parts. Uniformity when x grows with H requires a uniform Landau–Gonek formula [9, 8].

Heuristic 18.1 (Conditional Landau estimate for the ordinate kernel)**.** Assume RH and a uniform Landau–Gonek estimate in the range under consideration. With

$$L_H = \log(H/2\pi), \tag{18.1}$$

one expects at a prime power n

$$\Re M_H(n) \approx -\frac{\Lambda(n)}{\sqrt{n}\, L_H}. \tag{18.2}$$

Proposition 18.2 (Required termwise target)**.** *Individual transport of the signed area of mode n would require*

$$\Re M_H(n) \sim \pi v_n. \tag{18.3}$$

For a prime p,

$$\pi v_p = -\frac{(\log p)^2}{\sqrt{p}((\log p)^2 - \omega^2)} \sim -\frac{1}{\sqrt{p}}. \tag{18.4}$$

Proof. The increment identity (16.25) shows that exact mode-by-mode transport is equivalent to $\Re M_H(n)/(\pi v_n) \to 1$, which is (18.3). For a prime p, substitute $\Lambda(p) = \log p$ and $\ell_p = \log p$ into (14.12); division of numerator and denominator by $(\log p)^2$ gives (18.4). □

Heuristic 18.3 (Termwise ratio)**.** Combining (18.2) and (18.4) gives

$$\frac{\Re M_H(p)}{\pi v_p} \approx \frac{\log p}{L_H}. \tag{18.5}$$

If $p \asymp H^\theta$, this predicts

$$\frac{\Re M_H(p)}{\pi v_p} \approx \theta. \tag{18.6}$$

Thus termwise transport cannot be asymptotically exact at a fixed $\theta < 1$; it becomes plausible only for $p = H^{1-o(1)}$.

Warning 18.4. The fixed-p Landau formula alone does not prove (18.6). The latter is a falsifiable scaling prediction that requires uniformity in the joint range $p = p(H)$.

Conjecture 18.5 (Local transport near the height scale)**.** *Let*

$$Q_1, Q_2 = H^{1-o(1)}. \tag{18.7}$$

In a suitable uniform regime,

$$\Im\widehat{c}_{Q_2,H} - \Im\widehat{c}_{Q_1,H} \sim \pi\left[\Delta_\omega(Q_2) - \Delta_\omega(Q_1)\right]. \tag{18.8}$$

Proposition 18.6 (Exact local defect)**.** *The defect in Conjecture 18.5 is exactly*

$$\begin{aligned}&[\Im\widehat{c}_{Q_2,H} - \Im\widehat{c}_{Q_1,H}] - \pi\left[\Delta_\omega(Q_2) - \Delta_\omega(Q_1)\right] \\ &\quad = -\sum_{Q_1 < n \le Q_2} u_n\left[\Re M_H(n) - \pi v_n\right].\end{aligned} \tag{18.9}$$

Proof. Subtract (16.16) at Q_1 and Q_2. The constant C_H cancels. □

19. Mixed carrier area versus prime self-area

Proposition 19.1 (Mixed form measured by the centroid coefficient)**.** *The imaginary part of the raw carrier coefficient is*

$$\Im\widehat{c}_{Q,H} = \frac{1}{N(H)} \sum_{\gamma_k \le H} \Im\left[e^{-i\omega\gamma_k} R_Q(\gamma_k)\right] \tag{19.1}$$

$$= -\frac{1}{N(H)} \sum_{\gamma_k \le H} \Im\left[\overline{R_Q(\gamma_k)} e^{i\omega\gamma_k}\right]. \tag{19.2}$$

It is a mixed bilinear pairing between the residual position and the actual phase increment.

Proof. Take the imaginary part of the defining average (16.3). For the second equality, use

$$\Im(e^{-i\omega\gamma} R) = -\Im(\overline{R}\, e^{i\omega\gamma}), \tag{19.3}$$

which follows by complex conjugation. □

Definition 19.2. For a finite prime-power model G, define the formal self-area

$$\mathfrak{A}_H(G) = -\frac{\pi}{H} \int \Im\left(\overline{G(t)}\, \mathcal{J}G(t)\right) \mathrm{d}t, \tag{19.4}$$

where $\mathcal{J}$ is a specified frequency-sign or Hilbert-type operator.

Warning 19.3. The mixed pairing (19.2) is not identical to the self-pairing (19.4). Such an identification would require a justified replacement

$$e^{i\omega\gamma_k} \longleftrightarrow \mathcal{J}G_Q(\gamma_k). \tag{19.5}$$

The exact termwise defect of this replacement is encoded by $\Re M_H(n) - \pi v_n$.

20. Exact quadratic form for the sampled residual variance

The numerical variance is the squared norm of the *full residual* after jointly removing the constant and carrier. It is therefore not the projected norm of the prime-power model alone. The background vector, the mixed background–prime term, and the projected prime Gram form all enter exactly.

Definition 20.1 (Sampled inner product)**.** For sampled vectors on Γ_H, define

$$\langle \mathbf{f}, \mathbf{g} \rangle_H = \frac{1}{N(H)} \mathbf{f}^* \mathbf{g} = \frac{1}{N(H)} \sum_{\gamma_k \le H} \overline{f(\gamma_k)} g(\gamma_k). \tag{20.1}$$

Definition 20.2 (Sampled vectors and prime-mode matrix)**.** Let

$$\mathbf{1}_H = (1)_{\gamma_k \in \Gamma_H}, \tag{20.2}$$

$$\mathbf{a}_H = (e^{i\omega\gamma_k})_{\gamma_k \in \Gamma_H}, \tag{20.3}$$

$$\mathbf{y}_H = (Y(\gamma_k))_{\gamma_k \in \Gamma_H}. \tag{20.4}$$

For a prime power n and $\sigma \in \{-1, +1\}$, define the original, non-demodulated branch vector

$$\mathbf{w}_{n,\sigma} = \left(e^{i(\omega + \sigma \log n)\gamma_k}\right)_{\gamma_k \in \Gamma_H}. \tag{20.5}$$

For a cutoff Q, let

$$\mathcal{I}_Q = \{(n, \sigma) : n = p^r \le Q,\ \sigma \in \{-1, +1\}\}, \tag{20.6}$$

let V_Q be the matrix whose columns are the vectors $\mathbf{w}_{n,\sigma}$ in any fixed order, and let $\mathbf{c}_Q$ be the coefficient vector with entries c_n^σ. Then

$$\mathbf{p}_Q = V_Q \mathbf{c}_Q \tag{20.7}$$

is the sampled prime-power model, and

$$\mathbf{r}_Q = \mathbf{y}_H - V_Q \mathbf{c}_Q \tag{20.8}$$

is the sampled residual used by the joint fit.

Definition 20.3 (Joint projection and numerical variance)**.** Put

$$B_H = \begin{pmatrix} \mathbf{1}_H & \mathbf{a}_H \end{pmatrix}. \tag{20.9}$$

Assume $|M_H(e^\omega)| < 1$, equivalently that the two columns of B_H are linearly independent, and define

$$\Pi_H = I - B_H (B_H^* B_H)^{-1} B_H^*. \tag{20.10}$$

The variance reported after the joint constant–carrier fit is

$$\boxed{V(Q,H) = \frac{1}{N(H)} \mathbf{r}_Q^* \Pi_H \mathbf{r}_Q.} \tag{20.11}$$

Remark 20.4. If $|M_H(e^\omega)| = 1$, the same definition is made with the orthogonal projection onto the complement of $\operatorname{span}\{\mathbf{1}_H, \mathbf{a}_H\}$, or equivalently with the Moore–Penrose inverse. The data used in the numerical chapter are in the nondegenerate case.

Proposition 20.5 (Exact unprojected branch Gram kernel)**.** *For $j = (n,\sigma)$ and $k = (m,\tau)$ in $\mathcal{I}_Q$,*

$$K^{(0)}_{Q,H}(j,k) := \frac{1}{N(H)} \mathbf{w}^*_{n,\sigma} \mathbf{w}_{m,\tau} = M_H(m^\tau n^{-\sigma}). \tag{20.12}$$

Thus the four branch interactions between p^r and q^s involve

$$M_H(q^s/p^r), \quad M_H(q^s p^r), \quad M_H((q^s p^r)^{-1}), \quad M_H(p^r/q^s). \tag{20.13}$$

Proof. The common carrier cancels in the inner product. More explicitly,

$$\frac{1}{N(H)} \mathbf{w}^*_{n,\sigma} \mathbf{w}_{m,\tau} = \frac{1}{N(H)} \sum_{\gamma_k \le H} e^{-i(\omega+\sigma\log n)\gamma_k} e^{i(\omega+\tau\log m)\gamma_k} \tag{20.14}$$

$$= \frac{1}{N(H)} \sum_{\gamma_k \le H} (m^\tau n^{-\sigma})^{i\gamma_k} \tag{20.15}$$

$$= M_H(m^\tau n^{-\sigma}). \tag{20.16}$$

The four sign choices give (20.13). □

Proposition 20.6 (Exact projected branch Gram kernel)**.** *Put $q = e^\omega$ and*

$$G_H = \frac{1}{N(H)} B_H^* B_H = \begin{pmatrix} 1 & M_H(q) \\ M_H(q) & 1 \end{pmatrix}. \tag{20.17}$$

For $j = (n,\sigma)$ define

$$\mathbf{b}_j = \frac{1}{N(H)} B_H^* \mathbf{w}_{n,\sigma} = \begin{pmatrix} M_H(qn^\sigma) \\ M_H(n^\sigma) \end{pmatrix}. \tag{20.18}$$

Then

$$\boxed{\widetilde{K}_{Q,H}(j,k) := \frac{1}{N(H)} \mathbf{w}_j^* \Pi_H \mathbf{w}_k = K^{(0)}_{Q,H}(j,k) - \mathbf{b}_j^* G_H^{-1} \mathbf{b}_k.} \tag{20.19}$$

Proof. From (20.10),

$$\frac{1}{N(H)} V_Q^* \Pi_H V_Q = \frac{1}{N(H)} V_Q^* V_Q - \frac{1}{N(H)} V_Q^* B_H (B_H^* B_H)^{-1} B_H^* V_Q. \tag{20.20}$$

Since $B_H^* B_H = N(H) G_H$, the (j,k) entry of the second term is $\mathbf{b}_j^* G_H^{-1} \mathbf{b}_k$. The two components of $\mathbf{b}_j$ follow from

$$\frac{1}{N(H)} \mathbf{1}_H^* \mathbf{w}_{n,\sigma} = M_H(qn^\sigma), \tag{20.21}$$

$$\frac{1}{N(H)} \mathbf{a}_H^* \mathbf{w}_{n,\sigma} = M_H(n^\sigma). \tag{20.22}$$

□

Definition 20.7 (Projected background and mixed vector)**.** Define

$$V_Y(H) = \frac{1}{N(H)} \mathbf{y}_H^* \Pi_H \mathbf{y}_H, \tag{20.23}$$

$$\mathbf{d}_{Q,H} = \frac{1}{N(H)} V_Q^* \Pi_H \mathbf{y}_H, \tag{20.24}$$

$$\widetilde{K}_{Q,H} = \frac{1}{N(H)} V_Q^* \Pi_H V_Q. \tag{20.25}$$

Proposition 20.8 (Entrywise formula for the mixed vector)**.** *Let*

$$\mathbf{s}_{Y,H} = \frac{1}{N(H)} B_H^* \mathbf{y}_H. \tag{20.26}$$

For $j = (n, \sigma)$,

$$\boxed{d_{Q,H}(j) = \frac{1}{N(H)} \mathbf{w}_j^* \mathbf{y}_H - \mathbf{b}_j^* G_H^{-1} \mathbf{s}_{Y,H}.} \tag{20.27}$$

Proof. Insert (20.10) in (20.24). The normalization $B_H^* B_H = N(H) G_H$ gives the second term in (20.27). □

Theorem 20.9 (Exact decomposition of the numerical residual variance)**.** *For every finite cutoff Q,*

$$\boxed{V(Q,H) = V_Y(H) - 2\Re\left(\mathbf{c}_Q^* \mathbf{d}_{Q,H}\right) + \mathbf{c}_Q^* \widetilde{K}_{Q,H} \mathbf{c}_Q.} \tag{20.28}$$

Grouping the coefficient vector, mixed vector, and Gram matrix by prime base gives the exact identity

$$V(Q,H) = V_Y(H) - 2\Re \sum_p \mathbf{c}_p^* \mathbf{d}_{p,H} \tag{20.29}$$

$$+ \sum_p \mathbf{c}_p^* \widetilde{K}_{p,p;H} \mathbf{c}_p + \sum_{p \neq r} \mathbf{c}_p^* \widetilde{K}_{p,r;H} \mathbf{c}_r. \tag{20.30}$$

Proof. By (20.8) and (20.11),

$$V(Q,H) = \frac{1}{N(H)} (\mathbf{y}_H - V_Q \mathbf{c}_Q)^* \Pi_H (\mathbf{y}_H - V_Q \mathbf{c}_Q) \tag{20.31}$$

$$= \frac{1}{N(H)} \mathbf{y}_H^* \Pi_H \mathbf{y}_H - \frac{1}{N(H)} \mathbf{y}_H^* \Pi_H V_Q \mathbf{c}_Q$$

$$- \frac{1}{N(H)} \mathbf{c}_Q^* V_Q^* \Pi_H \mathbf{y}_H + \frac{1}{N(H)} \mathbf{c}_Q^* V_Q^* \Pi_H V_Q \mathbf{c}_Q. \tag{20.32}$$

The projection is Hermitian, so the two middle terms are complex conjugates. Definitions 20.7 therefore turn (20.32) into (20.28). Partitioning the finite index set by prime base yields (20.29)–(20.30). □

Corollary 20.10 (Prime model alone)**.** *The identity*

$$\frac{1}{N(H)}(V_Q\mathbf{c}_Q)^*\Pi_H(V_Q\mathbf{c}_Q)=\mathbf{c}_Q^*\widetilde{K}_{Q,H}\mathbf{c}_Q \tag{20.33}$$

holds for the projected prime model alone. It equals the numerical residual variance only when

$$V_Y(H)-2\Re(\mathbf{c}_Q^*\mathbf{d}_{Q,H})=0. \tag{20.34}$$

Proof. Equation (20.33) follows directly from (20.25). Comparison with (20.28) gives the necessary and sufficient condition (20.34). □

Warning 20.11 (What the diagonal question actually contains). The scalar Parseval sum

$$\sum_{p,r}\left(|c_{p^r}^-|^2+|c_{p^r}^+|^2\right) \tag{20.35}$$

is neither the full numerical variance nor the diagonal of the complete residual problem. Before any asymptotic assertion about a "diagonal" is made, one must control all four exact contributions in (20.29)– (20.30): the projected background energy, the background–prime mixed term, the within-tower prime blocks, and the cross-tower prime blocks. The fact that the arguments inside a fixed prime tower remain powers of that prime is exact; a Landau–Gonek interpretation of the ordinate-only kernel is an additional analytic step.

Question 20.12 (Exact variance components)**.** Determine, uniformly for a growing cutoff, the separate sizes of

$$V_Y(H),\qquad -2\Re\sum_p\mathbf{c}_p^*\mathbf{d}_{p,H},\qquad \sum_p\mathbf{c}_p^*\widetilde{K}_{p,p;H}\mathbf{c}_p,\qquad \sum_{p\neq r}\mathbf{c}_p^*\widetilde{K}_{p,r;H}\mathbf{c}_r. \tag{20.36}$$

Only their complete sum is the variance reported by the program.

21. Finite logarithmic shells in Bohr mean

Definition 21.1. For fixed $a>0$, define the finite multiplicative shell

$$\mathcal{S}_a(Q)=\{n=p^r : Q<n\le e^aQ\} \tag{21.1}$$

and its demodulated sum

$$G_{Q,a}(T)=\sum_{n\in\mathcal{S}_a(Q)}g_n(T). \tag{21.2}$$

Proposition 21.2 (Bohr energy of a fixed logarithmic shell)**.** *For fixed $a>0$,*

$$\lim_{X\to\infty}\frac{1}{X}\int_0^X|G_{Q,a}(T)|^2\,\mathrm{d}T=\frac{a}{2\pi^2\log Q}+o\left(\frac{1}{\log Q}\right) \tag{21.3}$$

as $Q\to\infty$.

Proof. Distinct frequencies $\pm\log n$ are orthogonal in Bohr mean. For primes,

$$|c_p^-|^2+|c_p^+|^2=\frac{1}{2\pi^2p}+O_\omega\left(\frac{1}{p(\log p)^2}\right). \tag{21.4}$$

The prime number theorem and partial summation give [2, 3]

$$\sum_{Q<p\le e^aQ}\frac{1}{p}=\frac{a}{\log Q}+o\left(\frac{1}{\log Q}\right). \tag{21.5}$$

Proper prime powers contribute a lower-order term. □

Corollary 21.3. *The choice $a=3/2$ gives*

$$\frac{3}{4\pi^2\log Q}+o\left(\frac{1}{\log Q}\right). \tag{21.6}$$

Proof. Substitute $a = 3/2$ in Proposition 21.2. □

Warning 21.4. Corollary 21.3 proves only the Bohr energy of a deliberately selected hard shell. It does not prove that the actual residual is supported in $(Q, e^{3/2}Q]$. Many transfer profiles have the same quadratic mass, and the projected zero-sampled variance is not the Bohr norm in Proposition 21.2.

22. Regularization and the remaining explicit-formula bridge

Definition 22.1. A regularized counting window is a family $H_{T,\eta}(u)$ tending to the midpoint indicator of $[0,T]$ as $\eta \to \infty$. A formal Gaussian example is

$$H_{T,\eta}(u) = \int_0^T \frac{\eta}{\sqrt{\pi}} e^{-\eta^2(u-v)^2}\,\mathrm{d}v. \tag{22.1}$$

Heuristic 22.2 (Fourier displacement)**.** For

$$h_{T,\eta,\omega}(u) = e^{i\omega u} H_{T,\eta}(u), \tag{22.2}$$

formal Fourier transformation gives

$$\widehat{h}_{T,\eta,\omega}(v) = e^{-(v-\omega)^2/(4\eta^2)} \frac{1 - e^{-i(v-\omega)T}}{i(v-\omega)}. \tag{22.3}$$

Insertion into the explicit formula generates the two branches in Definition 13.1, multiplied by explicit regularizing weights.

Warning 22.3. To turn Section 22 into a theorem one must specify a test class for which the explicit formula is valid, retain every archimedean and boundary term, and control jointly:

$$H, \qquad Q, \qquad \eta. \tag{22.4}$$

A formal Gaussian manipulation is not a proof of a sharp-window formula.

Conjecture 22.4 (Uniform explicit formula for the ordinate walk)**.** *There exist choices $Q = Q(H) \to \infty$ and $\eta = \eta(H) \to \infty$ for which the regularized prime-power formula, including its complete archimedean and endpoint terms, converges locally to the midpoint ordinate walk, with an error tending to zero in a topology strong enough to identify the atomic measure* (12.10).

Remark 22.5. The ordinary explicit formula reconstructs complete-zero weights, which retain $q^{\beta-1/2}$. Replacing that object by the unweighted ordinate walk is precisely where the positive defect $D_q(\tau)$ enters. Truncation control alone does not remove the horizontal zero information.

23. Energy exhaustion and distributional convergence

Conjecture 23.1 (Energy exhaustion)**.** *For a suitable joint regime $Q = Q(H) \to \infty$,*

$$\mathrm{Var}_{\gamma \le H}(F_{Q,H}) \le \varepsilon(Q,H), \qquad \varepsilon(Q(H),H) \to 0. \tag{23.1}$$

The variance is the full projected residual form (20.28)*, not the prime Gram term alone and not an unprojected Lebesgue L^2-norm.*

Theorem 23.2 (Abstract local distributional implication)**.** *Suppose a sequence F_j satisfies*

$$F_j - \frac{1}{2} \longrightarrow 0 \quad \text{in } L^2_{\mathrm{loc}}(\mathbb{R}), \tag{23.2}$$

and

$$\frac{\mathrm{d}}{\mathrm{d}T}\left[e^{i\omega T}\left(F_j(T) - \frac{1}{2}\right)\right] \longrightarrow \nu_q - \mu_q \quad \text{in } \mathcal{D}'(\mathbb{R}). \tag{23.3}$$

Then RH *holds.*

Proof. Define

$$G_j(T) = e^{i\omega T}\left(F_j(T) - \frac{1}{2}\right). \tag{23.4}$$

Multiplication by the bounded smooth phase preserves local L^2 convergence, so $G_j \to 0$ in L^2_{loc}. On every compact set, Cauchy–Schwarz then gives $G_j \to 0$ in L^1. For $\varphi \in C_c^\infty(\mathbb{R})$,

$$\langle G_j', \varphi\rangle = -\int_{\mathbb{R}} G_j(T)\varphi'(T)\,\mathrm{d}T \longrightarrow 0. \tag{23.5}$$

The assumed identification of the derivatives therefore implies $\nu_q - \mu_q = 0$ in $\mathcal{D}'$. A locally finite measure that is zero as a distribution is the zero measure. Multiplying by $\frac{1}{2}e^{-iT\log q}$ and applying Corollary 12.5 gives $D_q(\tau) = 0$ at every ordinate, hence RH. □

Warning 23.3. A normalized mean-square limit on intervals $[0, H]$ does not imply Theorem 23.2. A fixed compact defect, or even a sparse family of atomic defects, is diluted by division by H. Local or distributional convergence is indispensable.

24. Further analytic consequences of the finite Möbius interpolation

24.1. **Meromorphic structure and residues.**

Proposition 24.1 (Sources of poles)**.** *The meromorphic function Φ_N can have poles only at zeros of ζ and at zeros of the exponential polynomial P_N. At a simple zero ρ of ζ satisfying $P_N(\rho) \neq 0$, the residue is*

$$\operatorname*{Res}_{s=\rho} \Phi_N(s) = \frac{h_N}{\zeta'(\rho)P_N(\rho)}. \tag{24.1}$$

At a simple zero s_0 of P_N that is not a zero of ζ and satisfies $\zeta(s_0)^{-1} \neq A_N(s_0)$, the residue is

$$\operatorname*{Res}_{s=s_0} \Phi_N(s) = h_N \frac{\zeta(s_0)^{-1} - A_N(s_0)}{P_N'(s_0)}. \tag{24.2}$$

Proof. Both statements follow directly from the quotient in (6.10). At a simple zeta zero, $\zeta(s)^{-1}$ has principal part $1/(\zeta'(\rho)(s-\rho))$. At a simple zero of P_N, the denominator has principal part $P_N'(s_0)(s - s_0)$. □

Proposition 24.2 (Value at the pole of zeta)**.** *The reciprocal zeta function has a simple zero at $s = 1$, and therefore*

$$\Phi_N(1) = B_N - h_N \frac{A_N(1)}{P_N(1)}. \tag{24.3}$$

Proof. Use $\zeta(s)^{-1} \to 0$ as $s \to 1$ in (6.10). The denominator $P_N(1)$ is positive. □

24.2. **Eventually exact sign and quantitative rounding.**

Corollary 24.3 (Eventually exact sign)**.** *For sufficiently large real σ,*

$$\operatorname{sgn}\big(\Phi_N(\sigma) - M(N)\big) = \mu(a_N). \tag{24.4}$$

Proof. The leading asymptotic has the form

$$\Phi_N(\sigma) - M(N) = A_N(\sigma)(1 + o(1)), \qquad A_N(\sigma) = h_N\mu(a_N)\left(\frac{q_N}{a_N}\right)^{\sigma}. \tag{24.5}$$

The magnitude multiplying $\mu(a_N)$ is strictly positive. Choose σ_0 so that the relative error has modulus less than one half for $\sigma \geq \sigma_0$. Then the full expression cannot change the sign of its leading term, and its sign is $\mu(a_N)$. □

Proposition 24.4 (Existence of a finite rounding threshold)**.** *For every $N \geq 2$ there exists a finite $\sigma_N > 1$ such that*

$$\operatorname{nint} \Phi_N(\sigma) = M(N) \tag{24.6}$$

for all real $\sigma \geq \sigma_N$, where nint *denotes the nearest integer.*

Proof. Let

$$B_N(\sigma) = \frac{h_N N}{\sigma - 1} \left(\frac{q_N}{N} \right)^{\sigma}. \tag{24.7}$$

The base q_N/N lies strictly between zero and one. Hence its exponential power tends to zero, while $h_N N/(\sigma - 1)$ grows at most polynomially and in fact decreases after the constant factor is removed. Thus $B_N(\sigma) \to 0$. Choose σ_N so that $B_N(\sigma) < 1/2$ for every $\sigma \geq \sigma_N$. The rounding criterion then gives the conclusion uniformly for all such σ. □

Proposition 24.5 (A sufficient explicit threshold)**.** *Let*

$$\lambda_N = \log \frac{N}{q_N} > 0. \tag{24.8}$$

Every $\sigma > 1$ satisfying

$$\sigma \lambda_N > \log(2 h_N N) + \log \frac{1}{\sigma - 1} \tag{24.9}$$

is a valid rounding point. In particular, any $\sigma \geq 2$ satisfying

$$\sigma > \frac{\log(2 h_N N)}{\lambda_N} \tag{24.10}$$

is sufficient.

Proof. Rewrite the rounding condition as

$$\frac{h_N N}{\sigma - 1} e^{-\sigma \lambda_N} < \frac{1}{2}. \tag{24.11}$$

All factors are positive. Taking logarithms gives

$$\log(h_N N) - \log(\sigma - 1) - \sigma \lambda_N < -\log 2, \tag{24.12}$$

which rearranges to the implicit condition in the statement. When $\sigma \geq 2$, one has $\log(1/(\sigma - 1)) \leq 0$. The stronger but simpler inequality $\sigma \lambda_N > \log(2 h_N N)$ therefore implies the implicit condition and is sufficient. □

24.3. **Recursive recovery of the nonzero tail.**

Definition 24.6. Let

$$a_{N,1} < a_{N,2} < a_{N,3} < \cdots \tag{24.13}$$

be the increasing sequence of integers exceeding N for which $\mu(a_{N,j}) \neq 0$.

Theorem 24.7 (Recursive exponential extraction)**.** *The real-axis function $E_N(\sigma)$ determines the entire signed sequence*

$$\left(a_{N,j}, \mu(a_{N,j})\right)_{j \geq 1} \tag{24.14}$$

recursively. More precisely, once the first $J - 1$ terms are known, define

$$E_N^{(J)}(\sigma) = E_N(\sigma) - \frac{\sum_{j<J} \mu(a_{N,j}) a_{N,j}^{-\sigma}}{P_N(\sigma)}. \tag{24.15}$$

Then

$$a_{N,J} = q_N \lim_{\sigma\to\infty} |E_N^{(J)}(\sigma)|^{-1/\sigma}, \tag{24.16}$$

$$\mu(a_{N,J}) = \lim_{\sigma\to\infty} \operatorname{sgn} E_N^{(J)}(\sigma). \tag{24.17}$$

Proof. By definition,

$$E_N(\sigma) = \frac{1}{P_N(\sigma)} \sum_{j\geq 1} \mu(a_{N,j}) a_{N,j}^{-\sigma}. \tag{24.18}$$

Subtracting the first $J-1$ known terms leaves

$$E_N^{(J)}(\sigma) = \frac{1}{P_N(\sigma)} \sum_{j\geq J} \mu(a_{N,j}) a_{N,j}^{-\sigma}. \tag{24.19}$$

Factor the first remaining term and the least denominator term exactly as in Theorem 8.2. Dominated convergence gives

$$E_N^{(J)}(\sigma) = \mu(a_{N,J}) \left(\frac{q_N}{a_{N,J}}\right)^{\sigma} (1 + o(1)). \tag{24.20}$$

Taking the negative σ-th root recovers the ratio $a_{N,J}/q_N$, and eventual sign recovers $\mu(a_{N,J})$. Induction on J determines the entire signed support sequence uniquely. □

Remark 24.8. Theorem 24.7 is an exact uniqueness statement. It does not by itself yield a computational advantage, because evaluating E_N already requires the reciprocal zeta function with sufficient precision and control of exponentially small differences.

25. Additional structure of the positive horizontal defect

25.1. **Even-moment expansion.**

Proposition 25.1 (Power-series expansion in $\log q$)**.** *At a fixed ordinate τ, put*

$$\delta_\rho = \beta - \frac{1}{2} \tag{25.1}$$

for zeros with $\beta > 1/2$. Then

$$D_q(\tau) = \sum_{m=1}^{\infty} \frac{(\log q)^{2m}}{(2m)!} \sum_{\substack{\rho=\beta+i\tau \\ \beta>1/2}} \delta_\rho^{2m}. \tag{25.2}$$

Proof. At a fixed ordinate there are finitely many zeros, counted with finite multiplicity. Put $y = \log q$ and $\delta_\rho = \beta - 1/2 > 0$. The entire expansion

$$\cosh(\delta_\rho y) - 1 = \sum_{m=1}^{\infty} \frac{\delta_\rho^{2m} y^{2m}}{(2m)!} \tag{25.3}$$

converges absolutely for every complex y. Because the outer zero sum is finite at the fixed ordinate, it may be interchanged with the power series. Summing the coefficients gives the asserted even-moment expansion, with no uniform-convergence subtlety. □

Corollary 25.2 (Quadratic lower bound)**.** *For every $q > 1$,*

$$D_q(\tau) \geq \frac{(\log q)^2}{2} \sum_{\substack{\rho=\beta+i\tau \\ \beta>1/2}} \left(\beta - \frac{1}{2}\right)^2. \tag{25.4}$$

Proof. The Taylor series

$$\cosh x - 1 = \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots \tag{25.5}$$

has nonnegative terms for real x. Dropping every term except the first gives $\cosh x - 1 \geq x^2/2$. Apply this with $x = (\beta - 1/2)\log q$ to every off-line pair at the fixed ordinate. Since all summands remain nonnegative, summing preserves the inequality and produces the stated quadratic moment. □

Proposition 25.3 (Monotonicity and convexity)**.** *For fixed τ, the function*

$$t \longmapsto D_{e^t}(\tau), \qquad t \geq 0, \tag{25.6}$$

is nonnegative, increasing, and convex. It is identically zero if and only if every zero at ordinate τ lies on the critical line.

Proof. Set $t = \log q \geq 0$. For a horizontal displacement $\delta > 0$, define $f_\delta(t) = \cosh(\delta t) - 1$. Then

$$f_\delta'(t) = \delta \sinh(\delta t) \geq 0, \qquad f_\delta''(t) = \delta^2 \cosh(\delta t) > 0. \tag{25.7}$$

Thus every summand is increasing and strictly convex. Their finite sum has the same properties whenever at least one summand is present. If no off-line pair is present, the sum is identically zero. Conversely, an identically zero sum of nonnegative summands cannot contain a positive summand, so no off-line pair exists. □

Corollary 25.4 (One parameter and many parameters)**.** *One fixed $q > 1$ suffices for the logical criterion* (12.11)*. A continuum of values of q additionally encodes all even power sums of the horizontal displacements at each ordinate through* (25.2)*.*

Proof. The fixed-parameter assertion is Corollary 12.5. For the family assertion, Proposition 25.1 writes the analytic function of $y = \log q$ as

$$D_{e^y}(\tau) = \sum_{m \geq 1} \frac{y^{2m}}{(2m)!} P_{2m}(\tau), \tag{25.8}$$

where $P_{2m}(\tau)$ is the $2m$-th power sum of the positive horizontal displacements. Differentiating $2m$ times at zero gives $P_{2m}(\tau)$. Thus a neighborhood of one in the q-parameter encodes all even moments, whereas positivity makes a single nontrivial parameter enough merely to test whether they all vanish. □

25.2. **Jump formulation.**

Definition 25.5. Define the right-continuous cumulative functions

$$U_q(T) = \sum_{0 < \tau \leq T} m(\tau) e^{i\tau \log q}, \tag{25.9}$$

$$V_q(T) = \sum_{0 < \tau \leq T} W_q(\tau) e^{i\tau \log q}. \tag{25.10}$$

Proposition 25.6 (Exact jump defect)**.** *At every ordinate τ,*

$$\boxed{\frac{1}{2} e^{-i\tau \log q} \Delta(V_q - U_q)(\tau) = D_q(\tau).} \tag{25.11}$$

Proof. The jump of the complete-zero walk at τ is

$$\Delta V_q(\tau) = W_q(\tau) e^{i\tau \log q}, \tag{25.12}$$

whereas the jump of the ordinate-only walk is

$$\Delta U_q(\tau) = m(\tau) e^{i\tau \log q}. \tag{25.13}$$

Subtracting and using $W_q - m = 2D_q$ gives

$$\Delta(V_q - U_q)(\tau) = 2D_q(\tau)e^{i\tau \log q}. \tag{25.14}$$

Demodulation by $\frac{1}{2}e^{-i\tau \log q}$ leaves the nonnegative scalar $D_q(\tau)$. □

26. Regularized explicit formula with explicit prime-power weights

26.1. The explicit formula in the Fourier normalization used here.

Theorem 26.1 (Guinand–Weil explicit formula)**.** *Let h be holomorphic in a strip $|\Im z| \le 1/2 + \epsilon$, real-valued on $\mathbb{R}$, and suppose that, uniformly in the strip, $h(z) \ll (1 + |\Re z|)^{-1-\delta}$ for some $\delta > 0$. Define*

$$\widehat{h}(v) = \int_{\mathbb{R}} h(t)e^{-ivt}\, \mathrm{d}t. \tag{26.1}$$

Then, with zeros counted with multiplicity,

$$\sum_{\rho} h\Big(\frac{\rho - 1/2}{i}\Big) = h\Big(\frac{1}{2i}\Big) + h\Big(-\frac{1}{2i}\Big) - \frac{\widehat{h}(0)}{2\pi}\log\pi \tag{26.2}$$

$$+ \frac{1}{2\pi}\int_{-\infty}^{\infty} h(u)\Re\frac{\Gamma'}{\Gamma}\Big(\frac{1}{4} + \frac{iu}{2}\Big)\, \mathrm{d}u \tag{26.3}$$

$$- \frac{1}{2\pi}\sum_{n=2}^{\infty}\frac{\Lambda(n)}{\sqrt{n}}\Big[\widehat{h}(\log n) + \widehat{h}(-\log n)\Big]. \tag{26.4}$$

The formula is understood by the usual limiting procedure whenever the chosen test lies at the boundary of the stated class [10, 13].

Proof. Consider the completed function $\xi(s)$ from (2.7) and the contour integral

$$\frac{1}{2\pi i}\int_{\partial\mathcal{R}} \frac{\xi'}{\xi}(s)h\Big(\frac{s - 1/2}{i}\Big)\, \mathrm{d}s \tag{26.5}$$

over a rectangle whose vertical sides are $\Re s = 1/2 \pm a$, with $a > 1/2$, and whose horizontal sides tend to infinite height through values avoiding zeros. The decay assumptions on h make the horizontal integrals vanish. By the residue theorem, the enclosed zeros of ξ contribute the spectral sum on the left, while the factors $s(s-1)$ produce the two evaluations at $\pm 1/(2i)$.

On the right vertical side, decompose the logarithmic derivative as

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\frac{\Gamma'}{\Gamma}\Big(\frac{s}{2}\Big) + \frac{\zeta'}{\zeta}(s). \tag{26.6}$$

The functional equation $\xi(s) = \xi(1-s)$ transforms the left vertical side into the conjugate contribution of the right side. After the two sides are combined and parametrized by $s = 1/2 + iu$, the constant $-\frac{1}{2}\log\pi$ gives $-\widehat{h}(0)\log\pi/(2\pi)$, and the gamma logarithmic derivative gives the integral in (26.3).

For $\Re s > 1$, the absolutely convergent Dirichlet series

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n=2}^{\infty}\frac{\Lambda(n)}{n^s} \tag{26.7}$$

permits termwise integration. The contribution of n^{-s} on the two paired vertical sides is

$$\frac{\Lambda(n)}{\sqrt{n}}\Big[\widehat{h}(\log n) + \widehat{h}(-\log n)\Big], \tag{26.8}$$

with the prefactor $-1/(2\pi)$ arising from the contour normalization and the parametrization $\mathrm{d}s = i\, \mathrm{d}u$. Summing the contributions gives (26.4). Absolute convergence in the initial test class justifies all interchanges; the standard limiting argument extends the identity to the stated boundary cases. □

Corollary 26.2 (Complex-linear extension)**.** *The formula in Theorem 26.1 remains valid for complex-valued h satisfying the same analytic and decay conditions.*

Proof. Define

$$h_1(z) = \frac{1}{2}\left(h(z) + \overline{h(\overline{z})}\right), \tag{26.9}$$

$$h_2(z) = \frac{1}{2i}\left(h(z) - \overline{h(\overline{z})}\right). \tag{26.10}$$

Both functions are holomorphic in the strip, real-valued on the real axis, satisfy the same decay, and obey $h = h_1 + ih_2$. Apply the real formula separately to h_1 and h_2, then combine by complex linearity. □

Corollary 26.3 (Why only prime powers occur)**.** *The arithmetic side of the formula is supported exactly on prime powers, because*

$$\Lambda(n) \neq 0 \quad \Longleftrightarrow \quad n = p^r \tag{26.11}$$

for a prime p and integer $r \geq 1$.

Proof. This is the defining support property of the von Mangoldt function: $\Lambda(p^r) = \log p$ and $\Lambda(n) = 0$ otherwise. Applying it to the prime sum (26.4) removes every non-prime-power integer. □

26.2. **Gaussian convolution of a finite interval.**

Definition 26.4. For $\eta > 0$, define

$$g_\eta(t) = \frac{\eta}{\sqrt{\pi}} e^{-\eta^2 t^2}, \tag{26.12}$$

and

$$H_{T,\eta}(z) = \int_0^T g_\eta(z-u)\,\mathrm{d}u. \tag{26.13}$$

The oscillatory test is

$$h_{T,\eta,\omega}(z) = e^{i\omega z} H_{T,\eta}(z). \tag{26.14}$$

Proposition 26.5 (Fourier transform)**.** *With the convention*

$$\widehat{f}(v) = \int_{\mathbb{R}} f(t) e^{-ivt}\,\mathrm{d}t, \tag{26.15}$$

one has

$$\boxed{\widehat{h}_{T,\eta,\omega}(v) = e^{-(v-\omega)^2/(4\eta^2)} \frac{1 - e^{-i(v-\omega)T}}{i(v-\omega)},} \tag{26.16}$$

where the quotient is interpreted continuously as T at $v = \omega$.

Proof. With the Fourier convention $\widehat{f}(v) = \int_{\mathbb{R}} f(t) e^{-ivt}\,\mathrm{d}t$, the normalized Gaussian has transform

$$\widehat{g}_\eta(v) = e^{-v^2/(4\eta^2)}. \tag{26.17}$$

The regularized interval is the convolution $H_{T,\eta} = \mathbf{1}_{[0,T]} * g_\eta$, so the convolution theorem gives

$$\widehat{H}_{T,\eta}(v) = e^{-v^2/(4\eta^2)} \frac{1 - e^{-ivT}}{iv}, \tag{26.18}$$

with the quotient interpreted continuously at $v = 0$. Multiplication by $e^{i\omega t}$ translates the Fourier transform: $\widehat{e^{i\omega t} H}(v) = \widehat{H}(v - \omega)$. Substitution proves the formula. □

Proposition 26.6 (Regularized two-branch prime contribution)**.** *In the prime-power side of the Guinand–Weil explicit formula, the test* (26.14) *produces, apart from terms independent of T, the oscillatory sum*

$$\mathcal{P}^{\mathrm{osc}}_{\eta,\omega}(T) = \sum_{n=p^r} c_n^- e^{-(\ell_n-\omega)^2/(4\eta^2)} e^{i(\omega-\ell_n)T} \tag{26.19}$$

$$+ \sum_{n=p^r} c_n^+ e^{-(\ell_n+\omega)^2/(4\eta^2)} e^{i(\omega+\ell_n)T}. \tag{26.20}$$

Proof. By Corollary 26.2, the explicit formula applies to the complex oscillatory test. Insert $v = \ell_n$ and $v = -\ell_n$ into the exact Fourier transform. For the first argument,

$$\widehat{h}(\ell_n) = w_{n,-} \frac{1 - e^{-i(\ell_n-\omega)T}}{i(\ell_n - \omega)}, \tag{26.21}$$

where $w_{n,-} = e^{-(\ell_n-\omega)^2/(4\eta^2)}$. Multiplication by the prime-side factor $-\Lambda(n)/(2\pi\sqrt{n})$ separates a constant and the oscillatory term

$$w_{n,-} c_n^- e^{i(\omega-\ell_n)T}. \tag{26.22}$$

The argument $-\ell_n$ similarly gives

$$w_{n,+} c_n^+ e^{i(\omega+\ell_n)T}, \qquad w_{n,+} = e^{-(\ell_n+\omega)^2/(4\eta^2)}. \tag{26.23}$$

The remaining terms are independent of T and belong to the additive constant that must be retained or fitted before demodulation. □

26.3. **A uniform bound for the regularized prime tail.**

Theorem 26.7 (Uniform Gaussian tail)**.** *Assume $Q > e^{2\omega}$, put $L = \log Q$, and suppose $L > \omega + \eta^2$. Then*

$$\sup_{T\in\mathbb{R}} \left| \sum_{\substack{n=p^r \\ n>Q}} \left[c_n^- e^{-(\ell_n-\omega)^2/(4\eta^2)} e^{i(\omega-\ell_n)T} + c_n^+ e^{-(\ell_n+\omega)^2/(4\eta^2)} e^{i(\omega+\ell_n)T} \right] \right|$$

$$\ll_\omega Q^{-1/2} \exp\left[-\frac{(L-\omega)^2}{4\eta^2} \right] \tag{26.24}$$

$$+ \frac{\eta^2}{L-\omega-\eta^2} \exp\left(\frac{\eta^2}{4} - \frac{(L-\omega-\eta^2)^2}{4\eta^2} \right). \tag{26.25}$$

Proof. For $n > Q > e^{2\omega}$,

$$|\ell_n - \omega| \geq \frac{1}{2}\ell_n, \qquad \ell_n + \omega \geq \ell_n. \tag{26.26}$$

Since $\Lambda(n) \leq \log n$, the absolute value of the two oscillatory branches is bounded, up to a constant depending on ω, by

$$\sum_{n>Q} f(n), \qquad f(x) = x^{-1/2} \exp\left[-\frac{(\log x - \omega)^2}{4\eta^2} \right]. \tag{26.27}$$

The function f is decreasing for $\log x > \omega - \eta^2$, hence in particular on $[Q, \infty)$. The integral test therefore gives the endpoint-sensitive estimate

$$\sum_{n>Q} f(n) \leq f(Q) + \int_Q^\infty f(x)\, \mathrm{d}x. \tag{26.28}$$

The first term is exactly the expression in (26.24).

For the integral $I = \int_Q^\infty f(x)\,\mathrm{d}x$, put $u = \log x$. Completing the square gives

$$I = \int_L^\infty \exp\left[\frac{u}{2} - \frac{(u-\omega)^2}{4\eta^2}\right]\,\mathrm{d}u \tag{26.29}$$

$$= \exp\left(\frac{\omega}{2} + \frac{\eta^2}{4}\right)\int_L^\infty \exp\left[-\frac{(u-\omega-\eta^2)^2}{4\eta^2}\right]\,\mathrm{d}u. \tag{26.30}$$

For $r = L - \omega - \eta^2 > 0$,

$$\int_r^\infty e^{-v^2/(4\eta^2)}\,\mathrm{d}v \le \frac{2\eta^2}{r}e^{-r^2/(4\eta^2)}. \tag{26.31}$$

The factor $e^{\omega/2}$ is absorbed in the ω-dependent implied constant. This gives (26.25). No estimate depends on T, because all oscillatory exponentials have modulus one. □

Corollary 26.8 (One explicit joint choice)**.** *Assume* $\eta^2 > \omega/3$ *and*

$$\log Q = \omega + 3\eta^2. \tag{26.32}$$

Then the complete right-hand side of (26.24)– (26.25) *is*

$$O_\omega(e^{-3\eta^2/4}). \tag{26.33}$$

Proof. The condition $\eta^2 > \omega/3$ ensures $Q > e^{2\omega}$. Moreover,

$$L - \omega - \eta^2 = 2\eta^2. \tag{26.34}$$

The integral term has exponent $\eta^2/4 - \eta^2 = -3\eta^2/4$. The endpoint term has exponent

$$-\frac{L}{2} - \frac{(L-\omega)^2}{4\eta^2} = -\frac{\omega}{2} - \frac{15}{4}\eta^2, \tag{26.35}$$

and is therefore smaller. This proves the stated bound. □

Warning 26.9. Theorem 26.7 controls the prime side of a regularized explicit formula. It does not identify the complete-zero spectral side with the unweighted ordinate walk. That identification is exactly where the positive measure (12.10) appears.

27. Mean-square statements and their exact logical strength

27.1. Finite Dirichlet-polynomial tails in Lebesgue mean.

Proposition 27.1 (Finite-tail mean-value estimate)**.** *Let*

$$E_{Q,R}(T) = \sum_{\substack{Q<n\le R\\ n=p^r}} \left(c_n^- n^{-iT} + c_n^+ n^{iT}\right). \tag{27.1}$$

Then

$$\frac{1}{X}\int_0^X |E_{Q,R}(T)|^2\,\mathrm{d}T \ll_\omega \left(1 + \frac{R}{X}\right)\left[\log\frac{\log R}{\log Q} + \frac{1}{\log Q} + Q^{-1/2}\right] \tag{27.2}$$

for $e^{2\omega} < Q < R$.

Proof. Apply the Montgomery–Vaughan mean-value theorem for Dirichlet polynomials [4, Chapter 7] separately to

$$\sum_{Q<n\le R} c_n^- n^{-iT} \quad\text{and}\quad \sum_{Q<n\le R} c_n^+ n^{iT}. \tag{27.3}$$

It gives an upper bound of the form

$$\int_0^X \left| \sum_{n \le R} a_n n^{-iT} \right|^2 \, \mathrm{d}T \ll (X+R) \sum_{n \le R} |a_n|^2. \tag{27.4}$$

The inequality $|u+v|^2 \le 2|u|^2 + 2|v|^2$ combines the branches. For $n = p^r > e^{2\omega}$,

$$|c_n^-|^2 + |c_n^+|^2 \ll_\omega \frac{\Lambda(n)^2}{n(\log n)^2} = \frac{1}{r^2 p^r}. \tag{27.5}$$

The prime terms sum to $\log(\log R/\log Q) + O(1/\log Q)$ by Mertens' theorem for primes, while proper powers contribute $O(Q^{-1/2})$. Division by X yields the stated estimate. □

Corollary 27.2 (A finite logarithmic tail can be exhausted)**.** *Let*

$$R(X) = X^{1/2}, \qquad Q(X) = R(X)^{1-1/\sqrt{\log\log X}}. \tag{27.6}$$

Then

$$\frac{1}{X} \int_0^X |E_{Q(X),R(X)}(T)|^2 \, \mathrm{d}T \longrightarrow 0. \tag{27.7}$$

Proof. For the choice (27.6),

$$\frac{R(X)}{X} = X^{-1/2} \longrightarrow 0. \tag{27.8}$$

If $\delta_X = 1/\sqrt{\log\log X}$, then $\log Q = (1-\delta_X)\log R$, so

$$\log \frac{\log R}{\log Q} = -\log(1-\delta_X) = O(\delta_X) \longrightarrow 0. \tag{27.9}$$

Also $1/\log Q \to 0$ and $Q^{-1/2} \to 0$. Apply Proposition 27.1. □

Warning 27.3. Corollary 27.2 concerns a finite tail in Lebesgue measure. The numerical variance is sampled on zeta ordinates and contains the projected prime kernel (20.25); the two norms are not interchangeable.

27.2. **Normalized global convergence does not detect fixed atoms.**

Proposition 27.4 (Counterexample to global normalized sufficiency)**.** *There exist functions F_X such that*

$$\frac{1}{X} \int_0^X \left| F_X(T) - \frac{1}{2} \right|^2 \, \mathrm{d}T \longrightarrow 0, \tag{27.10}$$

but whose distributional derivatives retain a nonzero fixed atomic part.

Proof. Take $F_X(T) = 1/2 + \mathbf{1}_{[0,1]}(T)$. Then for every $X \ge 1$,

$$\frac{1}{X} \int_0^X \left| F_X(T) - \frac{1}{2} \right|^2 \, \mathrm{d}T = \frac{1}{X} \longrightarrow 0. \tag{27.11}$$

However, on the fixed compact interval $[0,1]$, the defect remains identically one. In the sense of distributions,

$$\frac{\mathrm{d}}{\mathrm{d}T} \mathbf{1}_{[0,1]} = \delta_0 - \delta_1, \tag{27.12}$$

which does not tend to zero. Thus normalization by the length of an expanding interval can erase a fixed local or atomic obstruction. □

Corollary 27.5. *Any route from energy exhaustion to the vanishing of the positive atomic measure* (12.10) *must use a local topology, atomwise convergence, or distributional convergence against compactly supported test functions.*

Proof. Proposition 27.4 shows that global normalized convergence alone permits a persistent compact atomic defect. Therefore any sufficient argument must contain additional local information of one of the stated forms. □

28. Weighted prime-power profiles

Definition 28.1. Let $W \in C_c(\mathbb{R})$ satisfy $\operatorname{supp} W \subset [0,\infty)$. Define

$$G_{Q,W}(T) = \sum_{n=p^r} W\left(\log \frac{n}{Q}\right) g_n(T). \tag{28.1}$$

Because $W(u) = 0$ for $u < 0$ and has compact support, the sum is finite and contains only prime powers in a fixed multiplicative interval above Q.

Theorem 28.2 (Weighted Bohr energy)**.** *As $Q \to \infty$,*

$$\lim_{X\to\infty} \frac{1}{X} \int_0^X |G_{Q,W}(T)|^2 \, \mathrm{d}T = \frac{1}{2\pi^2 \log Q} \int_0^\infty |W(u)|^2 \, \mathrm{d}u + o\left(\frac{1}{\log Q}\right). \tag{28.2}$$

Proof. Expand the squared modulus of the finite weighted sum. Every diagonal term contributes

$$|W(\log(n/Q))|^2 \left(|c_n^-|^2 + |c_n^+|^2\right). \tag{28.3}$$

Every unequal pair has a nonzero frequency difference and therefore zero Bohr mean. For primes, use

$$|c_p^-|^2 + |c_p^+|^2 = \frac{1}{2\pi^2 p} + O_\omega\left(\frac{1}{p(\log p)^2}\right). \tag{28.4}$$

Write $p = Qe^u$ and apply the prime number theorem by partial summation. For compactly supported continuous W,

$$\sum_p \frac{1}{p} \left| W\left(\log \frac{p}{Q}\right)\right|^2 = \frac{1}{\log Q} \int_0^\infty |W(u)|^2 \, \mathrm{d}u + o\left(\frac{1}{\log Q}\right). \tag{28.5}$$

The coefficient error and all proper prime powers are lower order. This proves the formula. □

Corollary 28.3 (Interpretation of the constant 3/4)**.** *A weighted Bohr model has leading energy $3/(4\pi^2 \log Q)$ precisely when*

$$\int_0^\infty |W(u)|^2 \, \mathrm{d}u = \frac{3}{2}. \tag{28.6}$$

The constant determines a quadratic profile mass, not a unique hard support.

Proof. Compare the leading term in Theorem 28.2 with $3/(4\pi^2 \log Q)$. Equality of the coefficients is equivalent to $\int_0^\infty |W(u)|^2 \, \mathrm{d}u = 3/2$. □

29. Detailed projected Gram algebra

29.1. Four branch expansion.

Proposition 29.1 (Full unprojected quadratic expansion)**.** *For a finite set $\mathcal{N}$ of prime powers, put*

$$G_{\mathcal{N}}(\gamma) = \sum_{n\in\mathcal{N}} \left(c_n^- n^{-i\gamma} + c_n^+ n^{i\gamma}\right). \tag{29.1}$$

Then

$$\|G_{\mathcal{N}}\|_H^2 = \sum_{m,n\in\mathcal{N}} \overline{c_m^-} c_n^- M_H(m/n) \tag{29.2}$$

$$+ \sum_{m,n\in\mathcal{N}} \overline{c_m^-} c_n^+ M_H(mn) \tag{29.3}$$

$$+ \sum_{m,n\in\mathcal{N}} \overline{c_m^+} c_n^- \overline{M_H(mn)} \tag{29.4}$$

$$+ \sum_{m,n\in\mathcal{N}} \overline{c_m^+} c_n^+ M_H(n/m). \tag{29.5}$$

Proof. Write the finite model as

$$G(\gamma) = \sum_n \left(c_n^- n^{-i\gamma} + c_n^+ n^{i\gamma} \right). \tag{29.6}$$

Multiplying by its conjugate produces four sign choices. For example,

$$\overline{c_m^- m^{-i\gamma}}\, c_n^- n^{-i\gamma} = \overline{c_m^-} c_n^- \left(\frac{m}{n} \right)^{i\gamma}, \tag{29.7}$$

whose sample average is $\overline{c_m^-} c_n^- M_H(m/n)$. The other three products give the arguments mn, $(mn)^{-1}$, and n/m. Summing all pairs yields the displayed expansion exactly. □

29.2. **Prime-tower blocks.**

Proposition 29.2 (Same-tower multiplicative closure)**.** *For $m = p^r$ and $n = p^s$, every argument of M_H in (29.2)–(29.5) is one of*

$$p^{s-r}, \qquad p^{r-s}, \qquad p^{r+s}, \qquad p^{-(r+s)}. \tag{29.8}$$

Thus the exact Gram arguments of a two-branch prime tower remain inside the multiplicative set generated by p; the block is not reduced to the scalar diagonal $r = s$.

Proof. Substitute $m = p^r$ and $n = p^s$ into the four multiplicative arguments. They become

$$p^{r-s}, \qquad p^{r+s}, \qquad p^{-(r+s)}, \qquad p^{s-r}. \tag{29.9}$$

Each is one, a prime power, or the reciprocal of a prime power. This is an exact algebraic closure statement. Interpreting these entries by the main term of a Landau–Gonek formula for the ordinate-only kernel M_H additionally requires the Riemann Hypothesis, or an independent replacement of $x^{i\gamma}$ by $x^{\rho-1/2}$. □

Proposition 29.3 (Projection correction)**.** *Let V be the matrix whose columns are the sampled prime-power modes, and let B be the two-column matrix with columns 1 and $e^{i\omega\gamma}$. Then the residual Gram matrix is*

$$\widetilde{K}_H = \frac{1}{N(H)} V^*V - \frac{1}{N(H)^2} V^*B \left(\frac{1}{N(H)} B^*B \right)^{-1} B^*V. \tag{29.10}$$

In particular, even the norm of a single frequency is altered by the constant–carrier projection.

Proof. Let B be the matrix whose columns are the sampled constant and carrier vectors. Since these columns are linearly independent when $|M_H(e^\omega)| < 1$, the orthogonal projection onto their span is

$$P_B = B(B^*B)^{-1}B^*. \tag{29.11}$$

The complementary projection is $I - P_B$. If V contains the sampled prime frequency vectors, the projected Gram matrix is

$$\frac{1}{N(H)} V^*(I - P_B)V = \frac{1}{N(H)} V^*V \tag{29.12}$$

$$- \frac{1}{N(H)} V^*B(B^*B)^{-1}B^*V. \tag{29.13}$$

Writing $B^*B = N(H)G_H$ gives the normalized Schur-complement form in the proposition. In particular, projection changes both diagonal and off-diagonal entries. □

30. Analytic targets isolated by the exact identities

Conjecture 30.1 (Collective law for the reported carrier)**.** *In a specified joint range $Q = Q(H) \to \infty$,*

$$\Im c_{Q,H} = \pi \Delta_\omega(Q) + o\left(\Delta_\omega(Q)\right). \tag{30.1}$$

By Corollary 17.6, this is equivalent to a collective cancellation involving the three exact kernel families

$$M_H(n), \qquad M_H(q/n), \qquad M_H(qn),$$

together with the nonorthogonality factor $1-|M_H(q)|^2$. The raw law for $\widehat{c}_{Q,H}$ is a different conjecture.

Conjecture 30.2 (Projected variance law)**.** *For a suitable joint range,*

$$V(Q,H) = \frac{3}{4}\Delta_\omega(Q) + o\left(\Delta_\omega(Q)\right). \tag{30.2}$$

The left-hand side is the full projected residual expression (20.28)*. Its prime Gram term is only one of three cutoff-dependent contributions in addition to the projected background energy.*

Question 30.3 (Background, mixed, tower, and cross-tower separation)**.** Can one prove, uniformly in a growing cutoff, asymptotic formulas for

$$-2\Re \sum_p \mathbf{c}_p^* \mathbf{d}_{p,H}, \tag{30.3}$$

$$\sum_p \mathbf{c}_p^* \widetilde{K}_{p,p;H} \mathbf{c}_p, \tag{30.4}$$

$$\sum_{p \neq r} \mathbf{c}_p^* \widetilde{K}_{p,r;H} \mathbf{c}_r \tag{30.5}$$

relative to the projected background energy $V_Y(H)$? These are the exact components that must be separated before any assertion about a "diagonal" can be made.

31. Exact routes from the exponential zero walk to the Riemann Hypothesis

This section records the routes that arise directly from the phase-walk construction. Each route is an exact implication or equivalence. What is missing in every incomplete route is stated separately; no numerical observation is used as a premise.

31.1. Route I: equality of the complete and ordinate-only atomic measures.

Theorem 31.1 (Fixed-q atomic criterion)**.** *For any fixed $q > 1$,*

$$\nu_q = \mu_q \quad \Longleftrightarrow \quad \text{RH}. \tag{31.1}$$

Equivalently, it is enough to prove equality of the jumps of the two walks at every positive ordinate.

Proof. The measure difference is given exactly by

$$\frac{1}{2} e^{-iT\log q}(\nu_q - \mu_q) = \sum_{\tau > 0} D_q(\tau)\delta_\tau. \tag{31.2}$$

Every coefficient on the right is nonnegative. Equality of the measures forces every coefficient to vanish, and Theorem 12.3 then places every zero on the critical line. Conversely, under RH, all complete-zero weights equal one and the measures are identical. Since a locally finite atomic measure is determined by its jumps, equality of jumps is equivalent to equality of measures. □

Corollary 31.2 (Midpoint residual criterion)**.** *Suppose an explicit-formula reconstruction is proved to converge locally to the complete-zero midpoint walk, and suppose its residual at every ordinate τ is shown to equal*

$$\frac{m(\tau)}{2}e^{i\tau\log q}. \tag{31.3}$$

Then RH *follows.*

Proof. The complete midpoint jump is $\frac{1}{2}W_q(\tau)e^{i\tau\log q}$. Equality with (31.3) gives $W_q(\tau)=m(\tau)$ for every τ. Apply Theorem 31.1. □

31.2. **Route II: local distributional convergence.**

Theorem 31.3 (Test-function criterion)**.** *Let η_j be a sequence of locally finite complex measures such that*

$$\eta_j \longrightarrow \nu_q-\mu_q \quad \text{in } \mathcal{D}'(\mathbb{R}). \tag{31.4}$$

If, for every $\varphi\in C_c^\infty(\mathbb{R})$,

$$\langle \eta_j,\varphi\rangle \longrightarrow 0, \tag{31.5}$$

then RH *holds.*

Proof. For every test function φ, convergence (31.4) gives

$$\langle \eta_j,\varphi\rangle \longrightarrow \langle \nu_q-\mu_q,\varphi\rangle. \tag{31.6}$$

The second hypothesis says that the same left-hand side tends to zero. Uniqueness of limits in $\mathbb{C}$ yields

$$\langle \nu_q-\mu_q,\varphi\rangle = 0 \tag{31.7}$$

for every $\varphi\in C_c^\infty$. Therefore the measure difference is the zero distribution, hence the zero locally finite measure. The fixed-q atomic criterion then implies RH. □

Remark 31.4. For the prime-power construction, the missing theorem is a joint (H,Q,η) limit that identifies the derivative of the regularized prime side with $\nu_q-\mu_q$, while the corresponding residual tends locally to zero. A normalized average over $[0,H]$ is not sufficient, because it can dilute fixed atoms.

31.3. **Route III: small-q curvature at every ordinate.**

Theorem 31.5 (Quadratic curvature criterion)**.** *Put $q=e^y$. At each ordinate τ,*

$$D_{e^y}(\tau)=\frac{y^2}{2}\sum_{\substack{\rho=\beta+i\tau\\ \beta>1/2}}\left(\beta-\frac12\right)^2+O_\tau(y^4)\qquad (y\to 0^+). \tag{31.8}$$

Consequently,

$$\text{RH}\quad\Longleftrightarrow\quad D_{e^y}(\tau)=o(y^2)\ \text{for every } \tau. \tag{31.9}$$

Proof. At a fixed ordinate, enumerate the positive horizontal displacements as $\delta_1,\dots,\delta_m$, repeated with multiplicity. Taylor's theorem at zero gives, uniformly over this finite set,

$$\cosh(\delta_j y)-1=\frac{\delta_j^2y^2}{2}+\frac{\delta_j^4y^4}{24}+O_\tau(y^6). \tag{31.10}$$

Summing proves the expansion. If the total is $o(y^2)$, division by y^2 and passage to the limit gives

$$\frac12\sum_{j=1}^m\delta_j^2=0. \tag{31.11}$$

Every term is nonnegative, so every displacement is zero. This holds at every ordinate and yields RH. Under RH, the defect is identically zero, proving the converse. □

Corollary 31.6 (Second-derivative formulation)**.** *The horizontal second moment at ordinate τ is recovered by*

$$\frac{\mathrm{d}^2}{\mathrm{d}y^2} D_{e^y}(\tau)\Big|_{y=0} = \sum_{\substack{\rho=\beta+i\tau \\ \beta>1/2}} \left(\beta - \frac{1}{2}\right)^2. \tag{31.12}$$

Proof. Differentiate the finite sum

$$D_{e^y}(\tau) = \sum_j \left[\cosh(\delta_j y) - 1\right] \tag{31.13}$$

twice. The second derivative of the j-th term is $\delta_j^2 \cosh(\delta_j y)$, which at $y = 0$ equals δ_j^2. Finite summation commutes with differentiation, proving the formula. □

31.4. **Route IV: large-q amplification.**

Definition 31.7. At a positive ordinate τ, define

$$\delta_{\max}(\tau) = \max_{\rho=\beta+i\tau} \left|\beta - \frac{1}{2}\right|, \tag{31.14}$$

with value zero when all zeros at the ordinate are on the critical line.

Theorem 31.8 (Recovery of the maximal horizontal displacement)**.** *For every positive ordinate τ,*

$$\boxed{\lim_{q\to\infty} \frac{\log\left(1 + D_q(\tau)\right)}{\log q} = \delta_{\max}(\tau).} \tag{31.15}$$

Proof. If $\delta_{\max} = 0$, then $D_q(\tau) = 0$ identically and the limit is zero. Otherwise, let $m_{\max} \geq 1$ denote the number of paired zeros with displacement $\delta_{\max}$. Since

$$\cosh(\delta \log q) - 1 = \frac{1}{2} q^{\delta} + \frac{1}{2} q^{-\delta} - 1, \tag{31.16}$$

we have the lower bound

$$D_q(\tau) \geq m_{\max} \left(\frac{1}{2} q^{\delta_{\max}} - 1\right). \tag{31.17}$$

If $M(\tau)$ is the finite number of off-line pairs at the ordinate, every summand is at most $q^{\delta_{\max}}$ for sufficiently large q, so

$$D_q(\tau) \leq M(\tau) q^{\delta_{\max}}. \tag{31.18}$$

Taking logarithms, dividing by $\log q$, and applying the squeeze theorem gives the result. □

Corollary 31.9 (Subpolynomial-growth route)**.** *If one can prove from the prime side that, for every ordinate τ,*

$$D_q(\tau) = q^{o(1)} \qquad (q \to \infty), \tag{31.19}$$

then RH *follows.*

Proof. The growth exponent in Theorem 31.8 must then be zero at every ordinate. □

31.5. **Route V: reconstruction from a family of frequencies.**

Theorem 31.10 (Finite exponential-polynomial reconstruction)**.** *At a fixed ordinate τ, the function*

$$\mathcal{W}_\tau(y) = W_{e^y}(\tau) = \sum_{\rho:\Im\rho=\tau} e^{(\beta-1/2)y} \tag{31.20}$$

is an entire exponential polynomial. Knowledge of $\mathcal{W}_\tau(y)$ on any real set having an accumulation point uniquely determines the multiset of horizontal displacements $\{\beta - 1/2\}$. In particular,

$$\mathcal{W}_\tau(y) \equiv m(\tau) \quad \Longleftrightarrow \quad \text{all zeros at } \tau \text{ lie on } \Re s = \frac{1}{2}. \tag{31.21}$$

Proof. The sum is finite, so $\mathcal{W}_\tau$ is entire. If two finite exponential polynomials agree on a set with an accumulation point, the identity theorem makes them equal everywhere. Their derivatives at zero are the power sums

$$\mathcal{W}_\tau^{(k)}(0) = \sum_{\rho:\Im\rho=\tau} \left(\beta - \frac{1}{2}\right)^k. \tag{31.22}$$

The finite multiset of exponents is determined by these power sums through Newton identities, after combining repeated values with multiplicity. If the function is the constant $m(\tau)$, its second derivative is zero; by symmetry and nonnegativity this forces every squared displacement to be zero. The converse is immediate. □

31.6. **Route VI: projected energy exhaustion plus local atomic control.**

Theorem 31.11 (A sufficient two-part energy route)**.** *Let F_j be regularized residuals and suppose:*

(i) *for every compact interval I,*

$$\int_I \left|F_j(T) - \frac{1}{2}\right|^2 \, \mathrm{d}T \longrightarrow 0; \tag{31.23}$$

(ii) *the distributional derivatives satisfy*

$$\frac{\mathrm{d}}{\mathrm{d}T}\left[e^{i\omega T}\left(F_j(T) - \frac{1}{2}\right)\right] \longrightarrow \nu_q - \mu_q. \tag{31.24}$$

Then RH *holds.*

Proof. Condition (i) is local L^2-convergence of $F_j - 1/2$ to zero. Multiplication by the smooth unit-modulus factor $e^{i\omega T}$ preserves that convergence. On compact sets, local L^2 implies local L^1 by Cauchy–Schwarz. Consequently, the distributional derivative of the product tends to zero when tested against a compactly supported smooth function. Condition (ii) identifies that same derivative limit with $\nu_q - \mu_q$. Hence the measure difference is zero, and the positive atomic criterion gives RH. □

Remark 31.12. The measured quantity $N(H)^{-1}\sum_{\gamma\le H}|F(\gamma)|^2$ is a global normalized energy. Even if it tends to zero, it does not imply (31.23). The route requires either local windows, compactly supported test functions, or an independent argument excluding a finite or sparse atomic defect.

31.7. **Route VII: a one-sided positive test family.**

Theorem 31.13 (Exhausting positive tests)**.** *Let*

$$\mathcal{D}_q = \sum_{\tau>0} D_q(\tau)\delta_\tau. \tag{31.25}$$

Suppose there is a sequence of nonnegative test functions $\varphi_j \in C_c^\infty(0,\infty)$ such that, for every compact set $K \subset (0,\infty)$,

$$\liminf_{j\to\infty} \inf_{t\in K} \varphi_j(t) > 0, \tag{31.26}$$

and

$$\langle \mathcal{D}_q, \varphi_j \rangle \longrightarrow 0. \tag{31.27}$$

Then RH *holds.*

Proof. Fix an ordinate $\tau > 0$ and choose a compact interval $K \subset (0,\infty)$ containing τ in its interior. By (31.26), there are constants $c_K > 0$ and j_K such that

$$\varphi_j(t) \geq c_K \qquad (t \in K,\ j \geq j_K). \tag{31.28}$$

Since $\mathcal{D}_q$ is a positive atomic measure,

$$0 \leq c_K D_q(\tau) \leq D_q(\tau)\varphi_j(\tau) \leq \langle \mathcal{D}_q, \varphi_j \rangle \tag{31.29}$$

for $j \geq j_K$. The right-hand side tends to zero, so $D_q(\tau) = 0$. Since τ was arbitrary, every mass of $\mathcal{D}_q$ vanishes, and Corollary 12.5 gives RH. □

Remark 31.14. The compact-set condition is compatible with compact support: for example, one may choose φ_j equal to one on $[1/j, j]$ and supported in $[1/(2j), 2j]$. The former condition over $(0, T]$ would be impossible for functions in $C_c^\infty(0,\infty)$, because every such function vanishes in a neighbourhood of zero.

Remark 31.15. The exact algebra does not choose among these routes. The analytic work still required is one of the following: local reconstruction of the ordinate-only measure, a sufficiently strong uniform bound in q, or a local projected-energy theorem accompanied by derivative identification. The finite numerical decay of a global variance is evidence for none of these implications by itself.

32. Open problems and falsifiable next computations

Question 32.1 (Raw and reported increment audit)**.** For every prime power n in a growing range, compute both exact ratios

$$\mathcal{I}_H^{\mathrm{raw}}(n) = \frac{\Re M_H(n)}{\pi v_n}, \tag{32.1}$$

$$\mathcal{I}_H^{\mathrm{joint}}(n) = \frac{J_H(n)}{\pi u_n v_n}. \tag{32.2}$$

Compare the raw ratio with the conditional leading prediction

$$\frac{\log n}{\log(H/2\pi)} \tag{32.3}$$

and measure separately the exact projection correction that converts the raw ratio into the reported joint ratio. This tests the local Landau mechanism without fitting a global law and prevents the two carrier statistics from being conflated.

Question 32.2 (Height–cutoff matrix)**.** Repeat the complete computation for a two-dimensional grid of (H, Q), including regimes

$$Q = H^\theta \tag{32.4}$$

with several fixed θ and a near-height regime $Q = H^{1-o(1)}$. Record raw and joint coefficients separately.

Question 32.3 (Numerical audit of the exact variance components)**.** For every cutoff and height, compute separately

$$V_Y(H), \quad -2\Re\sum_p \mathbf{c}_p^* \mathbf{d}_{p,H}, \quad \sum_p \mathbf{c}_p^* \widetilde{K}_{p,p;H} \mathbf{c}_p, \quad \sum_{p\neq r} \mathbf{c}_p^* \widetilde{K}_{p,r;H} \mathbf{c}_r, \tag{32.5}$$

and verify numerically that their sum equals the directly fitted residual variance in (20.11).

Question 32.4 (Local test-function convergence)**.** For compactly supported smooth φ, determine whether the regularized residual satisfies

$$\int \left(F_{Q,H}(T) - \frac{1}{2}\right) \varphi(T)\, \mathrm{d}T \longrightarrow 0 \tag{32.6}$$

under a joint limiting regime, while the prime-side derivative is identified with the complete spectral measure.

Part 4. Numerical support

33. Computational object and exact program identities

33.1. Input data and sampled height.

Experiment 33.1 (Zero data)**.** The full-data run used 2,001,052 positive ordinates from `zeross.bin`, with

$$\gamma_1 = 14.134725142, \qquad \gamma_{2{,}001{,}052} = 1{,}132{,}490.6587144111. \tag{33.1}$$

The phase parameter was

$$\omega = 1, \qquad q = e. \tag{33.2}$$

All ordinates were used; the stride was one.

33.2. Exact definition of the program.

Definition 33.2 (Program variables)**.** At each sampled ordinate, the code forms

$$Y(\gamma_k) = S_q(\gamma_k) - A_{\mathrm{smooth}}(\gamma_k), \tag{33.3}$$

and

$$R_Q(\gamma_k) = Y(\gamma_k) - P_Q^-(\gamma_k) - P_Q^+(\gamma_k), \tag{33.4}$$

where

$$P_Q^-(\gamma) = \sum_{\substack{n=p^r \\ n\le Q}} c_n^- e^{i(\omega-\ell_n)\gamma}, \tag{33.5}$$

$$P_Q^+(\gamma) = \sum_{\substack{n=p^r \\ n\le Q}} c_n^+ e^{i(\omega+\ell_n)\gamma}. \tag{33.6}$$

The raw carrier statistic is

$$\widehat{c}_{Q,H} = \frac{1}{N(H)} \sum_{\gamma_k \le H} e^{-i\omega\gamma_k} R_Q(\gamma_k). \tag{33.7}$$

Proposition 33.3 (Identity evaluated by the raw program)**.** *The program variables satisfy exactly*

$$\widehat{c}_{Q,H} = \widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} \left(c_n^- \overline{M_H(n)} + c_n^+ M_H(n)\right). \tag{33.8}$$

Its imaginary part is

$$\Im\widehat{c}_{Q,H} = \Im\widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} u_n \Re M_H(n). \tag{33.9}$$

Proof. The arrays used by the program are finite, so the abstract identity in Theorem 16.2 applies without any limiting operation. The code generates exactly the prime powers $n \le Q$, uses the coefficients in Definition 13.1, and forms the sample kernel by averaging $n^{i\gamma_k}$. Substitution of these arrays in the theorem gives the program identity entry by entry. □

33.3. **Joint constant–carrier fit used for the reported values.**

Definition 33.4. For every cutoff, the code fits

$$R_Q(\gamma_k) = \alpha_{Q,H} + c_{Q,H} e^{i\omega\gamma_k} + E_{Q,H}(\gamma_k) \tag{33.10}$$

by least squares in the sampled inner product (20.1). The reported variance is

$$V(Q,H) = \frac{1}{N(H)} \sum_{\gamma_k \le H} |E_{Q,H}(\gamma_k)|^2. \tag{33.11}$$

Proposition 33.5 (Closed form of the reported carrier)**.** *Let*

$$\mu_H = M_H(e^{\omega}). \tag{33.12}$$

Then

$$\boxed{c_{Q,H} = \frac{\langle e^{i\omega\gamma}, R_Q\rangle_H - \overline{\mu_H}\langle 1, R_Q\rangle_H}{1 - |\mu_H|^2}}. \tag{33.13}$$

The fitted constant is

$$\alpha_{Q,H} = \frac{\langle 1, R_Q\rangle_H - \mu_H\langle e^{i\omega\gamma}, R_Q\rangle_H}{1 - |\mu_H|^2}. \tag{33.14}$$

Proof. The fit minimizes

$$\sum_{k=1}^{N(H)} |R_k - a - c b_k|^2 \tag{33.15}$$

over $a, c \in \mathbb{C}$. Differentiating with respect to the conjugate parameters, or equivalently imposing orthogonality of the residual to the basis vectors 1 and b, gives the two normal equations. Their Gram matrix is $\left(\begin{smallmatrix} 1 & \mu_H \\ \overline{\mu}_H & 1 \end{smallmatrix}\right)$. Its inverse is explicit, and multiplication by the right-hand side gives the displayed formulas for the constant and carrier coefficients. □

33.4. **Prime-power generation and exact coefficients.**

Experiment 33.6 (Prime powers)**.** The cutoffs and counts were

Q	number of prime powers $n = p^r \le Q$
0	0
3	2
10	7
100	35
1000	193
10000	1280

Every coefficient was generated from (13.2)–(13.3); no frequency or amplitude was fitted individually.

34. Radial geometry and low prime-power resonances

34.1. **Direct radial audit.**

Experiment 34.1 (Original walk and theoretical subtractions)**.** After subtraction of the empirical complex center, the radial statistics were

case	mean radius	radial standard deviation	smoothed radial modes
original	1.9203104476	0.7265477726	1.35731059, 2.49789956
after $n = 3$	1.8079247921	0.3083120307	1.68017926
after $n = 2, 3$	1.7884459085	0.2214048337	1.81300495

The theoretical $n = 3$ coefficients were

$$c_3^- = -1.02370050729i, \qquad c_3^+ = 0.0481029537851i. \tag{34.1}$$

Subtracting $n = 3$ reduced the radial variance by approximately 81.99%; subtracting $n = 2, 3$ reduced it by approximately 90.71%. The finite conclusion is that the observed global double radial mode is dominated by the theoretical $n = 3$ oscillator.

34.2. **Geometric interpretation.**

Experiment 34.2 (Anisotropy)**.** At $Q = 10000$, the centered residual anisotropy was

$$0.981212. \tag{34.2}$$

This is consistent with the exact single-mode limit (14.21): after low frequencies are removed, large-n modes are strongly elongated in one direction.

35. Cutoff sweep and carrier stabilization

35.1. **Global cutoff table.**

Experiment 35.1 (Full-data cutoff sweep)**.** The jointly fitted carrier and residual energy were

Q	terms	$\Re c_{Q,H}$	$\Im c_{Q,H}$	$V(Q,H)$	$\sqrt{V(Q,H)}$
0	0	0.4999989123	0.1381673175	1.2420277147	1.1144630
3	2	0.4999998679	0.0956865241	0.1339324657	0.3659678
10	7	0.4999999862	0.0755224108	0.0596099186	0.2441514
100	35	0.5000000416	0.0545919879	0.0220396104	0.1484574
1000	193	0.5000000755	0.0426449212	0.0112822119	0.1062178
10000	1280	0.4999998710	0.0343125459	0.0082559398	0.0908622

At the largest cutoff the finite variance reduction relative to $Q = 0$ was

$$1 - \frac{0.0082559398}{1.2420277147} = 0.99335\ldots. \tag{35.1}$$

35.2. **Normalized $1/\log Q$ comparisons.**

Experiment 35.2 (Two normalized quantities)**.** For $Q \geq 3$, the finite normalized values were

Q	$\pi \log Q\, \Im c_{Q,H}$	$\frac{4\pi^2}{3} \log Q\, V(Q,H)$
3	0.33025	1.93628
10	0.54631	1.80623
100	0.78981	1.33564
1000	0.92545	1.02558
10000	0.99284	1.00065

At $Q = 10000$,

$$\frac{1}{\pi \log Q} = 0.0345601\ldots, \tag{35.2}$$

$$\frac{3}{4\pi^2 \log Q} = 0.00825061\ldots. \tag{35.3}$$

The corresponding measured values were 0.0343125459 and 0.0082559398. These are finite coincidences to be interpreted through the exact defect identities, not proofs of asymptotic laws.

36. Height dependence

36.1. **Twenty-four block audit.**

Experiment 36.1 (Block ranges)**.** For $Q = 10000$ and 24 consecutive height blocks,

$$\Re \operatorname{mean}_{\text{block}} F_Q \in [0.499994991361, 0.500003656384], \tag{36.1}$$

$$\Im \operatorname{mean}_{\text{block}} F_Q \in [0.033825249952, 0.035105891286], \tag{36.2}$$

$$V_{\text{block}} \in [0.0054744557, 0.0089851833]. \tag{36.3}$$

The standard deviation of the block means was

$$0.0003812589. \tag{36.4}$$

The carrier is stable across height at this scale, while the block variance has a visible drift. Consequently the global factor 3/4 cannot yet be regarded as height-uniform.

36.2. **Termwise increment scale.**

Experiment 36.2 (Present range of the raw Landau scale)**.** For

$$H = 1.13249 \times 10^6, \qquad Q = 10^4, \tag{36.5}$$

one has approximately

$$\frac{\log Q}{\log(H/2\pi)} \approx 0.76. \tag{36.6}$$

This number is the leading prediction for the *raw* increment ratio (16.25), under the additional uniform Landau hypothesis stated in Section 18. The printed coefficient is instead the joint coefficient $c_{Q,H}$, for which the exact prime-power increment is (17.22). Its global finite ratio is

$$\frac{\Im c_{Q,H}}{\pi \Delta_1(Q)} \approx 0.986. \tag{36.7}$$

No comparison of 0.986 with 0.76 is valid until the projection terms

$$M_H(q/n), \qquad M_H(qn), \qquad M_H(q)$$

in (17.9) are evaluated. The direct falsifiable test for the reported program is therefore (17.22), prime power by prime power.

37. Branch decomposition and omitted frequencies

37.1. **Branch energies.**

Experiment 37.1 (Minus and plus branches)**.** At $Q = 10000$,

$$\|P_Q^-\|^2 = 1.2433344613, \tag{37.1}$$

$$\|P_Q^+\|^2 = 0.0309962015, \tag{37.2}$$

$$2\Re\langle P_Q^-, P_Q^+\rangle = -0.0005963448. \tag{37.3}$$

The minus branch dominates, and direct plus–minus energy cancellation is small.

37.2. **Omitted prime-power shell.**

Experiment 37.2 (Exact-frequency omitted fit)**.** For the 56 omitted prime powers immediately above $Q = 10000$, the median amplitude-to-null ratios were

$$3.35349 \quad \text{and} \quad 2.67559 \tag{37.4}$$

for the minus and plus branches. The absolute correlation between the jointly fitted coefficient vector and the theoretical coefficient vector was

$$0.916474. \tag{37.5}$$

The narrow shell explained

$$0.0144446 \tag{37.6}$$

of the remaining residual energy. Thus the next prime-power frequencies are present, but this narrow shell does not account for the full residual.

38. Signed invariant and finite-area comparisons

38.1. **Direct evaluation of $\Delta_1(Q)$.**

Experiment 38.1 (Common invariant)**.** At $Q = 10000$,

$$\Delta_1(10000) \approx 0.01108. \tag{38.1}$$

Hence

$$\pi\Delta_1(10000) \approx 0.03481, \qquad \frac{3}{4}\Delta_1(10000) \approx 0.00831. \tag{38.2}$$

The same exact signed invariant therefore reproduces both observed scales to finite accuracy. The analytic claims that the centroid and projected variance equal these multiples remain Conjectures 30.1 and 30.2.

38.2. **Coefficient prefix sums.**

Experiment 38.2 (Prefix sums)**.** The coefficient combinations

$$A_Q = \sum_{n \le Q} (c_n^- + c_n^+), \qquad B_Q = \sum_{n \le Q} (c_n^- - c_n^+) \tag{38.3}$$

were, at $Q = 10000$,

$$A_Q = -2.5552137617i, \qquad B_Q = -10.6932626383i. \tag{38.4}$$

They do not account for the real carrier value $1/2$.

39. Finite-part computations for Φ_N

39.1. **Direct algebraic verification.**

Experiment 39.1 (Finite-part identity)**.** A numerical comparison between the direct tail expression and the value reconstructed from Proposition 10.3 gave

$$|R_{\text{from FP}} - R_{\text{direct}}| = 4.043 \times 10^{-10}. \tag{39.1}$$

This verifies the implemented identity at the working precision.

39.2. **Scale scan.**

Experiment 39.2 (Normalized finite-part differences)**.** For

$$N = 2000, 4000, 8000, 16000, 32000, 64000, \tag{39.2}$$

the successive magnitudes

$$|\mathcal{F}_{2N}(\rho) - \mathcal{F}_N(\rho)| \tag{39.3}$$

were approximately

$$1.12, \quad 0.75, \quad 0.80, \quad 0.75, \quad 0.44. \tag{39.4}$$

No monotone cancellation law is established by these six scales.

40. Numerical tests dictated by the analytic identities

40.1. **Prime-power increment audit.**

Experiment 40.1 (Required output)**.** For every prime power n in the selected range, compute and store

$$\mathcal{I}_H(n) = \frac{\Re M_H(n)}{\pi v_n}, \tag{40.1}$$

together with

$$\theta_H(n) = \frac{\log n}{\log(H/2\pi)}. \tag{40.2}$$

The comparison is pointwise and requires no global fitting.

40.2. **Two-parameter height–cutoff scan.**

Experiment 40.2 (Required grid)**.** For several heights H_j and exponents θ, set

$$Q_{j,\theta} = H_j^{\theta} \tag{40.3}$$

and record separately:

$$\widehat{c}_{Q,H}, \quad c_{Q,H}, \quad V(Q,H), \quad \Delta_\omega(Q), \quad \sum_p a_p^* \widetilde{K}_{p,p;H} a_p, \quad \sum_{p \neq q} a_p^* \widetilde{K}_{p,q;H} a_q. \tag{40.4}$$

This directly separates the raw carrier, the joint fitted carrier, the projected background energy, the mixed background–prime term, the tower blocks, and the cross-tower interactions.

41. Complete MATLAB implementation

41.1. **Self-contained implementation.**

Remark 41.1. The following program is the complete implementation used for the prime-power audit. It reads only the original binary ordinate file, generates all prime powers internally, retains all figures, performs the joint constant–carrier fit, and exports every table used in the preceding numerical sections.

Listing 1. Complete MATLAB implementation of the prime-power phase audit.

```
%% ZETA_PRIME_POWER_ANALYTIC_AUDIT
%
% Script autocontenido para los seis tests de la descomposicion
%
%   S_q(T) = sum_{gamma_j <= T} exp(i*gamma_j*log(q))
%
% No usa ningun .mat, .csv ni resultado anterior.
```

```
% El unico dato externo necesario es el fichero original de ordenadas:
%
%   zeross.bin
%
% Si no esta en la carpeta actual, MATLAB abre un selector para localizarlo.
%
% No resta 0.5*exp(i*T). Ajusta conjuntamente una constante aditiva y
% el coeficiente de exp(i*log(q)*T). La constante se elimina antes de
% demodular.
% Todos los resumenes principales se muestran con fprintf/disp.
% No se cierra ninguna figura.
%
% TESTS
% -----
% 1. Separacion P_Q^- y P_Q^+ y termino cruzado.
% 2. Identidades de c_n^-, c_n^+ y sumas A_Q, B_Q.
% 3. Busqueda de telescopado dentro de p,p^2,...,p^R.
% 4. Funcionales positivos y matrices de Gram por bloques.
% 5. Ecuacion funcional del residuo demodulado F_Q.
% 6. Unicidad: bloques, anisotropia, gaps y frecuencias omitidas.
%
% Requiere una version de MATLAB que permita funciones locales al final
% de un script (R2016b o posterior).

clear;
clc;
format long g;

%% ========================================================================
% CONFIGURACION
% ========================================================================

zeroBinaryFile = 'zeross.bin';

% q=e reproduce exp(i*gamma).
qPhase = exp(1);
omega = log(qPhase);

% stride=10: usa todos los ceros para construir S_q, pero analiza 1 de cada 10.
% stride=1: analiza los 2,001,052 puntos completos, mucho mas lento para Q=10000.
stride = 1;

Qlist = [0, 3, 10, 100, 1000, 10000];
Qlist = unique(sort(Qlist(:).'));
Qmax = max(Qlist);

numHeightBlocks = 24;
numGramBlocks = 20;
maxGramPoints = 60000;

omittedTestWidth = 500;
maxSpectralPoints = 50000;
numFrequencyGridPoints = 401;
```

```
gapMedianWindow = 1001;
resonanceTolerance = 1e-12;
minimumVariance = 1e-30;

% Si se pone true, guarda tablas y MAT al final. No son necesarios para correr.
saveResults = false;
outputDirectory = 'ZETA_PRIME_POWER_STRUCTURE_RESULTS';

%% =========================================================================
% LOCALIZAR Y LEER zeross.bin
% =========================================================================

if ~isfile(zeroBinaryFile)
    [selectedFile,selectedPath] = uigetfile( ...
        {'*.bin','Binary doubles (*.bin)';'*.*','Todos los archivos'}, ...
        'Selecciona zeross.bin');

    if isequal(selectedFile,0)
        error(['No se encontro zeross.bin y no se selecciono ningun archivo. ' ...
            'Este es el unico fichero de datos necesario.']);
    end

    zeroBinaryFile = fullfile(selectedPath,selectedFile);
end

fileID = fopen(zeroBinaryFile,'rb');
if fileID == -1
    error('No se pudo abrir %s.',zeroBinaryFile);
end

cleanupObject = onCleanup(@() fclose(fileID));
gammaAll = fread(fileID,Inf,'double=>double');
clear cleanupObject;

gammaAll = gammaAll(:);
gammaAll = gammaAll(isfinite(gammaAll) & gammaAll>0);

if numel(gammaAll)<2
    error('El fichero no contiene suficientes ordenadas validas.');
end

if any(diff(gammaAll)<=0)
    error('Las ordenadas no estan estrictamente ordenadas.');
end

Nfull = numel(gammaAll);

fprintf('\n============================================================\n');
fprintf('ZETA PRIME-POWER STRUCTURE: ANALYTIC AUDIT\n');
fprintf('============================================================\n');
fprintf('Archivo de ceros       : %s\n',zeroBinaryFile);
fprintf('Ceros disponibles      : %d\n',Nfull);
fprintf('q                      : %.17g\n',qPhase);
fprintf('omega=log(q)           : %.17g\n',omega);
```

```
fprintf('stride                 : %d\n',stride);
fprintf('Qlist                  :');
fprintf(' %g',Qlist);
fprintf('\n');

%% ========================================================================
% CONSTRUIR S_q CON TODOS LOS CEROS; MUESTREAR DESPUES
% ========================================================================

carrierAll = exp(1i*omega*gammaAll);
SphaseAll = cumsum(carrierAll);

% Parte suave analitica, salvo constante aditiva:
%
% int exp(i*omega*T)/(2*pi)*log(T/(2*pi)) dT
%
% = [exp(i omega T)log(T/2pi)-Ei(i omega T)]/(2*pi*i*omega).
%
% MATLAB: Ei(z) = -expint(-z), salvo la constante de rama, que solo
% modifica una constante aditiva y se absorbe en los ajustes.
Ei_iomegaT_all = -expint(-1i*omega*gammaAll);
AsmoothAll = (carrierAll.*log(gammaAll/(2*pi))-Ei_iomegaT_all) ...
    /(2*pi*1i*omega);

indices = (1:stride:Nfull).';
gamma = gammaAll(indices);
carrier = carrierAll(indices);
Sphase = SphaseAll(indices);
Asmooth = AsmoothAll(indices);

gapBeforeAll = [NaN;diff(gammaAll)];
gapLocalMedianAll = movmedian(gapBeforeAll,gapMedianWindow, ...
    'omitmissing','Endpoints','shrink');
gapResidualAll = gapBeforeAll-gapLocalMedianAll;

gapBefore = gapBeforeAll(indices);
gapResidual = gapResidualAll(indices);

clear carrierAll SphaseAll AsmoothAll Ei_iomegaT_all;
clear gapBeforeAll gapLocalMedianAll gapResidualAll;

Y = Sphase-Asmooth;
N = numel(gamma);

fprintf('Puntos analizados       : %d\n',N);
fprintf('gamma inicial           : %.15g\n',gamma(1));
fprintf('gamma final             : %.15g\n',gamma(end));

%% ========================================================================
% GENERAR TODAS LAS POTENCIAS PRIMAS <= Qmax
% ========================================================================

[qVec,basePrimeVec,powerExponentVec,lambdaVec] = makePrimePowers(Qmax);
numTerms = numel(qVec);
```

```
ellVec = log(qVec);
deltaMinusVec = omega-ellVec;
deltaPlusVec = omega+ellVec;

resonantIndex = find(abs(deltaMinusVec)<=resonanceTolerance,1);
if ~isempty(resonantIndex)
    error(['qPhase es resonante con n=%d. Esta version no oculta la ' ...
        'singularidad: debe introducirse el termino resonante regularizado.'], ...
        qVec(resonantIndex));
end

cMinusVec = 1i*lambdaVec ./ ...
    (2*pi*sqrt(qVec).*deltaMinusVec);

cPlusVec = 1i*lambdaVec ./ ...
    (2*pi*sqrt(qVec).*deltaPlusVec);

%% ========================================================================
% TEST 2: IDENTIDADES ALGEBRAICAS DE LOS COEFICIENTES
% ========================================================================

sumIdentityRHS = 1i*lambdaVec.*omega ./ ...
    (pi*sqrt(qVec).*(omega^2-ellVec.^2));

differenceIdentityRHS = 1i*lambdaVec.*ellVec ./ ...
    (pi*sqrt(qVec).*(omega^2-ellVec.^2));

ratioIdentityRHS = (omega+ellVec)./(omega-ellVec);

sumIdentityError = abs((cMinusVec+cPlusVec)-sumIdentityRHS);
differenceIdentityError = abs((cMinusVec-cPlusVec)-differenceIdentityRHS);
ratioIdentityError = abs((cMinusVec./cPlusVec)-ratioIdentityRHS);

AQ = cumsum(cMinusVec+cPlusVec);
BQ = cumsum(cMinusVec-cPlusVec);

fprintf('\n============================================================\n');
fprintf('TEST 2: IDENTIDADES DE COEFICIENTES\n');
fprintf('============================================================\n');
fprintf('Potencias primas <=Qmax                  : %d\n',numTerms);
fprintf('max |(c^-+c^+)-formula|                : %.6e\n', ...
    max(sumIdentityError));
fprintf('max |(c^--c^+)-formula|                : %.6e\n', ...
    max(differenceIdentityError));
fprintf('max |c^-/c^+-formula|                  : %.6e\n', ...
    max(ratioIdentityError));

coefficientCutoffRows = cell(numel(Qlist),9);

for jj = 1:numel(Qlist)
    currentQ = Qlist(jj);
    lastIndex = find(qVec<=currentQ,1,'last');
```

```
    if isempty(lastIndex)
        currentA = 0;
        currentB = 0;
        numberTerms = 0;
    else
        currentA = AQ(lastIndex);
        currentB = BQ(lastIndex);
        numberTerms = lastIndex;
    end

    coefficientCutoffRows(jj,:) = { ...
        currentQ,numberTerms, ...
        real(currentA),imag(currentA),abs(currentA-0.5), ...
        real(currentB),imag(currentB),abs(currentB-0.5), ...
        abs(currentB+0.5)};
end

coefficientCutoffDisplay = cell2table(coefficientCutoffRows, ...
    'VariableNames',{ ...
    'Q','num_terms','Re_AQ','Im_AQ','abs_AQ_minus_half', ...
    'Re_BQ','Im_BQ','abs_BQ_minus_half','abs_BQ_plus_half'});

disp(coefficientCutoffDisplay);


%% ========================================================================
% AUDITORIA RADIAL DIRECTA: ORIGINAL, n=3, n=2+3
% ========================================================================

fprintf('\n============================================================\n');
fprintf('AUDITORIA RADIAL DIRECTA DE LA BIMODALIDAD\n');
fprintf('============================================================\n');

numRadialBins = 220;
radialSmoothWidth = 9;

index3 = find(qVec==3,1);
index23 = find(qVec<=3);

if isempty(index3)
    error('No se encontro n=3 entre las potencias primas.');
end

ell3 = ellVec(index3);
P3minus = cMinusVec(index3).*carrier.*exp(-1i*ell3*gamma);
P3plus = cPlusVec(index3).*carrier.*exp(1i*ell3*gamma);
P3 = P3minus+P3plus;

P23 = complex(zeros(N,1));
for auditIndex = index23(:).'
    ellAudit = ellVec(auditIndex);
    P23 = P23 ...
        +cMinusVec(auditIndex).*carrier.*exp(-1i*ellAudit*gamma) ...
        +cPlusVec(auditIndex).*carrier.*exp(1i*ellAudit*gamma);
```

```
end

[radialOriginal,radialCentersOriginal,radialDensityOriginal, ...
    radialPeaksOriginal] = radialDistributionSummary( ...
    Sphase,numRadialBins,radialSmoothWidth);

[radialAfter3,radialCentersAfter3,radialDensityAfter3, ...
    radialPeaksAfter3] = radialDistributionSummary( ...
    Sphase-P3,numRadialBins,radialSmoothWidth);

[radialAfter23,radialCentersAfter23,radialDensityAfter23, ...
    radialPeaksAfter23] = radialDistributionSummary( ...
    Sphase-P23,numRadialBins,radialSmoothWidth);

radialAuditTable = table( ...
    ["original";"after_n3";"after_n2_n3"], ...
    [mean(radialOriginal);mean(radialAfter3);mean(radialAfter23)], ...
    [std(radialOriginal);std(radialAfter3);std(radialAfter23)], ...
    [numel(radialPeaksOriginal);numel(radialPeaksAfter3); ...
        numel(radialPeaksAfter23)], ...
    [joinNumericVector(radialPeaksOriginal); ...
        joinNumericVector(radialPeaksAfter3); ...
        joinNumericVector(radialPeaksAfter23)], ...
    'VariableNames',{'case_name','mean_radius','std_radius', ...
    'num_detected_peaks','detected_peak_radii'});

disp(radialAuditTable);

fprintf('Coeficiente teorico total n=3:\n');
fprintf('  c3^- = %.12g %+.12gi\n',real(cMinusVec(index3)), ...
    imag(cMinusVec(index3)));
fprintf('  c3^+ = %.12g %+.12gi\n',real(cPlusVec(index3)), ...
    imag(cPlusVec(index3)));

figure('Name','Auditoria radial: original y sustracciones pequenas', ...
    'Color','w');
plot(radialCentersOriginal,radialDensityOriginal,'LineWidth',1.3);
hold on;
plot(radialCentersAfter3,radialDensityAfter3,'LineWidth',1.3);
plot(radialCentersAfter23,radialDensityAfter23,'LineWidth',1.3);
grid on;
xlabel('Radio respecto del centro medio');
ylabel('Densidad suavizada');
legend('Original','Despues de n=3','Despues de n=2,3', ...
    'Location','best');
title('Prueba directa de la bimodalidad radial');

%% ========================================================================
% PREPARAR MATRIZ DE GRAM POR BLOQUES LOGARITMICOS DE n
% ========================================================================

gramIndices = unique(round(linspace(1,N,min(N,maxGramPoints)))).';
numGramPoints = numel(gramIndices);
```

```
logEdges = linspace(log(2),log(Qmax)+100*eps(log(Qmax)), ...
    numGramBlocks+1);

gramBlockOfTerm = discretize(ellVec,logEdges);
gramBlockOfTerm(isnan(gramBlockOfTerm)) = numGramBlocks;

gramMinus = complex(zeros(numGramPoints,numGramBlocks,'single'));
gramPlus = complex(zeros(numGramPoints,numGramBlocks,'single'));

%% ========================================================================
% BUCLE PRINCIPAL: TESTS 1, 4, 5 Y PARTE DE 6
% ========================================================================

Pminus = complex(zeros(N,1));
Pplus = complex(zeros(N,1));

nextTerm = 1;
maxIncrementIdentityError = 0;
maxConstantCancellationIdentityError = 0;

summaryRows = cell(numel(Qlist),43);
positivityRows = cell(numel(Qlist),17);

blockRows = cell(numel(Qlist)*numHeightBlocks,16);
blockRow = 0;

[constantY,cY] = fitConstantAndCarrier(Y,carrier);

RbothAtQmax = [];
FAtQmax = [];
EAtQmax = [];
PminusAtQmax = [];
PplusAtQmax = [];
cBothAtQmax = NaN;

for qi = 1:numel(Qlist)

    Q = Qlist(qi);

    while nextTerm<=numTerms && qVec(nextTerm)<=Q

        ell = ellVec(nextTerm);
        z = exp(1i*ell*gamma);

        termMinus = cMinusVec(nextTerm).*carrier.*conj(z);
        termPlus = cPlusVec(nextTerm).*carrier.*z;

        oldCombinedResidual = Y-Pminus-Pplus;

        Pminus = Pminus+termMinus;
        Pplus = Pplus+termPlus;

        newCombinedResidual = Y-Pminus-Pplus;
```

```
    incrementError = max(abs( ...
        (newCombinedResidual-oldCombinedResidual) ...
        +termMinus+termPlus));

    maxIncrementIdentityError = max( ...
        maxIncrementIdentityError,incrementError);

    gramBlock = gramBlockOfTerm(nextTerm);

    gramMinus(:,gramBlock) = gramMinus(:,gramBlock) ...
        +single(termMinus(gramIndices));

    gramPlus(:,gramBlock) = gramPlus(:,gramBlock) ...
        +single(termPlus(gramIndices));

    nextTerm = nextTerm+1;
end

numTermsUsed = nextTerm-1;

% Cuatro residuos pedidos.
RminusOnlyRaw = Y-Pminus;
RplusOnlyRaw = Y-Pplus;
RbothRaw = Y-Pminus-Pplus;

% CORRECCION: la parte suave esta definida salvo constante aditiva.
% Se ajustan [constante + portadora] conjuntamente en cada residuo.
[constantMinus,cMinusResidual] = ...
    fitConstantAndCarrier(RminusOnlyRaw,carrier);

[constantPlus,cPlusResidual] = ...
    fitConstantAndCarrier(RplusOnlyRaw,carrier);

[jointConstant,cBothResidual] = ...
    fitConstantAndCarrier(RbothRaw,carrier);

RminusOnly = RminusOnlyRaw-constantMinus;
RplusOnly = RplusOnlyRaw-constantPlus;
Rboth = RbothRaw-jointConstant;

E = Rboth-cBothResidual*carrier;
F = conj(carrier).*Rboth;
Fc = F-cBothResidual;

[constantPminus,projectionMinus] = ...
    fitConstantAndCarrier(Pminus,carrier);

[constantPplus,projectionPlus] = ...
    fitConstantAndCarrier(Pplus,carrier);

carrierCancellationIdentityError = abs( ...
    cBothResidual-(cY-projectionMinus-projectionPlus));

constantCancellationIdentityError = abs( ...
```

```
    jointConstant-(constantY-constantPminus-constantPplus));

maxConstantCancellationIdentityError = max( ...
    maxConstantCancellationIdentityError, ...
    constantCancellationIdentityError);

% TEST 1: energias y termino cruzado.
energyMinus = mean(abs(Pminus).^2);
energyPlus = mean(abs(Pplus).^2);
crossMinusPlus = 2*real(mean(conj(Pminus).*Pplus));
energyCombined = mean(abs(Pminus+Pplus).^2);

crossEnergyIdentityError = abs( ...
    energyCombined-(energyMinus+energyPlus+crossMinusPlus));

% Forma centrada.
PminusCentered = Pminus-mean(Pminus);
PplusCentered = Pplus-mean(Pplus);

centeredEnergyMinus = mean(abs(PminusCentered).^2);
centeredEnergyPlus = mean(abs(PplusCentered).^2);

centeredCross = 2*real(mean( ...
    conj(PminusCentered).*PplusCentered));

centeredEnergyCombined = mean(abs( ...
    PminusCentered+PplusCentered).^2);

% TEST 5: ecuacion funcional discreta y diferencial.
dgamma = diff(gamma);

derivativeDefect = diff(Rboth)./dgamma ...
    -1i*omega*0.5*(Rboth(2:end)+Rboth(1:end-1));

discreteDefect = Rboth(2:end) ...
    -exp(1i*omega*dgamma).*Rboth(1:end-1);

Fvariance = mean(abs(Fc).^2);
FmaxDeviation = max(abs(Fc));
FmeanAbsIncrement = mean(abs(diff(F)));
FtotalVariation = sum(abs(diff(F)));
Fanisotropy = complexAnisotropy(Fc,minimumVariance);

corrGapReal = safeCorrelation(real(Fc),gapBefore);
corrGapImag = safeCorrelation(imag(Fc),gapBefore);
corrGapAbs = safeCorrelation(abs(Fc),gapBefore);

corrGapResidualReal = safeCorrelation(real(Fc),gapResidual);
corrGapResidualImag = safeCorrelation(imag(Fc),gapResidual);
corrGapResidualAbs = safeCorrelation(abs(Fc),gapResidual);

% Bloques de altura.
heightEdges = round(linspace(1,N+1,numHeightBlocks+1));
blockMeans = complex(NaN(numHeightBlocks,1));
```

```
for hb = 1:numHeightBlocks

    idx = heightEdges(hb):(heightEdges(hb+1)-1);

    Fb = F(idx);
    Rb = Rboth(idx);
    blockMean = mean(Fb);
    blockMeans(hb) = blockMean;
    Fbc = Fb-blockMean;

    if numel(idx)>=2
        dgb = diff(gamma(idx));

        discreteBlock = Rb(2:end) ...
            -exp(1i*omega*dgb).*Rb(1:end-1);

        derivativeBlock = diff(Rb)./dgb ...
            -1i*omega*0.5*(Rb(2:end)+Rb(1:end-1));

        meanAbsDeltaBlock = mean(abs(diff(Fb)));

        variationPerHeight = sum(abs(diff(Fb))) ...
            /max(gamma(idx(end))-gamma(idx(1)),eps);

        discreteBlockRMS = sqrt(mean(abs(discreteBlock).^2));
        derivativeBlockRMS = sqrt(mean(abs(derivativeBlock).^2));
    else
        meanAbsDeltaBlock = NaN;
        variationPerHeight = NaN;
        discreteBlockRMS = NaN;
        derivativeBlockRMS = NaN;
    end

    blockRow = blockRow+1;

    blockRows(blockRow,:) = { ...
        Q,hb,indices(idx(1)),indices(idx(end)), ...
        gamma(idx(1)),gamma(idx(end)), ...
        real(blockMean),imag(blockMean),mean(abs(Fbc).^2), ...
        max(abs(Fb-cBothResidual)),max(abs(Fbc)), ...
        meanAbsDeltaBlock,variationPerHeight, ...
        complexAnisotropy(Fbc,minimumVariance), ...
        discreteBlockRMS,derivativeBlockRMS};
end

FblockMeanStd = sqrt(mean(abs(blockMeans-mean(blockMeans)).^2));

% TEST 4: positividad elemental/Gram.
normE2 = mean(abs(E).^2);
normCarrier2 = mean(abs(carrier).^2);

innerCarrierE = mean(conj(carrier).*E);
innerEMinus = mean(conj(E).*Pminus);
```

```
innerEPlus = mean(conj(E).*Pplus);
innerMinusPlus = mean(conj(Pminus).*Pplus);

carrierSchur = mean(abs(Rboth).^2) ...
    -abs(mean(conj(carrier).*Rboth))^2/normCarrier2;

cauchyDetEMinus = normE2*energyMinus-abs(innerEMinus)^2;
cauchyDetEPlus = normE2*energyPlus-abs(innerEPlus)^2;
cauchyDetMinusPlus = energyMinus*energyPlus ...
    -abs(innerMinusPlus)^2;

QEplusMinus = mean(abs(E+Pminus).^2);
QEminusMinus = mean(abs(E-Pminus).^2);

polarizationErrorMinus = abs( ...
    0.5*(QEplusMinus+QEminusMinus)-normE2-energyMinus);

QEplusPlus = mean(abs(E+Pplus).^2);
QEminusPlus = mean(abs(E-Pplus).^2);

polarizationErrorPlus = abs( ...
    0.5*(QEplusPlus+QEminusPlus)-normE2-energyPlus);

positivityRows(qi,:) = { ...
    Q,normE2,carrierSchur, ...
    abs(innerCarrierE),real(innerCarrierE),imag(innerCarrierE), ...
    real(innerEMinus),real(innerEPlus), ...
    cauchyDetEMinus,cauchyDetEPlus,cauchyDetMinusPlus, ...
    QEplusMinus,QEminusMinus,polarizationErrorMinus, ...
    QEplusPlus,QEminusPlus,polarizationErrorPlus};

if numTermsUsed>0
    currentA = AQ(numTermsUsed);
    currentB = BQ(numTermsUsed);
else
    currentA = 0;
    currentB = 0;
end

summaryRows(qi,:) = { ...
    Q,numTermsUsed, ...
    real(cMinusResidual),imag(cMinusResidual), ...
    real(cPlusResidual),imag(cPlusResidual), ...
    real(cBothResidual),imag(cBothResidual),abs(cBothResidual), ...
    abs(cBothResidual-0.5), ...
    real(jointConstant),imag(jointConstant), ...
    real(cBothResidual),imag(cBothResidual),abs(cBothResidual-0.5), ...
    real(projectionMinus),imag(projectionMinus), ...
    real(projectionPlus),imag(projectionPlus), ...
    carrierCancellationIdentityError, ...
    sqrt(mean(abs(RminusOnly).^2)), ...
    sqrt(mean(abs(RplusOnly).^2)), ...
    sqrt(mean(abs(Rboth).^2)),sqrt(normE2), ...
    energyMinus,energyPlus,crossMinusPlus,energyCombined, ...
```

```
        crossEnergyIdentityError, ...
        centeredEnergyMinus,centeredEnergyPlus,centeredCross, ...
        centeredEnergyCombined, ...
        Fvariance,FmaxDeviation,FmeanAbsIncrement,FtotalVariation, ...
        FblockMeanStd,Fanisotropy, ...
        sqrt(mean(abs(discreteDefect).^2)), ...
        sqrt(mean(abs(derivativeDefect).^2)), ...
        real(currentA),imag(currentA)};

    fprintf(['Q=%-6g terms=%-5d  c^-res=%.7f%+.7fi  ' ...
        'c^+res=%.7f%+.7fi  c^both=%.7f%+.7fi  ' ...
        'Var(F)=%.4e  RMS(E)=%.4e\n'], ...
        Q,numTermsUsed, ...
        real(cMinusResidual),imag(cMinusResidual), ...
        real(cPlusResidual),imag(cPlusResidual), ...
        real(cBothResidual),imag(cBothResidual), ...
        Fvariance,sqrt(normE2));

    if Q==Qmax
        RbothAtQmax = Rboth;
        FAtQmax = F;
        EAtQmax = E;
        PminusAtQmax = Pminus;
        PplusAtQmax = Pplus;
        cBothAtQmax = cBothResidual;
    end
end

summaryVariableNames = { ...
    'Q','num_prime_power_terms', ...
    'c_after_minus_real','c_after_minus_imag', ...
    'c_after_plus_real','c_after_plus_imag', ...
    'c_after_both_real','c_after_both_imag','c_after_both_abs', ...
    'distance_c_after_both_to_half', ...
    'joint_constant_real','joint_constant_imag', ...
    'joint_carrier_real','joint_carrier_imag', ...
    'distance_joint_carrier_to_half', ...
    'projection_Pminus_real','projection_Pminus_imag', ...
    'projection_Pplus_real','projection_Pplus_imag', ...
    'carrier_cancellation_identity_error', ...
    'rms_centered_Y_minus_Pminus','rms_centered_Y_minus_Pplus', ...
    'rms_centered_Y_minus_both', ...
    'rms_after_both_fitted_carrier', ...
    'energy_Pminus','energy_Pplus', ...
    'cross_term_2Re_inner_Pminus_Pplus', ...
    'energy_Pminus_plus_Pplus','cross_energy_identity_error', ...
    'centered_energy_Pminus','centered_energy_Pplus', ...
    'centered_cross_term','centered_energy_combined', ...
    'F_variance','F_max_abs_deviation_from_mean', ...
    'F_mean_abs_increment','F_total_variation', ...
    'F_block_mean_std','F_anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms', ...
    'A_Q_real','A_Q_imag'};
```

```
summaryTable = cell2table(summaryRows, ...
    'VariableNames',summaryVariableNames);

positivityVariableNames = { ...
    'Q','norm_E2','carrier_schur_complement', ...
    'carrier_projection_abs','carrier_projection_real', ...
    'carrier_projection_imag','Re_inner_E_Pminus', ...
    'Re_inner_E_Pplus','cauchy_det_E_Pminus', ...
    'cauchy_det_E_Pplus','cauchy_det_Pminus_Pplus', ...
    'Q_E_plus_Pminus','Q_E_minus_Pminus', ...
    'polarization_error_minus','Q_E_plus_Pplus', ...
    'Q_E_minus_Pplus','polarization_error_plus'};

positivityTable = cell2table(positivityRows, ...
    'VariableNames',positivityVariableNames);

blockVariableNames = { ...
    'Q','block','k_start','k_end','gamma_start','gamma_end', ...
    'F_mean_real','F_mean_imag','F_variance', ...
    'sup_abs_F_minus_global_c','sup_abs_F_minus_block_c', ...
    'mean_abs_delta_F','variation_per_height','anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms'};

blockTable = cell2table(blockRows(1:blockRow,:), ...
    'VariableNames',blockVariableNames);

%% ========================================================================
% DISPLAY PRINCIPAL: TEST 1 Y TEST 5
% ========================================================================

fprintf('\n============================================================\n');
fprintf('TEST 1: RAMAS; TEST 5: ECUACION FUNCIONAL\n');
fprintf('============================================================\n');

mainDisplayVariables = { ...
    'Q','num_prime_power_terms', ...
    'c_after_minus_real','c_after_minus_imag', ...
    'c_after_plus_real','c_after_plus_imag', ...
    'c_after_both_real','c_after_both_imag', ...
    'distance_c_after_both_to_half', ...
    'rms_centered_Y_minus_Pminus','rms_centered_Y_minus_Pplus', ...
    'rms_centered_Y_minus_both', ...
    'rms_after_both_fitted_carrier', ...
    'energy_Pminus','energy_Pplus', ...
    'cross_term_2Re_inner_Pminus_Pplus', ...
    'F_variance','F_max_abs_deviation_from_mean', ...
    'F_block_mean_std','F_anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms'};

disp(summaryTable(:,mainDisplayVariables));

fprintf('max error R_j-R_{j-1}=-P_j                  : %.6e\n', ...
    maxIncrementIdentityError);
```

```
fprintf('max error ley de energia de ramas           : %.6e\n', ...
    max(summaryTable.cross_energy_identity_error));

fprintf('max error identidad de proyeccion portadora : %.6e\n', ...
    max(summaryTable.carrier_cancellation_identity_error));
fprintf('max error identidad de constantes            : %.6e\n', ...
    maxConstantCancellationIdentityError);

%% ========================================================================
% TEST 3: TELESCOPADO DENTRO DE CADA PRIMO p
% ========================================================================

[groupSummaryTable,groupPartialTable] = primePowerTelescopingTests( ...
    qVec,basePrimeVec,powerExponentVec,cMinusVec,cPlusVec, ...
    gamma,carrier);

fprintf('\n============================================================\n');
fprintf('TEST 3: TELESCOPADO EN p,p^2,...,p^R\n');
fprintf('============================================================\n');

if isempty(groupSummaryTable)
    fprintf('No hay primos con al menos dos potencias <=Qmax.\n');
else
    [~,orderTelescoping] = sort( ...
        groupSummaryTable.boundary_fit_relative_rms_combined,'ascend');

    numDisplay = min(20,height(groupSummaryTable));

    telescopingDisplayVariables = { ...
        'base_prime','num_powers','largest_power', ...
        'combined_energy_ratio_sum_over_individual', ...
        'combined_rms_over_sum_term_rms', ...
        'boundary_fit_relative_rms_combined', ...
        'boundary_fit_relative_rms_minus', ...
        'boundary_fit_relative_rms_plus', ...
        'boundary_alpha_first_real','boundary_alpha_first_imag', ...
        'boundary_alpha_last_real','boundary_alpha_last_imag', ...
        'tail_primitive_start_abs_minus', ...
        'tail_primitive_start_abs_plus'};

    disp(groupSummaryTable( ...
        orderTelescoping(1:numDisplay), ...
        telescopingDisplayVariables));

    fprintf(['boundary_fit_relative_rms_combined cercano a cero seria ' ...
        'evidencia de colapso a los dos bordes del bloque p^r.\n']);
end

%% ========================================================================
% TEST 4: FUNCIONALES POSITIVOS Y MATRICES DE GRAM
% ========================================================================

Gminus = double(gramMinus);
Gplus = double(gramPlus);
```

```
GcombinedCentered = Gminus+Gplus;
GcombinedCentered = GcombinedCentered-mean(GcombinedCentered,1);

GbranchCentered = [ ...
    Gminus-mean(Gminus,1), ...
    Gplus-mean(Gplus,1)];

gramCombined = (GcombinedCentered'*GcombinedCentered)/numGramPoints;
gramBranch = (GbranchCentered'*GbranchCentered)/numGramPoints;

gramCombined = 0.5*(gramCombined+gramCombined');
gramBranch = 0.5*(gramBranch+gramBranch');

eigCombined = sort(real(eig(gramCombined)));
eigBranch = sort(real(eig(gramBranch)));

fprintf('\n============================================================\n');
fprintf('TEST 4: POSITIVIDAD Y GRAM\n');
fprintf('============================================================\n');

positivityDisplayVariables = { ...
    'Q','norm_E2','carrier_schur_complement', ...
    'carrier_projection_abs','Re_inner_E_Pminus', ...
    'Re_inner_E_Pplus','cauchy_det_E_Pminus', ...
    'cauchy_det_E_Pplus','cauchy_det_Pminus_Pplus', ...
    'polarization_error_minus','polarization_error_plus'};

disp(positivityTable(:,positivityDisplayVariables));

fprintf('Gram combinada: min=%.6e, max=%.6e, negativos<-1e-10=%d\n', ...
    min(eigCombined),max(eigCombined),nnz(eigCombined<-1e-10));

fprintf('Gram por ramas: min=%.6e, max=%.6e, negativos<-1e-10=%d\n', ...
    min(eigBranch),max(eigBranch),nnz(eigBranch<-1e-10));

%% ========================================================================
% TEST 5 POR BLOQUES PARA Qmax
% ========================================================================

blockQmax = blockTable(blockTable.Q==Qmax,:);

fprintf('\n============================================================\n');
fprintf('TEST 5: F_Q POR BLOQUES DE ALTURA, Q=%g\n',Qmax);
fprintf('============================================================\n');

blockDisplayVariables = { ...
    'block','gamma_start','gamma_end', ...
    'F_mean_real','F_mean_imag','F_variance', ...
    'sup_abs_F_minus_global_c','sup_abs_F_minus_block_c', ...
    'mean_abs_delta_F','variation_per_height','anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms'};

disp(blockQmax(:,blockDisplayVariables));
```

```
fprintf('Rango Re media por bloque : [%.12g, %.12g]\n', ...
    min(blockQmax.F_mean_real),max(blockQmax.F_mean_real));

fprintf('Rango Im media por bloque : [%.12g, %.12g]\n', ...
    min(blockQmax.F_mean_imag),max(blockQmax.F_mean_imag));

fprintf('Max sup global por bloque : %.6e\n', ...
    max(blockQmax.sup_abs_F_minus_global_c));

fprintf('Max sup local por bloque  : %.6e\n', ...
    max(blockQmax.sup_abs_F_minus_block_c));

%% ========================================================================
% TEST 6: FRECUENCIAS OMITIDAS
% ========================================================================

spectralIndices = unique(round(linspace(1,N, ...
    min(N,maxSpectralPoints)))).';

Ts = gamma(spectralIndices);
Fs = FAtQmax(spectralIndices)-cBothAtQmax;
Es = EAtQmax(spectralIndices);

testStart = Qmax+1;
testEnd = Qmax+omittedTestWidth;
nCandidates = (testStart:testEnd).';

[qTestAll,pTestAll,rTestAll,lambdaTestAll] = makePrimePowers(testEnd);

[isPrimePowerCandidate,locPP] = ismember(nCandidates,qTestAll);

candidateLambda = zeros(size(nCandidates));
candidateBasePrime = zeros(size(nCandidates));
candidateExponent = zeros(size(nCandidates));

candidateLambda(isPrimePowerCandidate) = ...
    lambdaTestAll(locPP(isPrimePowerCandidate));

candidateBasePrime(isPrimePowerCandidate) = ...
    pTestAll(locPP(isPrimePowerCandidate));

candidateExponent(isPrimePowerCandidate) = ...
    rTestAll(locPP(isPrimePowerCandidate));

omittedRows = cell(numel(nCandidates),19);

for jj = 1:numel(nCandidates)

    n = nCandidates(jj);
    ell = log(n);

    basisMinusF = exp(-1i*ell*Ts);
    basisPlusF = exp(1i*ell*Ts);
```

```
    fitMinusF = fitCarrier(Fs,basisMinusF);
    fitPlusF = fitCarrier(Fs,basisPlusF);

    basisMinusR = exp(1i*(omega-ell)*Ts);
    basisPlusR = exp(1i*(omega+ell)*Ts);

    fitMinusR = fitCarrier(Es,basisMinusR);
    fitPlusR = fitCarrier(Es,basisPlusR);

    theoryMinus = 0;
    theoryPlus = 0;

    if isPrimePowerCandidate(jj)
        lam = candidateLambda(jj);

        theoryMinus = 1i*lam / ...
            (2*pi*sqrt(n)*(omega-ell));

        theoryPlus = 1i*lam / ...
            (2*pi*sqrt(n)*(omega+ell));
    end

    omittedRows(jj,:) = { ...
        n,isPrimePowerCandidate(jj),candidateBasePrime(jj), ...
        candidateExponent(jj),candidateLambda(jj), ...
        real(fitMinusF),imag(fitMinusF),abs(fitMinusF), ...
        real(fitPlusF),imag(fitPlusF),abs(fitPlusF), ...
        real(fitMinusR),imag(fitMinusR), ...
        real(fitPlusR),imag(fitPlusR), ...
        real(theoryMinus),imag(theoryMinus), ...
        real(theoryPlus),imag(theoryPlus)};
end

omittedVariableNames = { ...
    'n','is_prime_power','base_prime','power_exponent','Lambda_n', ...
    'fit_F_minus_real','fit_F_minus_imag','fit_F_minus_abs', ...
    'fit_F_plus_real','fit_F_plus_imag','fit_F_plus_abs', ...
    'fit_R_minus_real','fit_R_minus_imag', ...
    'fit_R_plus_real','fit_R_plus_imag', ...
    'theory_minus_real','theory_minus_imag', ...
    'theory_plus_real','theory_plus_imag'};

omittedTable = cell2table(omittedRows, ...
    'VariableNames',omittedVariableNames);

ppMask = omittedTable.is_prime_power;
nullMask = ~ppMask;

medianPPMinus = median(omittedTable.fit_F_minus_abs(ppMask));
medianNullMinus = median(omittedTable.fit_F_minus_abs(nullMask));

medianPPPlus = median(omittedTable.fit_F_plus_abs(ppMask));
medianNullPlus = median(omittedTable.fit_F_plus_abs(nullMask));
```

```
omittedPPMask = qTestAll>Qmax & qTestAll<=testEnd;
omittedPP = qTestAll(omittedPPMask);
omittedPPLambda = lambdaTestAll(omittedPPMask);

Xomitted = complex(zeros(numel(Ts),2*numel(omittedPP)));

for jj = 1:numel(omittedPP)
    ell = log(omittedPP(jj));
    Xomitted(:,jj) = exp(-1i*ell*Ts);
    Xomitted(:,numel(omittedPP)+jj) = exp(1i*ell*Ts);
end

if isempty(Xomitted)
    fittedOmittedCoefficients = [];
    FsAfterOmittedFit = Fs;
    omittedExplainedFraction = NaN;
    fittedTheoryCorrelation = NaN;
    relativeTheoryCoefficientError = NaN;
else
    fittedOmittedCoefficients = Xomitted\Fs;
    FsAfterOmittedFit = Fs-Xomitted*fittedOmittedCoefficients;

    omittedExplainedFraction = 1 ...
        -sum(abs(FsAfterOmittedFit).^2)/sum(abs(Fs).^2);

    theoryOmittedCoefficients = complex(zeros(2*numel(omittedPP),1));

    for jj = 1:numel(omittedPP)

        n = omittedPP(jj);
        ell = log(n);
        lam = omittedPPLambda(jj);

        theoryOmittedCoefficients(jj) = 1i*lam / ...
            (2*pi*sqrt(n)*(omega-ell));

        theoryOmittedCoefficients(numel(omittedPP)+jj) = 1i*lam / ...
            (2*pi*sqrt(n)*(omega+ell));
    end

    fittedTheoryCorrelation = complexVectorCorrelation( ...
        fittedOmittedCoefficients,theoryOmittedCoefficients);

    relativeTheoryCoefficientError = norm( ...
        fittedOmittedCoefficients-theoryOmittedCoefficients) ...
        /max(norm(theoryOmittedCoefficients),eps);
end

nuMax = log(testEnd);
nuGrid = linspace(-nuMax,nuMax,numFrequencyGridPoints).';
frequencyAmplitude = zeros(size(nuGrid));

for jj = 1:numel(nuGrid)
```

```
    basis = exp(1i*nuGrid(jj)*Ts);
    frequencyAmplitude(jj) = abs(fitCarrier(Fs,basis));
end

[sortedAmplitude,sortOrder] = sort(frequencyAmplitude,'descend');
topCount = min(30,numel(sortOrder));

topFrequencyTable = table( ...
    (1:topCount).', ...
    nuGrid(sortOrder(1:topCount)), ...
    sortedAmplitude(1:topCount), ...
    'VariableNames',{'rank','demodulated_frequency','fit_amplitude'});

uniquenessTable = table( ...
    Qmax,numel(omittedPP), ...
    medianPPMinus,medianNullMinus,medianPPMinus/medianNullMinus, ...
    medianPPPlus,medianNullPlus,medianPPPlus/medianNullPlus, ...
    omittedExplainedFraction, ...
    real(fittedTheoryCorrelation),imag(fittedTheoryCorrelation), ...
    abs(fittedTheoryCorrelation),relativeTheoryCoefficientError, ...
    mean(abs(Fs).^2),mean(abs(FsAfterOmittedFit).^2), ...
    complexAnisotropy(Fs,minimumVariance), ...
    safeCorrelation(real(Fs),gapBefore(spectralIndices)), ...
    safeCorrelation(imag(Fs),gapBefore(spectralIndices)), ...
    safeCorrelation(abs(Fs),gapBefore(spectralIndices)), ...
    'VariableNames',{ ...
    'Qmax','num_omitted_prime_powers_joint_fit', ...
    'median_minus_amplitude_prime_power', ...
    'median_minus_amplitude_null', ...
    'minus_prime_power_to_null_ratio', ...
    'median_plus_amplitude_prime_power', ...
    'median_plus_amplitude_null', ...
    'plus_prime_power_to_null_ratio', ...
    'joint_omitted_explained_fraction', ...
    'fit_theory_correlation_real','fit_theory_correlation_imag', ...
    'fit_theory_correlation_abs','relative_theory_coefficient_error', ...
    'F_energy_before_omitted_fit','F_energy_after_omitted_fit', ...
    'F_anisotropy','corr_gap_realF','corr_gap_imagF','corr_gap_absF'});

fprintf('\n============================================================\n');
fprintf('TEST 6: UNICIDAD Y FRECUENCIAS OMITIDAS\n');
fprintf('============================================================\n');

disp(uniquenessTable);

omittedPrimePowerOnly = omittedTable(omittedTable.is_prime_power,:);

if ~isempty(omittedPrimePowerOnly)

    omittedPrimePowerOnly.max_branch_fit_abs = max( ...
        omittedPrimePowerOnly.fit_F_minus_abs, ...
        omittedPrimePowerOnly.fit_F_plus_abs);

    omittedPrimePowerOnly = sortrows( ...
```

```
        omittedPrimePowerOnly,'max_branch_fit_abs','descend');

    numDisplay = min(20,height(omittedPrimePowerOnly));

    fprintf('\nPotencias primas omitidas con mayor amplitud:\n');

    disp(omittedPrimePowerOnly(1:numDisplay,{ ...
        'n','base_prime','power_exponent','Lambda_n', ...
        'fit_F_minus_real','fit_F_minus_imag','fit_F_minus_abs', ...
        'fit_F_plus_real','fit_F_plus_imag','fit_F_plus_abs', ...
        'theory_minus_real','theory_minus_imag', ...
        'theory_plus_real','theory_plus_imag', ...
        'max_branch_fit_abs'}));
end

fprintf('\nFrecuencias no nulas dominantes de F_Q-mean(F_Q):\n');
disp(topFrequencyTable);

fprintf('\nDIAGNOSTICO FINAL PARA Q=%g\n',Qmax);
fprintf('mean(F_Q)                          = %.12g %+.12gi\n', ...
    real(cBothAtQmax),imag(cBothAtQmax));
fprintf('|mean(F_Q)-1/2|                    = %.6e\n', ...
    abs(cBothAtQmax-0.5));
fprintf('Var(F_Q)                           = %.6e\n', ...
    mean(abs(FAtQmax-cBothAtQmax).^2));
fprintf('sup |F_Q-mean(F_Q)|                = %.6e\n', ...
    max(abs(FAtQmax-cBothAtQmax)));
fprintf('RMS E_Q                            = %.6e\n', ...
    sqrt(mean(abs(EAtQmax).^2)));
fprintf('razon omitidas potencia/null (-) = %.6f\n', ...
    medianPPMinus/medianNullMinus);
fprintf('razon omitidas potencia/null (+) = %.6f\n', ...
    medianPPPlus/medianNullPlus);
fprintf('fraccion explicada ajuste omitido= %.6f\n', ...
    omittedExplainedFraction);

%% ========================================================================
% FIGURAS: NO SE CIERRA NINGUNA
% ========================================================================

figure('Name','Coeficientes de portadora por rama','Color','w');
tiledlayout(2,1,'TileSpacing','compact','Padding','compact');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_minus_real,'-o','LineWidth',1.2);
hold on;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_plus_real,'-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_both_real,'-o','LineWidth',1.2);
yline(0.5,':');
grid on;
xlabel('Q');
```

```
ylabel('Parte real');
legend('Y-P^-','Y-P^+','Y-P^--P^+','1/2','Location','best');
title('Coeficiente ajustado de exp(i omega T)');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_minus_imag,'-o','LineWidth',1.2);
hold on;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_plus_imag,'-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_both_imag,'-o','LineWidth',1.2);
yline(0,':');
grid on;
xlabel('Q');
ylabel('Parte imaginaria');
legend('Y-P^-','Y-P^+','Y-P^--P^+','0','Location','best');

figure('Name','Energia y termino cruzado de ramas','Color','w');
semilogx(max(summaryTable.Q,1), ...
    summaryTable.energy_Pminus,'-o','LineWidth',1.2);
hold on;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.energy_Pplus,'-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.cross_term_2Re_inner_Pminus_Pplus, ...
    '-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.energy_Pminus_plus_Pplus,'-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('Energia');
legend('||P^-||^2','||P^+||^2','2Re<P^-,P^+>', ...
    '||P^-+P^+||^2','Location','best');
title('Cancelacion entre ramas');

figure('Name','Ecuacion funcional del residuo','Color','w');
tiledlayout(2,2,'TileSpacing','compact','Padding','compact');

nexttile;
semilogx(max(summaryTable.Q,1),summaryTable.F_variance, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('Var(F_Q)');
title('Dependencia residual de T');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.distance_c_after_both_to_half, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('|mean(F_Q)-1/2|');
```

```
title('Distancia del coeficiente a 1/2');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.carrier_discrete_defect_rms, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('RMS defecto discreto');
title('R_{k+1}-e^{i omega Delta T}R_k');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.F_block_mean_std, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('STD de medias por bloque');
title('Deriva por altura');

figure('Name','Sumas parciales A_Q y B_Q','Color','w');
tiledlayout(2,1,'TileSpacing','compact','Padding','compact');

nexttile;
plot(qVec,real(AQ),'LineWidth',1);
hold on;
plot(qVec,imag(AQ),'LineWidth',1);
yline(0.5,':');
yline(-0.5,':');
set(gca,'XScale','log');
grid on;
xlabel('Q');
ylabel('A_Q');
legend('Re A_Q','Im A_Q','1/2','-1/2','Location','best');

nexttile;
plot(qVec,real(BQ),'LineWidth',1);
hold on;
plot(qVec,imag(BQ),'LineWidth',1);
yline(0.5,':');
yline(-0.5,':');
set(gca,'XScale','log');
grid on;
xlabel('Q');
ylabel('B_Q');
legend('Re B_Q','Im B_Q','1/2','-1/2','Location','best');

blockSupMatrix = reshape( ...
    blockTable.sup_abs_F_minus_global_c, ...
    numHeightBlocks,numel(Qlist)).';

figure('Name','Desviacion suprema por bloques','Color','w');
imagesc(1:numHeightBlocks,1:numel(Qlist),log10(blockSupMatrix));
colorbar;
```

```
yticks(1:numel(Qlist));
yticklabels(string(Qlist));
xlabel('Bloque de altura');
ylabel('Q');
title('log10 sup_{T en I_j}|F_Q(T)-c_Q|');

figure('Name','Eigenvalores de Gram','Color','w');
tiledlayout(1,2,'TileSpacing','compact','Padding','compact');

nexttile;
semilogy(max(eigCombined,eps),'-o');
grid on;
xlabel('Indice');
ylabel('Eigenvalor');
title('Gram combinada');

nexttile;
semilogy(max(eigBranch,eps),'-o');
grid on;
xlabel('Indice');
ylabel('Eigenvalor');
title('Gram por ramas');

figure('Name','Frecuencias omitidas','Color','w');
semilogy(omittedTable.n,omittedTable.fit_F_minus_abs,'.');
hold on;
semilogy(omittedTable.n(ppMask), ...
    omittedTable.fit_F_minus_abs(ppMask),'o');
semilogy(omittedTable.n,omittedTable.fit_F_plus_abs,'.');
semilogy(omittedTable.n(ppMask), ...
    omittedTable.fit_F_plus_abs(ppMask),'s');
grid on;
xlabel('n');
ylabel('Amplitud ajustada');
legend('Rama - todos','Rama - potencias primas', ...
    'Rama + todos','Rama + potencias primas','Location','best');

figure('Name','Espectro no nulo de F_Q','Color','w');
plot(nuGrid,frequencyAmplitude,'LineWidth',1);
grid on;
xlabel('nu en exp(i nu T)');
ylabel('Amplitud');
title(sprintf('F_Q-mean(F_Q), Q=%g',Qmax));

%% ========================================================================
% GUARDADO OPCIONAL; NUNCA ES NECESARIO PARA EJECUTAR
% ========================================================================

if saveResults

    if ~isfolder(outputDirectory)
        mkdir(outputDirectory);
    end
```

```
    writetable(summaryTable, ...
        fullfile(outputDirectory,'SIX_TESTS_SUMMARY.csv'));

    writetable(coefficientCutoffDisplay, ...
        fullfile(outputDirectory,'COEFFICIENT_CUTOFF_SUMMARY.csv'));

    writetable(groupSummaryTable, ...
        fullfile(outputDirectory,'TELESCOPING_SUMMARY.csv'));

    writetable(groupPartialTable, ...
        fullfile(outputDirectory,'TELESCOPING_PARTIALS.csv'));

    writetable(positivityTable, ...
        fullfile(outputDirectory,'POSITIVITY_FUNCTIONALS.csv'));

    writetable(blockTable, ...
        fullfile(outputDirectory,'BLOCKWISE_FUNCTIONAL_EQUATION.csv'));

    writetable(omittedTable, ...
        fullfile(outputDirectory,'OMITTED_FREQUENCIES.csv'));

    writetable(uniquenessTable, ...
        fullfile(outputDirectory,'UNIQUENESS_SUMMARY.csv'));

    save(fullfile(outputDirectory,'ALL_RESULTS.mat'), ...
        'summaryTable','coefficientCutoffDisplay','radialAuditTable', ...
        'groupSummaryTable','groupPartialTable', ...
        'positivityTable','blockTable','omittedTable', ...
        'uniquenessTable','topFrequencyTable', ...
        'gramCombined','gramBranch','eigCombined','eigBranch', ...
        'RbothAtQmax','FAtQmax','EAtQmax', ...
        'PminusAtQmax','PplusAtQmax','cBothAtQmax','-v7.3');

    fprintf('\nResultados guardados en %s\n',outputDirectory);
end

fprintf('\n============================================================\n');
fprintf('TERMINADO\n');
fprintf('Tablas disponibles directamente en el workspace:\n');
fprintf('  radialAuditTable\n');
fprintf('  summaryTable\n');
fprintf('  coefficientCutoffDisplay\n');
fprintf('  groupSummaryTable\n');
fprintf('  groupPartialTable\n');
fprintf('  positivityTable\n');
fprintf('  blockQmax\n');
fprintf('  omittedTable\n');
fprintf('  uniquenessTable\n');
fprintf('  topFrequencyTable\n');
fprintf('============================================================\n');

%% ========================================================================
% FUNCIONES LOCALES
% ========================================================================
```

```
function c = fitCarrier(z,b)

    z = z(:);
    b = b(:);

    denominator = b'*b;

    if abs(denominator)<=eps
        c = NaN;
    else
        c = (b'*z)/denominator;
    end
end

function [a,c] = fitConstantAndCarrier(z,b)

    z = z(:);
    b = b(:);

    X = [ones(size(b)),b];
    coefficients = X\z;

    a = coefficients(1);
    c = coefficients(2);
end

function value = complexAnisotropy(z,minimumVariance)

    z = z(:);
    z = z-mean(z,'omitnan');

    denominator = mean(abs(z).^2,'omitnan');

    if ~isfinite(denominator) || denominator<=minimumVariance
        value = NaN;
    else
        value = abs(mean(z.^2,'omitnan'))/denominator;
    end
end

function r = safeCorrelation(x,y)

    x = x(:);
    y = y(:);

    valid = isfinite(x) & isfinite(y);

    if nnz(valid)<3
        r = NaN;
        return;
    end

    x = x(valid);
```

```
    y = y(valid);

    x = x-mean(x);
    y = y-mean(y);

    denominator = sqrt(sum(abs(x).^2)*sum(abs(y).^2));

    if denominator<=eps
        r = NaN;
    else
        r = real((x'*y)/denominator);
    end
end

function r = complexVectorCorrelation(x,y)

    x = x(:);
    y = y(:);

    valid = isfinite(real(x)) & isfinite(imag(x)) ...
        & isfinite(real(y)) & isfinite(imag(y));

    x = x(valid);
    y = y(valid);

    denominator = norm(x)*norm(y);

    if denominator<=eps
        r = NaN;
    else
        r = (x'*y)/denominator;
    end
end

function [qVec,basePrimeVec,powerExponentVec,lambdaVec] = ...
    makePrimePowers(Qmax)

    if Qmax<2
        qVec = zeros(0,1);
        basePrimeVec = zeros(0,1);
        powerExponentVec = zeros(0,1);
        lambdaVec = zeros(0,1);
        return;
    end

    pList = primes(Qmax);

    qList = zeros(0,1);
    baseList = zeros(0,1);
    exponentList = zeros(0,1);
    lambdaList = zeros(0,1);

    for jj = 1:numel(pList)
```

```
        p = pList(jj);
        currentPower = p;
        exponent = 1;

        while currentPower<=Qmax

            qList(end+1,1) = currentPower; %#ok<AGROW>
            baseList(end+1,1) = p; %#ok<AGROW>
            exponentList(end+1,1) = exponent; %#ok<AGROW>
            lambdaList(end+1,1) = log(p); %#ok<AGROW>

            if currentPower>floor(Qmax/p)
                break;
            end

            currentPower = currentPower*p;
            exponent = exponent+1;
        end
    end

    [qVec,order] = sort(qList);
    basePrimeVec = baseList(order);
    powerExponentVec = exponentList(order);
    lambdaVec = lambdaList(order);

    [qVec,uniqueIndex] = unique(qVec,'stable');
    basePrimeVec = basePrimeVec(uniqueIndex);
    powerExponentVec = powerExponentVec(uniqueIndex);
    lambdaVec = lambdaVec(uniqueIndex);
end

function [summaryTable,partialTable] = primePowerTelescopingTests( ...
    qVec,basePrimeVec,powerExponentVec,cMinusVec,cPlusVec, ...
    gamma,carrier)

    uniquePrimes = unique(basePrimeVec);

    keep = false(size(uniquePrimes));

    for jj = 1:numel(uniquePrimes)
        keep(jj) = nnz(basePrimeVec==uniquePrimes(jj))>=2;
    end

    uniquePrimes = uniquePrimes(keep);

    summaryNames = { ...
        'base_prime','num_powers','largest_power', ...
        'combined_energy_ratio_sum_over_individual', ...
        'minus_energy_ratio_sum_over_individual', ...
        'plus_energy_ratio_sum_over_individual', ...
        'combined_rms_over_sum_term_rms', ...
        'boundary_fit_relative_rms_combined', ...
        'boundary_fit_relative_rms_minus', ...
        'boundary_fit_relative_rms_plus', ...
```

```
    'boundary_alpha_first_real','boundary_alpha_first_imag', ...
    'boundary_alpha_last_real','boundary_alpha_last_imag', ...
    'tail_primitive_error_minus','tail_primitive_error_plus', ...
    'tail_primitive_start_abs_minus', ...
    'tail_primitive_start_abs_plus'};

partialNames = { ...
    'base_prime','power_exponent','n_prime_power', ...
    'term_rms_minus','term_rms_plus','term_rms_combined', ...
    'partial_rms_minus','partial_rms_plus','partial_rms_combined', ...
    'partial_coefficient_minus_real', ...
    'partial_coefficient_minus_imag', ...
    'partial_coefficient_plus_real', ...
    'partial_coefficient_plus_imag'};

if isempty(uniquePrimes)
    summaryTable = cell2table(cell(0,numel(summaryNames)), ...
        'VariableNames',summaryNames);

    partialTable = cell2table(cell(0,numel(partialNames)), ...
        'VariableNames',partialNames);

    return;
end

totalPartialRows = 0;

for jj = 1:numel(uniquePrimes)
    totalPartialRows = totalPartialRows ...
        +nnz(basePrimeVec==uniquePrimes(jj));
end

summaryRows = cell(numel(uniquePrimes),numel(summaryNames));
partialRows = cell(totalPartialRows,numel(partialNames));

partialRow = 0;

for jp = 1:numel(uniquePrimes)

    p = uniquePrimes(jp);
    idxTerms = find(basePrimeVec==p);

    [~,order] = sort(powerExponentVec(idxTerms));
    idxTerms = idxTerms(order);

    R = numel(idxTerms);

    sumMinus = complex(zeros(size(gamma)));
    sumPlus = complex(zeros(size(gamma)));

    sumEnergyMinus = 0;
    sumEnergyPlus = 0;
    sumEnergyCombined = 0;
    sumTermRMSCombined = 0;
```

```
firstMinus = [];
firstPlus = [];
lastMinus = [];
lastPlus = [];

cumulativeCoefficientMinus = 0;
cumulativeCoefficientPlus = 0;

for rr = 1:R

    j = idxTerms(rr);
    ell = log(qVec(j));
    z = exp(1i*ell*gamma);

    termMinus = cMinusVec(j).*carrier.*conj(z);
    termPlus = cPlusVec(j).*carrier.*z;
    termCombined = termMinus+termPlus;

    if rr==1
        firstMinus = termMinus;
        firstPlus = termPlus;
    end

    if rr==R
        lastMinus = termMinus;
        lastPlus = termPlus;
    end

    sumMinus = sumMinus+termMinus;
    sumPlus = sumPlus+termPlus;

    termEnergyMinus = mean(abs(termMinus).^2);
    termEnergyPlus = mean(abs(termPlus).^2);
    termEnergyCombined = mean(abs(termCombined).^2);

    sumEnergyMinus = sumEnergyMinus+termEnergyMinus;
    sumEnergyPlus = sumEnergyPlus+termEnergyPlus;
    sumEnergyCombined = sumEnergyCombined+termEnergyCombined;

    sumTermRMSCombined = sumTermRMSCombined ...
        +sqrt(termEnergyCombined);

    cumulativeCoefficientMinus = cumulativeCoefficientMinus ...
        +cMinusVec(j);

    cumulativeCoefficientPlus = cumulativeCoefficientPlus ...
        +cPlusVec(j);

    partialRow = partialRow+1;

    partialRows(partialRow,:) = { ...
        p,powerExponentVec(j),qVec(j), ...
        sqrt(termEnergyMinus),sqrt(termEnergyPlus), ...
```

```
        sqrt(termEnergyCombined), ...
        sqrt(mean(abs(sumMinus).^2)), ...
        sqrt(mean(abs(sumPlus).^2)), ...
        sqrt(mean(abs(sumMinus+sumPlus).^2)), ...
        real(cumulativeCoefficientMinus), ...
        imag(cumulativeCoefficientMinus), ...
        real(cumulativeCoefficientPlus), ...
        imag(cumulativeCoefficientPlus)};
end

sumCombined = sumMinus+sumPlus;

energyRatioCombined = mean(abs(sumCombined).^2) ...
    /max(sumEnergyCombined,eps);

energyRatioMinus = mean(abs(sumMinus).^2) ...
    /max(sumEnergyMinus,eps);

energyRatioPlus = mean(abs(sumPlus).^2) ...
    /max(sumEnergyPlus,eps);

rmsOverSumRMS = sqrt(mean(abs(sumCombined).^2)) ...
    /max(sumTermRMSCombined,eps);

firstCombined = firstMinus+firstPlus;
lastCombined = lastMinus+lastPlus;

[relativeCombined,alphaCombined] = boundaryFit( ...
    sumCombined,firstCombined,lastCombined);

[relativeMinus,~] = boundaryFit( ...
    sumMinus,firstMinus,lastMinus);

[relativePlus,~] = boundaryFit( ...
    sumPlus,firstPlus,lastPlus);

cM = cMinusVec(idxTerms);
cP = cPlusVec(idxTerms);

% Toda sucesion admite esta primitiva de cola. La identidad es
% tautologica; lo informativo es su tamano y si existe forma cerrada.
aM = [-flipud(cumsum(flipud(cM)));0];
aP = [-flipud(cumsum(flipud(cP)));0];

primitiveErrorM = max(abs(cM-(aM(2:end)-aM(1:end-1))));
primitiveErrorP = max(abs(cP-(aP(2:end)-aP(1:end-1))));

summaryRows(jp,:) = { ...
    p,R,max(qVec(idxTerms)), ...
    energyRatioCombined,energyRatioMinus,energyRatioPlus, ...
    rmsOverSumRMS, ...
    relativeCombined,relativeMinus,relativePlus, ...
    real(alphaCombined(1)),imag(alphaCombined(1)), ...
    real(alphaCombined(2)),imag(alphaCombined(2)), ...
```

```
            primitiveErrorM,primitiveErrorP, ...
            abs(aM(1)),abs(aP(1))};
    end

    summaryTable = cell2table(summaryRows, ...
        'VariableNames',summaryNames);

    partialTable = cell2table(partialRows(1:partialRow,:), ...
        'VariableNames',partialNames);
end

function [relativeRMS,alpha] = boundaryFit(target,firstTerm,lastTerm)

    target = target(:);
    firstTerm = firstTerm(:);
    lastTerm = lastTerm(:);

    X = [firstTerm,lastTerm];
    alpha = X\target;

    residual = target-X*alpha;

    relativeRMS = sqrt(mean(abs(residual).^2)) ...
        /max(sqrt(mean(abs(target).^2)),eps);
end


function [radius,centers,densitySmooth,peakRadii] = ...
    radialDistributionSummary(z,numBins,smoothWidth)

    z = z(:);
    z = z(isfinite(real(z)) & isfinite(imag(z)));

    center = mean(z);
    radius = abs(z-center);

    rMin = min(radius);
    rMax = max(radius);

    if rMax<=rMin
        rMax = rMin+eps(rMin+1);
    end

    edges = linspace(rMin,rMax,numBins+1);
    density = histcounts(radius,edges,'Normalization','pdf');
    centers = ((edges(1:end-1)+edges(2:end))/2).';

    densitySmooth = smoothdata(density(:),'gaussian',smoothWidth);

    candidates = find( ...
        densitySmooth(2:end-1)>densitySmooth(1:end-2) & ...
        densitySmooth(2:end-1)>=densitySmooth(3:end))+1;

    if isempty(candidates)
```

```
        peakRadii = zeros(0,1);
        return;
    end

    candidates = candidates( ...
        densitySmooth(candidates)>=0.12*max(densitySmooth));

    if isempty(candidates)
        peakRadii = zeros(0,1);
        return;
    end

    % Conserva maximos separados al menos seis bins, ordenados por altura.
    [~,heightOrder] = sort(densitySmooth(candidates),'descend');
    selected = zeros(0,1);

    for j = 1:numel(heightOrder)
        candidate = candidates(heightOrder(j));

        if isempty(selected) || all(abs(candidate-selected)>=6)
            selected(end+1,1) = candidate; %#ok<AGROW>
        end

        if numel(selected)>=4
            break;
        end
    end

    peakRadii = sort(centers(selected));
end

function s = joinNumericVector(v)

    v = v(:).';

    if isempty(v)
        s = "";
        return;
    end

    pieces = strings(size(v));

    for j = 1:numel(v)
        pieces(j) = sprintf('%.9g',v(j));
    end

    s = strjoin(pieces,', ');
end
```

42. Complete console output of the full-data run

42.1. **Coefficient identities, radial audit, branches, towers, blocks, and omitted frequencies.**

Remark 42.1. The console transcript below is included as numerical support. The analytic claims of the manuscript are established in the theorem–proof sections and do not depend on numerical rounding in this transcript.

Listing 2. Complete console output for the corrected full-data run.

```
============================================================
ZETA PRIME-POWER STRUCTURE: AUDITORIA ANALITICA
============================================================
Archivo de ceros       : zeross.bin
Ceros disponibles      : 2001052
q                      : 2.7182818284590451
omega=log(q)           : 1
stride                 : 1
Qlist                  : 0 3 10 100 1000 10000
Puntos analizados       : 2001052
gamma inicial           : 14.134725142
gamma final             : 1132490.65871441

============================================================
TEST 2: IDENTIDADES DE COEFICIENTES
============================================================
Potencias primas <=Qmax                 : 1280
max |(c^-+c^+)-formula|                : 2.220446e-16
max |(c^--c^+)-formula|                : 2.220446e-16
max |c^-/c^+-formula|                  : 1.776357e-15
      Q      num_terms    Re_AQ          Im_AQ           abs_AQ_minus_half    Re_BQ          Im_BQ
     abs_BQ_minus_half    abs_BQ_plus_half
    -----    ---------    -----    ------------------    -----------------    -----    ------------------
     -----------------    -----------------

        0         0         0                       0                  0.5      0                       0
                  0.5                 0.5
        3         2         0      -0.675311086221108    0.840264876793701      0      -0.863660742917475
     0.997952844004547    0.997952844004547
       10         7         0       -1.07698676185356      1.1873923046777      0       -1.54062594080643
      1.6197309311999      1.6197309311999
      100        35         0       -1.47655654519435     1.55891604365222      0       -2.87744845271495
     2.92056665700882     2.92056665700882
     1000       193         0       -1.89626714054451      1.9610785472053      0       -5.31082516388603
     5.33431006985581     5.33431006985581
    10000      1280         0       -2.55521376172467     2.60367382137378      0       -10.6932626383477
     10.7049458593999     10.7049458593999


============================================================
AUDITORIA RADIAL DIRECTA DE LA BIMODALIDAD
============================================================
      case_name        mean_radius          std_radius        num_detected_peaks      detected_peak_radii
    -------------    -----------------    -----------------    ------------------    ------------------------

    "original"       1.92031044763071    0.726547772615351            2            "1.35731059, 2.49789956"
    "after_n3"       1.80792479214104    0.308312030736288            1            "1.68017926"
    "after_n2_n3"    1.78844590851816     0.22140483373138            1            "1.81300495"

Coeficiente teorico total n=3:
  c3^- = 0 -1.02370050729i
  c3^+ = 0 +0.0481029537851i
```

```
Q=0      terms=0      c^-res=0.4999989+0.1381673i  c^+res=0.4999989+0.1381673i  c^both=0.4999989+0.1381673i
     Var(F)=1.2420e+00  RMS(E)=1.1145e+00
Q=3      terms=2      c^-res=0.4999999+0.0909043i  c^+res=0.4999989+0.1429495i  c^both=0.4999999+0.0956865i
     Var(F)=1.3393e-01  RMS(E)=3.6597e-01
Q=10     terms=7      c^-res=0.5000000+0.0636593i  c^+res=0.4999989+0.1500304i  c^both=0.5000000+0.0755224i
     Var(F)=5.9610e-02  RMS(E)=2.4415e-01
Q=100    terms=35     c^-res=0.5000000+0.0191267i  c^+res=0.4999989+0.1736326i  c^both=0.5000000+0.0545920i
     Var(F)=2.2040e-02  RMS(E)=1.4846e-01
Q=1000   terms=193    c^-res=0.5000000-0.0206343i  c^+res=0.4999990+0.2014465i  c^both=0.5000001+0.0426449i
     Var(F)=1.1282e-02  RMS(E)=1.0622e-01
Q=10000  terms=1280   c^-res=0.4999999-0.0581400i  c^+res=0.4999989+0.2306199i  c^both=0.4999999+0.0343125i
     Var(F)=8.2559e-03  RMS(E)=9.0862e-02

============================================================
TEST 1: RAMAS; TEST 5: ECUACION FUNCIONAL
============================================================
      Q      num_prime_power_terms    c_after_minus_real    c_after_minus_imag     c_after_plus_real
     c_after_plus_imag    c_after_both_real    c_after_both_imag     distance_c_after_both_to_half
     rms_centered_Y_minus_Pminus    rms_centered_Y_minus_Pplus    rms_centered_Y_minus_both
     rms_after_both_fitted_carrier     energy_Pminus         energy_Pplus
     cross_term_2Re_inner_Pminus_Pplus        F_variance         F_max_abs_deviation_from_mean
     F_block_mean_std        F_anisotropy       carrier_discrete_defect_rms    derivative_defect_rms
    -----    ---------------------    ------------------    --------------------    -----------------
     -----------------    ------------------    ------------------    ------------------------------
     ----------------------------    ---------------------------    -------------------------
     ------------------------------    ----------------    --------------------
     ----------------------------------    --------------------    ------------------------------
     --------------------    -----------------    ----------------------------    ---------------------

        0               0             0.499998912312768       0.138167317526428    0.499998912312768
     0.138167317526428    0.499998912312768     0.138167317526428          0.138167317530709
     1.22927492233399              1.22927492233399              1.22927492233399
     1.11446297144985                          0                         0                                0
         1.24202771473284           2.53363749717284            0.0101613448739479    0.164704934429685
       0.828969921883291           2.13681541452671
        3               2              0.49999988687813      0.0909043288314701    0.499998893284938
     0.142949512755117    0.499999867850236    0.0956865240601598          0.095686524060251
     0.630307565358949              1.22653992517659              0.626967498686433
     0.365967847866614          1.11258876067543    0.00443650350346841             0.00251726408927867
         0.133932465672121            1.3998447731532           0.00612569360136624    0.670875082815371
        0.476587597924875            1.7823665130755
       10               7              0.49999998247139      0.0636593469198101    0.499998916032037
     0.150030381454066    0.499999986190665     0.075522410847451          0.0755224108474523
     0.568023295911874              1.22456854935429              0.561527861557534
     0.244151425501691          1.18492844347144    0.00871311004796696             0.00421177723983395
        0.0596099185745075           1.05195331714233             0.0042120343511029    0.886135136342853
        0.398840669251055           1.69391674505877
      100              35             0.500000017346496      0.0191267412460307     0.49999893653376
     0.173632564206007     0.50000004156745     0.054591987925598          0.0545919879256138
     0.536555599882361              1.22197814824027              0.524423432995709
     0.148457436196022          1.21916077552125     0.0180717973642778             0.00501426487870665
        0.0220396103618959          0.799206246858361             0.0022297937207615    0.958121438757457
          0.2682174006908           1.45913395069404
     1000             193             0.500000036693381     -0.0206342853236017    0.499998951118377
     0.201446524072374    0.500000075498899    0.0426449212223482          0.042644921222415
     0.526364711348993              1.22166555561873              0.51293359872215
     0.10621775695736          1.23384826802886     0.0252917244776332             0.00469116700005195
       0.0112822118930529          0.668993854159613             0.0011112380638497    0.970189423496525
       0.179876374959657           1.21197809635573
```

```
    10000            1280             0.499999922217885     -0.0581399985655881     0.499998861049832
     0.230619861962331     0.499999870954991     0.0343125458703112          0.0343125458705538
     0.523988825848724              1.22334869951959              0.509345817216466
     0.0908622020249825          1.24333446127337      0.030996201483447          -0.000596344803520444
        0.00825593975682874           0.61202824502835          0.000381258918696766     0.981211983063741
         0.150986530997687            1.0471736199134

max error R_j-R_{j-1}=-P_j                 : 8.962228e-16
max error ley de energia de ramas          : 2.220446e-16
max error identidad de proyeccion portadora : 3.559044e-14
max error identidad de constantes           : 5.446388e-14

============================================================
TEST 3: TELESCOPADO EN p,p^2,...,p^R
============================================================
    base_prime    num_powers    largest_power    combined_energy_ratio_sum_over_individual
     combined_rms_over_sum_term_rms    boundary_fit_relative_rms_combined    boundary_fit_relative_rms_minus
       boundary_fit_relative_rms_plus    boundary_alpha_first_real    boundary_alpha_first_imag
     boundary_alpha_last_real    boundary_alpha_last_imag    tail_primitive_start_abs_minus
     tail_primitive_start_abs_plus
    ----------    ----------    -------------    ----------------------------------------
     ------------------------------    ----------------------------------    ------------------------------
       ------------------------------    -------------------------    -------------------------
     ------------------------    ------------------------    ------------------------------
     ----------------------------

        89            2             7921                       0.99578526600946
     0.951568779802395                  3.25731221427626e-15               1.66529459300868e-14
      5.68490958536155e-15             0.999999999999997           5.38298239788951e-18
     0.999999999999988        -6.13165759929884e-17             0.022712415828171
     0.0146012082752434
        37            2             1369                      0.992325345063945
     0.929272152959258                  3.32342530365488e-15               6.90975846079265e-15
      2.57414422667853e-14             0.999999999999997          -8.74885970690982e-18
     1.00000000000001        -8.01331294276757e-17            0.0386826763307904
     0.0223795089575229
        41            2             1681                      0.992798804843908
     0.932179807015267                  4.01764974378634e-15               5.71123669041389e-15
      6.00240878492108e-15                              1          -4.41766278752356e-18
     0.999999999999997         6.86611361451242e-17            0.0362585450928221
     0.0212931769572583
        23            2              529                       0.98981553486963
     0.914722655159467                  5.29829056019486e-15               3.01909112408219e-14
      5.06477325930078e-14             0.999999999999995            1.5256443527521e-17
     1.00000000000002        -4.12721429521546e-17             0.052842635596283
     0.0281454412919937
        67            2             4489                      0.994867301216485
     0.945073421083208                  6.62713306705871e-15               2.92396921729543e-14
      3.16762417963749e-14             0.999999999999993           4.06385013591999e-18
     0.999999999999977         -6.6991869985725e-17            0.0268592065905553
     0.0167695322528614
        73            2             5329                      0.995124247078512
     0.947082293993756                  8.02253098923419e-15               4.21876931539434e-14
      2.84327775551962e-14             0.999999999999992           2.29352846836488e-17
     0.999999999999967         1.25402372809613e-16            0.0255226923919741
     0.0160830067737586
```

```
   29            2             841                    0.991065442073751
0.921970578232658                1.69958093184349e-14              3.92022883362443e-15
 4.85298540260265e-14            1.00000000000002          5.31348409790742e-18
0.999999999999986       -1.67495027840586e-16           0.0452613167826162
0.0251764129412383
   59            2            3481                    0.994356231992412
0.941904651888199                1.82233584563793e-14              4.44612460135524e-14
  3.3145884657823e-14            1.00000000000002          3.31967944946604e-17
0.999999999999965        2.13895102136555e-16           0.0289902084009457
0.0178408950865232
   61            2            3721                    0.994484921893845
0.942760057482987                1.89061413493395e-14              4.06345157893362e-14
 1.56552838293448e-14            0.999999999999981         -1.56580841679002e-17
0.999999999999966       -2.22783722808597e-17           0.0284133680209016
0.0175536598492679
   43            2            1849                    0.993018430417279
0.933512335891677                1.91767251608662e-14              9.19221287192386e-15
 2.67694821919958e-14            1.00000000000002         -5.47518533793213e-18
0.999999999999988        1.24241990600724e-16           0.0351952587338588
0.0208067465987886
   97            2            9409                    0.996034159322401
0.953412551798485                2.19836316719303e-14              2.48682193279047e-15
 4.67252549818149e-15            0.999999999999978         -1.96285882336457e-17
0.999999999999975       -2.20623599182162e-16           0.0216013618259659
0.0140005339242615
   71            2            5041                    0.995055469368263
0.94645716703403                 2.45606582770506e-14               1.40537772877289e-14
1.35601376253121e-14            1.00000000000002         -2.17658103478551e-17
1.00000000000007        -3.81439794611675e-16           0.0259471363764858
0.0163022731943209
   83            2            6889                    0.995556221672419
0.950034257670606                2.80380400407119e-14              7.38122722146823e-15
 1.69133036510308e-14            1.00000000000003         -1.68047878446504e-17
1.00000000000006        -3.60217187758929e-16           0.0236603972161033
0.0151069817398424
   53            2            2809                    0.993923584391844
0.939166861275507                3.64856201837887e-14              5.33595501442798e-14
 6.78833363346336e-15            1.00000000000004         -1.38986297087105e-17
0.999999999999991        -4.48286028808063e-16            0.030939494184714
0.0187966802009105
   47            2            2209                     0.99342415440416
0.935964632967202                4.02588932254384e-14              4.69976317545872e-15
 3.70119612870561e-14            1.00000000000004          3.64557105503385e-17
0.999999999999993        -5.00636376597972e-16           0.0333062074556399
0.0199271852036153
   79            2            6241                    0.995407211972277
0.948922007921855                4.26585420767673e-14              6.84664346328644e-16
 1.95720970351145e-14            0.999999999999957          4.32727341964049e-17
0.999999999999994        6.93817715660186e-16           0.0243581235750551
0.015475353217867
   31            2             961                     0.99142834108407
0.924010046881153                4.48560611982839e-14              1.73461849505888e-14
 4.95966705898557e-15            1.00000000000004          1.34960651075476e-18
1.00000000000001        -1.47682556498033e-16           0.0433336826232223
0.0243790159002065
```

```
        3            8             6561                      0.994136424160185
    0.925379114250609                    0.0526293055865214                  0.0488679460726364
         0.38539545977793             0.997125200778133           1.04928001769985e-08
    -0.373319182377488          0.00313372076449291              1.09725325128044
    0.080998072339286
       19            3             6859                      0.988180958947934
    0.896291448276131                    0.099571908742451                   0.0910436270892582
        0.130813323815928             0.994234206127615           8.21480982437346e-07
    0.587653567802991        -6.66788850988072e-05            0.0610579674917809
    0.0314116469586121
       17            3             4913                      0.987462564808847
    0.891547371679716                    0.104055647033926                   0.0950571818522644
        0.138802345385611             0.993904379532631           7.82474768756128e-08
    0.603159942163849        -1.05608473149603e-05            0.0661990286058359
    0.0331867205015699

boundary_fit_relative_rms_combined cercano a cero seria evidencia de colapso a los dos bordes del bloque p^r.

============================================================
TEST 4: POSITIVIDAD Y GRAM
============================================================
      Q            norm_E2          carrier_schur_complement    carrier_projection_abs     Re_inner_E_Pminus
           Re_inner_E_Pplus       cauchy_det_E_Pminus     cauchy_det_E_Pplus     cauchy_det_Pminus_Pplus
     polarization_error_minus    polarization_error_plus
    -----    --------------------    ------------------------    ----------------------    --------------------
         ----------------------    --------------------    --------------------    ------------------------
     ------------------------    ------------------------

        0       1.24202771473284         1.24202771473284         4.1753563397406e-15                         0
                              0                        0                          0                           0
                         0                        0
        3      0.133932465672121        0.133932465672121        3.19449329162395e-14     -0.00516035229366167
          0.000339021642957032     0.148985126760466     0.000594076917452501      0.00493441978003228
     2.22044604925031e-16       8.67361737988404e-19
       10     0.0596099185745075       0.0596099185745074        7.25002166702646e-15     -0.00596341179707172
          0.000207836720997232    0.0705979257513303     0.000519344584387362       0.0103199771600532
     2.22044604925031e-16       3.46944695195361e-18
      100     0.0220396103618959       0.0220396103618958        1.12998339769851e-14     -0.00697316806144494
         -0.000663780178893137     0.026821203386263     0.000397854767628292        0.022026140776441
                         0                        0
     1000     0.0112822118930529       0.0112822118930528        3.76207946660896e-14      -0.0090127255333809
          -0.00296783860935025    0.0138393083795104     0.000276538526093234       0.0312006486796008
     2.22044604925031e-16       1.38777878078145e-17
    10000    0.00825593975682874      0.00825593975682881        2.86661244710573e-14     -0.00982808606142384
          -0.00476027747414303     0.010168303134064     0.000233242529391743       0.0385385565639377
     2.22044604925031e-16       3.46944695195361e-18

Gram combinada: min=2.291421e-03, max=1.119809e+00, negativos<-1e-10=0
Gram por ramas: min=4.858227e-04, max=1.112574e+00, negativos<-1e-10=0

============================================================
TEST 5: F_Q POR BLOQUES DE ALTURA, Q=10000
============================================================
    block      gamma_start          gamma_end           F_mean_real          F_mean_imag            F_variance
              sup_abs_F_minus_global_c    sup_abs_F_minus_block_c     mean_abs_delta_F
     variation_per_height       anisotropy        carrier_discrete_defect_rms    derivative_defect_rms
    -----    ----------------    ----------------    -----------------    ------------------
     --------------------    ------------------------    ------------------------    ------------------
     --------------------    -----------------    --------------------------    --------------------
```

```
 1          14.134725142       63699.04754739    0.499994991361003     0.0350523539924386
0.0054744556993458        0.542532351404617         0.542510726459913        0.0963670633500536
0.126163323741996       0.979858632599489         0.121147135176878            0.538564667956422
 2        63699.700151835     118462.736278749    0.500001451274137      0.035105891285452
0.00701927614984981        0.530144044699919          0.530116241500027         0.110272510195026
0.167888441917521       0.980474241382446         0.137953328199234            0.726396110177662
 3       118463.232066064     170651.787459452    0.499998562254379     0.0349210363561646
0.00745311407766993        0.549893284088027          0.549871555991103         0.113755262822731
0.181736637024686       0.980873421275848         0.142306171066992            0.7574604344515
 4        170652.38272511     221291.519378741    0.500000256946116     0.0348042049768492
0.00767564743723714        0.581090952553066          0.581063883641327         0.116011219222976
0.191009406027093       0.980878273487911         0.144913294713216            0.880192519706131
 5       221292.295838797     270836.328162977    0.500000955646856     0.0347080955882823
0.007831121437243         0.57962624512367           0.579600207778524         0.117381041928519
0.197536641502951       0.980905060378457         0.146681991054392            0.923600652953341
 6       270836.931684803     319539.773103185    0.499999624306421     0.0346475825508576
0.00808884143363537        0.534809090007345          0.53479827792399          0.118850165982451
0.203463517740728       0.981360633102241         0.14914505740182             1.16913063854399
 7       319540.570678831     367564.435949345    0.49999957921803      0.0345578903969988
0.00818709113446676         0.52294466209729           0.52292439742989          0.119910420246151
0.208180893856153       0.981406498730984         0.150029861940657            0.880453279774671
 8       367565.006156693     415022.283240227    0.500000487984706     0.0344519123654481
0.00814253081563704        0.600282121372596          0.600276208543475         0.119491391046747
0.209930169452775       0.981010897479751         0.149551484403089            1.00807618824186
 9        415023.15758963     461995.409615676    0.500002125859323     0.0343892982332912
0.00818298949111592        0.572654690283211          0.572652885861422          0.1200026729604
0.213007944730247       0.980937348210029         0.150172616772631            1.08111614297039
10       461995.761482843     508544.337297334    0.499999128775315     0.0343361215632085
0.00835629092630199        0.542768812320236          0.542768199088721          0.12137168582189
0.217396246824282       0.981335496066421         0.152037594340513            0.934750513408792
11       508544.765796632     554719.129189251    0.49999902683957      0.0342841671661181
0.00838655885140866        0.555269896680713          0.55526995649336          0.121722656912353
0.219791838956823       0.981261093718569         0.152335939237595            1.12844994151658
12       554719.634077261     600557.928334761    0.499999052950246     0.0342521074641879
0.00848384933678091        0.606087165112338          0.606091467206371          0.12213899474892
0.222160553553316       0.981322097240396         0.153080139635091            1.07065793214649
13       600558.535755246     646093.102175532    0.500000235514038     0.0342171059277988
0.00852631056710716        0.524846602477766          0.524849827921532         0.122203119197641
0.223759839331275       0.981196827744373         0.153515460380432            1.14290197565882
14        646094.01946502     691351.826509571    0.499999459797555     0.0341425285063966
0.00852459159575678         0.61202824502835           0.612034526321802         0.122717190800213
0.226075215930959        0.98130473863714          0.153711972752659            1.52515702182085
15       691352.526827545     736357.304575928     0.50000063261387     0.0340996079052541
0.00863390467854271        0.603869638677834          0.603880275368038         0.123435919580137
0.228680535306117         0.9813459888377           0.154802894954523            1.13198994875413
16       736358.196419715     781127.536440619    0.500000507064813     0.0340763607134163
0.00863949315690915        0.551934114882366          0.551943379429299         0.123479710718261
0.229961941722585       0.981411615889858         0.154800438591316            1.02064882439406
17       781128.075976077     825679.648395047    0.499999867045727     0.0340594494960409
0.00867051311819898        0.586536403755823          0.586546166766736         0.123341062550961
0.230826520207627       0.981193765149418         0.155214716157765            1.15057126782006
18       825680.493113046     870029.136314775    0.499999157278646     0.0340083355422251
0.00880860286374675        0.564338258958265          0.564345300438399         0.124310351932475
0.233705005485217       0.981676906683931         0.156532439348918            1.09637171176575
19       870029.689029286     914187.909332815    0.499999527336173     0.0339700206865135
0.00876154865120847         0.56850486610471           0.568522179523045         0.124185981373753
0.234477981944182       0.981561079878874          0.15588197711697            0.995676199295695
```

```
 20       914188.612117913     958168.018537517      0.50000099473625     0.0339345816929237
0.00878369871741152        0.599171890474147           0.599196831871485         0.124030304858425
0.235136204413765        0.98142509043805          0.156206512272264             1.06669516660454
 21        958168.60666595     1001979.98016229     0.499998363067321     0.0338951222311845
0.0088529271020219         0.597221450310013           0.597233477557632         0.125000771021225
0.237887755021112        0.981656183204604          0.157057387478477             1.16995875593502
 22       1001980.68512795     1045632.21311476     0.500003656384172     0.0338638859862856
0.00874571864686109        0.562103795049353           0.562114143907445         0.124273605310392
0.237367088718873        0.98131312028741          0.156023921934697             1.01328485216897
 23       1045632.45556665     1089133.21595891     0.499998860220289     0.0338981896499074
0.00898518332990823        0.571629074187526           0.571636102485886           0.1254657058582
0.240474616933205        0.981670300817732          0.158067171395785             1.12663969590805
 24       1089133.56438868     1132490.65871441     0.500000398441743     0.0338252499519786
0.00892480514817264        0.590816935924078           0.590833242133321         0.125308756813391
0.240969628397672        0.981703647581976          0.157535630788163             1.18442415402658

Rango Re media por bloque : [0.499994991361, 0.500003656384]
Rango Im media por bloque : [0.033825249952, 0.0351058912855]
Max sup global por bloque : 6.120282e-01
Max sup local por bloque  : 6.120345e-01

============================================================
TEST 6: UNICIDAD Y FRECUENCIAS OMITIDAS
============================================================
    Qmax     num_omitted_prime_powers_joint_fit    median_minus_amplitude_prime_power
     median_minus_amplitude_null    minus_prime_power_to_null_ratio    median_plus_amplitude_prime_power
     median_plus_amplitude_null    plus_prime_power_to_null_ratio    joint_omitted_explained_fraction
     fit_theory_correlation_real    fit_theory_correlation_imag    fit_theory_correlation_abs
     relative_theory_coefficient_error    F_energy_before_omitted_fit    F_energy_after_omitted_fit
     F_anisotropy        corr_gap_realF        corr_gap_imagF        corr_gap_absF
    -----    ----------------------------------    ----------------------------------
     ----------------------------    ----------------------------------    ----------------------------------
     ----------------------------    ----------------------------------    ----------------------------------
     ----------------------------    ----------------------------    ----------------------------
     ----------------------------------    ----------------------------    ----------------------------
     -----------------    -----------------    -----------------    -----------------

    10000                     56                              0.0011038260045886
     0.000329157644250013               3.35348737564237                  0.000885575009662317
     0.000330982677195599              2.6755932279168                   0.0144445968386994
     0.916473061428038             0.00108815005327049            0.916473707420905
     0.480753428806987               0.00822729788430414           0.00810845788329348
     0.981063724279348    0.187253690050176    0.0933065219921283    -0.139580596671858


Potencias primas omitidas con mayor amplitud:
      n      base_prime    power_exponent        Lambda_n          fit_F_minus_real         fit_F_minus_imag
           fit_F_minus_abs         fit_F_plus_real         fit_F_plus_imag          fit_F_plus_abs
     theory_minus_real     theory_minus_imag      theory_plus_real     theory_plus_imag
     max_branch_fit_abs
    -----    ----------    --------------    -----------------    ----------------------    --------------------
        --------------------    ----------------------    --------------------    --------------------
     -----------------    --------------------    -----------------    --------------------
     --------------------

    10301      10301             1           9.23999625688369      0.000371892244369111     -0.00161813748988026
         0.00166032309433371       0.00033865779123921       0.00138200597601097       0.00142289480190808
        0            -0.00175843054430363            0             0.0014149869530774     0.00166032309433371
```

```
10169	10169	1	9.22709915579114	-0.000115283050974615	-0.00156964135711816
0.00157386917239577	-5.07240436672432e-05	0.00133670966276089	0.0013376717277136
0	-0.00177010677368571	0	0.00142394668533194	0.00157386917239577
10181	10181	1	9.2282785171072	0.000278847600421771	-0.00152895592997706
0.00155417573654751	0.000286682906630887	0.00132354059079306	0.00135423291365672
0	-0.00176903580369307	0	0.00142312504251575	0.00155417573654751
10039	10039	1	9.21423278669153	0.00041374673843581	-0.00144753771339835
0.00150550715550501	0.000421812244560221	0.00124369263450663	0.00131327717515647
0	-0.00178183332924528	0	0.00143294107929247	0.00150550715550501
10333	10333	1	9.24309793621435	6.14996145285231e-05	-0.00149251220327071
0.00149377872507917	5.12963409860825e-05	0.00123167205653074	0.00123273978172087
0	-0.00175563411722169	0	0.00141284053501554	0.00149377872507917
10331	10331	1	9.24290436284945	-0.000230890640791369	-0.00147441795772886
0.0014923869478384	-0.000347632129299537	0.00129628666277216	0.00134209061147238
0	-0.00175580850836064	0	0.00141297439683093	0.0014923869478384
10211	10211	1	9.23122084955554	-0.000633103531925687	-0.00134320284045877
0.00148492893862073	-0.000631071689296745	0.00106646662668294	0.00123919431198676
0	-0.00176636674804379	0	0.00142107721241217	0.00148492893862073
10369	10369	1	9.24657586455828	4.67547186174683e-05	-0.00147559974971223
0.00147634028091894	7.99910522234896e-05	0.00124576018232505	0.00124832567878033
0	-0.00175250381884864	0	0.00141043758286627	0.00147634028091894
10453	10453	1	9.25464429753271	-0.000304478868200946	-0.00141117850295559
0.00144365236410464	-0.000223525782210025	0.00124767153432	0.00126753620574129
0	-0.00174526357211357	0	0.0014048785677242	0.00144365236410464
10271	10271	1	9.23707966916581	0.000151926677381929	-0.00139593924366338
0.00140418235542959	0.000199437478543235	0.00117395228955498	0.00119077255846742
0	-0.00176106420552525	0	0.00141700823205664	0.00140418235542959
10141	10141	1	9.224341891612	-2.69378011998342e-05	-0.00139213634163686
0.00139239694011425	1.72266295677679e-05	0.00116438745087312	0.00116451487432195
0	-0.001772613186228	0	0.00142586946325406	0.00139239694011425
10177	10177	1	9.22788555119193	0.000562556014238549	-0.00123881678217888
0.00136056469488371	0.000525626337530265	0.00105890848795867	0.00118218874659523
0	-0.00176939258053915	0	0.00142339876359955	0.00136056469488371
10061	10061	1	9.21642184229205	-0.000271184796372421	-0.0012905239456047
0.00131870893223739	-0.000199026354140983	0.00104720828811089	0.00106595341752386
0	-0.00177983266568738	0	0.00143140682870411	0.00131870893223739
10247	10247	1	9.23474025879972	0.000296226420166288	-0.00127916398853374
0.00131301584208496	0.000225798114775531	0.00100579696158174	0.00103083088650042
0	-0.00176317956235639	0	0.0014186315781825	0.00131301584208496
10457	10457	1	9.25502688959859	-0.00038302950951308	-0.00124755231657541
0.00130502811760916	-0.000336465701900575	0.0010635359416545	0.00111548996756868
0	-0.00174492100357133	0	0.00140461550806041	0.00130502811760916
10223	10223	1	9.23239536275701	4.37551338344118e-05	-0.00126550404521048
0.00126626024188591	6.46513618466203e-05	0.00112056448509538	0.00112242797713069
0	-0.00176530245864271	0	0.00142026058016567	0.00126626024188591
10099	10099	1	9.22019168802656	3.77615682350026e-05	-0.0012349486588066
0.00123552584996178	0.000122474628046823	0.00095475772183488	0.000962581083295603
0	-0.00177639258075131	0	0.00142876845882169	0.00123552584996178
10429	10429	1	9.25234566612121	0.000104275707715879	-0.00122341220360067
0.00122784805376671	0.000135594574873729	0.000939541687326359	0.000949275761282911
0	-0.00174732317276288	0	0.00140646007086086	0.00122784805376671
10343	10343	1	9.2440652413778	8.65573758937093e-05	-0.00120755464889939
0.00121065288559526	0.000151903149928664	0.000998017665518561	0.00100951167781525
0	-0.00175476293000098	0	0.00141217180262058	0.00121065288559526
10111	10111	1	9.2213792190914	9.1842043748573e-05	-0.00120508425512231
0.00120857892706419	5.50085507262679e-05	0.000972297490544918	0.00097385232493071
0	-0.00177531032055336	0	0.00142793835000022	0.00120857892706419
```

```
Frecuencias no nulas dominantes de F_Q-mean(F_Q):
    rank    demodulated_frequency       fit_amplitude
    ----    ---------------------    --------------------

      1          7.22212181819358      0.00102998542892402
      2        -0.509252179488009     0.000972670281121882
      3         -7.22212181819358     0.000962787079088038
      4         0.509252179488009     0.000944393206974707
      5         -7.26841747087431     0.000923900818886456
      6         0.694434790210921     0.000912309613268334
      7        -0.694434790210921     0.000907278710300437
      8          7.26841747087431     0.000828631152433364
      9         -2.82403481352441     0.000809343213775063
     10         -6.89805224942848     0.000799201794454691
     11          6.89805224942848     0.000798947722046012
     12          8.42580878789251     0.000771802707211477
     13         0.879617400933833     0.000761874424348131
     14        0.0462956526807281     0.000760539159031102
     15        -0.879617400933833     0.000742831699098273
     16         -5.32400005828373     0.000727744418239324
     17         -8.42580878789251     0.000703624825291971
     18         -8.56469574593469     0.000703431607127999
     19         -8.61099139861542     0.000697831699989827
     20          8.61099139861542     0.000697444147794987
     21          8.93506096738052     0.000693242426624136
     22         -9.02765227274197     0.000691823571238767
     23       -0.0462956526807281     0.000690811206825879
     24          8.56469574593469     0.000687798036575012
     25          9.02765227274197     0.000680600662464099
     26                         0      0.00067560438002473
     27         -8.33321748253105     0.000673228670039579
     28          3.84253917250043     0.000672369683136021
     29         -3.84253917250043       0.0006713463801668
     30         -6.24991311189829     0.000667528070597226


DIAGNOSTICO FINAL PARA Q=10000
mean(F_Q)                          = 0.499999870955 +0.0343125458703i
|mean(F_Q)-1/2|                   = 3.431255e-02
Var(F_Q)                           = 8.255940e-03
sup |F_Q-mean(F_Q)|                = 6.120282e-01
RMS E_Q                            = 9.086220e-02
razon omitidas potencia/null (-) = 3.353487
razon omitidas potencia/null (+) = 2.675593
fraccion explicada ajuste omitido= 0.014445

============================================================
TERMINADO
Tablas disponibles directamente en el workspace:
  radialAuditTable
  summaryTable
  coefficientCutoffDisplay
  groupSummaryTable
  groupPartialTable
  positivityTable
  blockQmax
  omittedTable
  uniquenessTable
  topFrequencyTable
```

==========
>>

43. Conclusion

The finite Möbius interpolation is completely oriented by

$$\Phi_N(s) = M(N) + h_N \frac{R_N(s)}{P_N(s)}. \tag{43.1}$$

Its value at zero, its integer limit, its exponential rate, its rounding criterion, and its recovery of a_N are mutually consistent. The absolutely convergent half-plane does not select $M(N)$ among feasible terminal values.

For the zeta phase walk, the following components are exact:

(i) the endpoint conversion and triangular half identity;
(ii) the positive atomic horizontal defect $D_q(\tau)$;
(iii) the two-branch coefficient algebra;
(iv) the convergent signed invariant $\Delta_\omega(Q) \sim \omega/(\pi^2 \log Q)$;
(v) the raw zero-kernel identity for the carrier coefficient;
(vi) the exact raw global and local defects (16.16) and (18.9);
(vii) the exact joint constant–carrier formulas (17.8)–(17.22), which apply to the coefficient reported by the program;
(viii) the exact four-term decomposition of the projected sampled residual variance in (20.28) and (20.29)– (20.30).

The finite computation shows strong prime-power structure, including the disappearance of the radial double mode after subtraction of $n = 3$, a 99.335% finite variance reduction through $Q = 10000$, stable real carrier coefficient near $1/2$, and detectable omitted prime-power frequencies. None of these finite observations proves uniform energy exhaustion.

The possible routes to the Riemann Hypothesis are now explicit: equality of atomic jumps for one fixed $q > 1$, local distributional convergence, small-q quadratic curvature, large-q subpolynomial growth, or local projected-energy exhaustion accompanied by derivative identification. The main unresolved statements are not ordinary Parseval estimates. They are uniform estimates for the projected zero-sampled kernel, together with a local explicit-formula limit strong enough to identify the positive atomic measure

$$\sum_{\tau} D_q(\tau)\delta_\tau. \tag{43.2}$$

Under the additional uniform Landau–Gonek approximation used in Heuristic 18.3, the *raw* termwise ratio is predicted to approach $\log n/\log(H/2\pi)$, rather than one, away from the near-height regime. This is a conditional obstruction, not an unconditional theorem. For the reported joint carrier there are further exact projection terms, displayed in (17.9). Consequently, any stable global relation between the reported centroid and $\Delta_\omega(Q)$ must be analysed through the collective joint defect (17.19).

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